Clear Current Technologies — Complete Research Compendium

Prepared for: Dan Schreiber (CPO) & John Reuter (CEO) **Team:** Clayton Hester | Jason Wisnieski | Adam Case | Victor Pan **CDL Texas MBA Consulting | April 2026**

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DOCUMENT 1: Clear Current Technologies — Master Research Synthesis (Primary + Secondary)

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Clear Current Technologies

Master Research Synthesis: Primary + Secondary Evidence Brief

Prepared for: Dan Schreiber (CPO) & John Reuter (CEO) **Team:** Clayton Hester | Jason Wisnieski | Adam Case | Victor Pan **CDL Texas MBA Consulting | April 2026 Version:** 2.0 — Updated to incorporate all 16 primary interviews

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This brief consolidates the full body of primary research conducted for Clear Current Technologies — 16 interviews across higher education, healthcare, QSR, hospitality, channel partners, and market advisors — and maps each finding to the secondary research assembled across six deep research sessions, three university databases (IBISWorld, ABI/Inform, Orbis), and a full claim verification audit. The goal is to show Dan and John where primary and secondary sources converge, where they diverge, and where gaps remain.

16 Interviews | **6** Research Sessions | **3** Database Sources | **12+** Stats Verified | **5** Deliverables

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01 — Full Interviewee Roster

#	Interviewee	Organization	Title	Vertical	Date
1	Andee Chamberlain, Carl Teague & James Norton	Texas State University	Energy & Water Conservation Manager + Team	Higher Ed	March 2026
2	Andi Ault	UT Austin	Sr. Associate Director, Finance & Administrative Services	Higher Ed	April 1, 2026
3	Troy Nevells	UT Austin	Director of Engineering & Technical Services	Higher Ed	April 14, 2026
4	Don Johnson	Trane Technologies	CEM, Sr. Account Manager – Pacific Southwest	Channel Partner	March 2026
5	Phil Combs	Trane Technologies	Sr. Digital Services Engineer, CMVP, CEM	Channel Partner	March 31, 2026
6	Marc Spieler	NVIDIA	Sr. Managing Director, Global Energy Industry	Strategic Advisor	April 3, 2026
7	Adam Zavadsky	ENGIE Impact	Senior Account Manager	Ecosystem/ Competitor	April 14, 2026
8	Jamare Bates	JLL	VP, Clean Energy & Infrastructure Advisory	CRE Advisory	April 2026
9	John Culver & Scott Sukits	AdventHealth Corporate	Corporate Sustainability + Director, Control Systems & Energy Mgmt	Healthcare	April 15, 2026
10	Ann Walston	Bon Secours Mercy Health	VP, Acute Care Facilities Management	Healthcare	April 14, 2026
11	Ross Snow, Matt Wilson & Bart Peacock	Intermountain Health	Director of Energy Management + Team	Healthcare	April 14, 2026
12	Scott Czubkowski	Medxcel (Ascension Health)	National Director, Energy & Facility Performance	Healthcare	April 14, 2026
13	Sean Sevy	Houston Methodist	Director of Operations, System Facilities	Healthcare	April 14, 2026
14	Roger Goldstein	Panda Restaurant Group	Executive Director, Facilities & Energy	QSR	April 2026
15	Walt Taylor	Panda Restaurant Group	Energy Manager (Contractor, GWT2Energy)	QSR	April 16, 2026
16	Randy Dawes	Hyatt Hotels Corporation	Director, Openings Engineering	Hospitality	April 2026

Vertical breakdown: Higher Ed (5 people, 2 institutions) · Healthcare (8 people, 5 systems) · QSR (2 people, 1 company) · Hospitality (1 person, 1 company) · Channel Partners (2 people) · Advisors / Ecosystem (3 people)

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02 — Primary Research: Who We Spoke To and Why It Matters

#	Interviewee	Organization	Role	Vertical	Key Contribution
1	Andee Chamberlain, Carl Teague, James Norton	Texas State University	Energy & Water Conservation Manager + Team	Higher Ed	Active EnergyCAP users; confirmed district energy metering gap and chargeback complexity
2	Andi Ault	UT Austin	Sr. Associate Director, Finance	Higher Ed	Confirmed dual-audience problem from finance side; surfaced budget-timing regulatory intelligence use case
3	Troy Nevells	UT Austin	Director of Engineering & Technical Services	Higher Ed	Engineering production lens; redirected ideal buyer profile toward multi-site commercial/ QSR
4	John Culver & Scott Sukits	AdventHealth Corporate	Corporate Sustainability + Director, Control Systems & Energy Mgmt	Healthcare	Surfaced ghost meter/address reconciliation problem; BrainBox AI stuck in IT review 2.5+ years
5	Ann Walston	Bon Secours Mercy Health	VP, Acute Care Facilities Management	Healthcare	Most detailed billing error recovery account; confirmed finance-side reactive pattern across 36 hospitals
6	Ross Snow, Matt Wilson, Bart Peacock	Intermountain Health	Director of Energy Management + Team	Healthcare	14-person energy team; third party missed $40K→$140K spike; two-layer optimization/ capital calendar
7	Scott Czubkowski	Medxcel (Ascension Health)	National Director, Energy & Facility Performance	Healthcare	$265M utility spend; dead meter detection; polar vortex gas nomination ROI ($7.2M exposure); Peer AI use cases
8	Sean Sevy	Houston Methodist	Director of Operations, System Facilities	Healthcare	35-year historical arc; billing-first wedge is right entry point; 3–5 year horizon for unified ops+billing
9	Roger Goldstein	Panda Restaurant Group	Executive Director, Facilities & Energy	QSR	Baseline drift problem — paid neighbor's electricity for a year; threshold systems structurally blind to slow errors
10	Walt Taylor			QSR	

#	Interviewee	Organization	Role	Vertical	Key Contribution
		GWT2Energy (Panda contractor)	Energy Manager (Contractor)		PSC rate case experience; confirmed ENGIE Impact/ Arcadia data relationship; no AI in Panda stack
11	Randy Dawes	Hyatt Hotels Corporation	Director, Openings Engineering	Hospitality	Owner-operator structure means corporate ≠ buyer; real buyer is asset manager or mgmt company
12	Don Johnson	Trane Technologies	CEM, Sr. Account Manager – Pacific Southwest	Channel Partner	Utility-side + solutions-side lens; validated fake energy manager persona and reactive buyer pattern
13	Phil Combs	Trane Technologies	Sr. Digital Services Engineer, CMVP, CEM	Channel Partner	Excel is primary daily tool; regulatory intelligence is "killer feature nobody has"
14	Marc Spieler	NVIDIA	Sr. MD, Global Energy Industry	Strategic Advisor	Three GTM tracks; Arcadia named as primary threat; property tax protest pricing model
15	Jamare Bates	JLL	VP, Clean Energy & Infrastructure Advisory	CRE Advisory	Energy becoming #1 site selection factor; legacy building inefficiency now a financial liability
16	Adam Zavadsky	ENGIE Impact	Senior Account Manager	Ecosystem	ENGIE's moat is the data (20 yrs history), not analytics; AI used internally, not client-facing

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Higher Education (3 institutions, 5 interviewees)

Texas State University — Andee Chamberlain, Carl Teague, James Norton. Active EnergyCAP users managing ~1,500 meters across 98 buildings in a district energy model with central steam and chilled water plants. This interview produced the richest on-the-ground EnergyCAP critique in the research program, confirmed the district energy metering problem, and validated the finance/ops dual-audience tension from the operations side.

UT Austin — Finance — Andi Ault, Sr. Associate Director for Finance. Finance-side perspective at a PEER Platinum district energy system with ~$20M annual gas spend. Confirmed the dual-audience problem from the finance chair. Identified the budget timing gap (budget locked in March before Austin finalizes rates) as the clearest real-world use case for regulatory intelligence.

UT Austin — Engineering — Troy Nevells, Director of Engineering & Technical Services. Production-side engineering perspective: UT generates ~100% of its own electricity via combined cycle, making external utility invoice volume low. Troy immediately redirected the value proposition toward multi-site commercial operators and QSR — specifically naming Panda — as the highest-fit market, validating Clear Current's beachhead thesis from an insider who understands the billing problem deeply.

Healthcare (5 systems, 8 interviewees)

AdventHealth — John Culver (Sustainability) & Scott Sukits (Director, Control Systems & Energy Management, Central Florida). 55 hospitals across 9 states; 73 different electric utilities; 1,000–2,000 utility bills processed daily. Surfaced the "ghost meter / address reconciliation" problem as the #1 automation wish. BrainBox AI via Trane stuck in IT/cybersecurity review for 2.5+ years — the clearest evidence of the IT security gate problem outside of Trane's own testimony.

Bon Secours Mercy Health — Ann Walston, VP Acute Care Facilities Management. 36 fully owned hospitals; all electricity and gas purchased from the grid; Utilities Procurement Steering Committee sets hedging strategy quarterly. Provided the most detailed account of a major billing error recovery in the entire research program: a billing error totaling $5–6M in a single event, with only $1M ultimately recovered — a partial, board-level outcome that required months of utility dispute and illustrated the "identified vs. realized" gap in the clearest terms of any interview. Uses third-party processor Get Choice; Energy Star as the primary benchmarking framework. Finance posture is reactive except for hedging; rate trend visibility would be the second proactive lever her team currently lacks.

Intermountain Health — Ross Snow (Director of Energy Management), Matt Wilson, Bart Peacock. 14-person energy management team — the largest internal hospital energy team encountered in this research program. 14 billion BAS data points in SkySpark. Third party missed a bill spike from ~$40K to ~$140K; internal accountant caught it. Described the two-layer model (optimization vs. capital replacement on opposite seasonal schedules) that is the most important operational calendar framework for healthcare.

Ascension / Medxcel — Scott Czubkowski, PE, CHC, CEM. Formerly managed $265M annual utility spend for Ascension; now at Medxcel. Built a three-team supply side / engineering / facilities performance structure. Transitioned to Peer AI (Constellation Energy) nine months prior. Provided the most sophisticated real-world AI use cases: dead meter detection ($32/month exact charge identified as inactive), polar vortex gas nomination alerting (single event cost $7.2M; proactive nomination estimated to be worth several million annually), and hospital-to-hospital benchmarking by square footage and climate zone.

Houston Methodist — Sean Sevy, Director of Operations, System Facilities. 8 hospitals + 300+ outpatient locations across Houston. Provided the best 35-year historical arc of energy management evolution (pneumatic → DDC → BAS → AI). Uses Energy Print for billing and Energy Star; Prism Energy as broker. Candidly assessed that a unified billing-plus-operations platform is likely 3–5 years from widespread adoption, but described Clear Current's current positioning as exactly right for the billing-first wedge.

QSR (1 company, 2 interviewees with distinct vantage points)

Roger Goldstein, Panda Restaurant Group — Executive Director, Facilities & Energy. Internal perspective on a 2,300-location chain outsourcing bill processing to NG Insight (third party). Produced the clearest real-world validation of the "baseline drift" problem: Panda paid a neighbor's electricity for an entire year because the charge was consistent enough to establish a new baseline — never flagged by NG Insight's threshold rules.

Walt Taylor, GWT2Energy (Panda contractor) — External energy consultant with unusual candor. One of the few interviewees with hands-on PSC rate case experience (successfully argued a utility's calculation methodology was inequitable). Confirmed the ENGIE Impact / Arcadia relationship, which has direct implications for the data access problem in the QSR vertical. No AI in use anywhere in Panda's current energy management process.

Hospitality (1 company, 1 interviewee)

Randy Dawes, Hyatt Hotels Corporation — Director, Openings Engineering. Hyatt's owner-operator structure means corporate does not touch utility invoices — billing responsibility falls to individual property owners, asset managers, and management companies. The right buyer in hospitality is not a brand's corporate team. Surfaced metrics fragmentation (different units across properties) and invoice automation (late arrivals causing late fees) as the two most clearly articulated pain points.

Channel Partners (2 interviewees)

Don Johnson, Trane Technologies — CEM with 9 years at El Paso Electric and 20 years on the solutions side. Dual-lens perspective on how utilities construct bills and how commercial customers misunderstand them. Validated the "fake energy manager" persona, the reactive buyer pattern, and billing as the entry point into the relationship.

Phil Combs, Trane Technologies — Sr. Digital Services Engineer, CMVP. Described Excel as the primary tool for daily energy analysis, even within Trane's own technology portfolio. Identified regulatory intelligence as the "killer feature nobody has" — the strongest language used about any feature in the entire research program. Identified bill format variation as the core technical barrier to previous AI energy tools.

Strategic Advisors & Ecosystem (3 interviewees)

Marc Spieler, NVIDIA — Sr. MD, Global Energy Industry; Board of EPRI. Identified three distinct GTM tracks (direct enterprise, sell to utilities, license to ERP platforms). Named Arcadia as the primary AI-native competitive threat. Proposed the property tax protest model as the right pricing analogy — evergreen % of savings, zero fee if nothing found. Described the identified-vs-realized error problem (recovered credits often go uncollected).

Jamare Bates, JLL — VP, Clean Energy & Infrastructure Advisory. Provided the most expansive macro framing: energy is becoming the #1 site selection factor, legacy buildings are now financial liabilities, and the office vacancy crisis compounds the energy inefficiency problem across CRE. Works with Fortune 200/500 clients and public sector / higher ed institutions.

Adam Zavadsky, ENGIE Impact — Senior Account Manager. Described ENGIE Impact's competitive moat as the data itself — 20 years of client utility data in their system — not the dashboard or analytics. Confirmed that ENGIE Impact uses AI primarily for internal bill processing efficiency, not client-facing analytics. Identified the structural data access problem: a new platform entering a relationship where ENGIE already holds all the historical data must either partner with ENGIE or displace them entirely.

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03 — Primary + Secondary Correlations: Where All 16 Interviews Map

The table below shows where primary interview findings are confirmed, quantified, or expanded by secondary research. "STRONG" means both streams agree with specific evidence from multiple interviews. "MODERATE" means directional agreement but one stream is weaker or coverage is partial. "EMERGING" means primary evidence is building but secondary data is limited.

Finding 1: The Reactive Buyer Is the Norm, Not the Exception

Primary Evidence (11 interviews): Don Johnson (Trane): "Most customers are still reactive — even hospitals." Phil Combs: "Tyranny of the urgent." Roger Goldstein (Panda): exception-only monitoring — if a $2,000 average bill jumps to $12,000, it gets flagged; otherwise, not. Walt Taylor (Panda): "management by exception — look at the top 20% of outliers and investigate those." Adam Zavadsky (ENGIE Impact): most clients reactive; ENGIE's value prop is labor avoidance, not proactive optimization. Ann Walston (Bon Secours): reactive except for hedging, which is the one proactive lever. Andi Ault (UT Austin): finance is reactive; hedging is the one proactive tool. Scott Sukits (AdventHealth): described wanting to get ahead of problems, not just respond to them. Sean Sevy (Houston Methodist): historically reactive; beginning to incorporate proactive BAS analytics. Jamare Bates (JLL): most clients arriving only after energy costs have already become a crisis.

Secondary Corroboration: EnergyCAP State of Utilities 2026: 76% of commercial energy buyers expect utility costs to rise but 69% remain reactive on cost reduction. ABI/Inform: 30%+ of facility managers report a deferred-maintenance mindset. Verdantix 2023: all 15 leading EMS vendors incorporating AI because buyers want automation over manual workflows.

Six-Thread Secondary Brief: fewer than 30% of multi-site commercial operators have a named, dedicated Energy Manager — meaning the reactive buyer is often also a part-time buyer, juggling energy alongside facilities, operations, or finance. At a 200-site operator, this translates to 400–1,000+ monthly invoices landing with no dedicated reviewer. Top 7 documented buying triggers (ranked by frequency in secondary research): (1) a large,

unexpected bill; (2) a CFO-level cost reduction mandate; (3) sustainability/ESG reporting requirement; (4) a new hire or external consultant identifying a gap; (5) a competitor adoption signal; (6) a regulatory compliance deadline; (7) a utility rate increase announcement. The fully loaded cost of a dedicated Energy Manager is $120K–$160K salary plus benefits, or $200K+ total — creating a strong internal incentive to delay hiring and tolerate reactive management. FTE benchmarks from secondary research: best-in-class operators run 200–500 locations per energy FTE; average commercial operators run far higher ratios with no dedicated FTE at all.

Deliverable Link: Engagement Trigger Map; User Journey Maps; Product Module Recommendations (push dashboard is not optional — the user will not go looking for what's wrong; it has to find them)

Strength: STRONG

Finding 2: The Dual-Audience Problem Is Structural and Unsolved

Primary Evidence (6 interviews): Andee Chamberlain (Texas State): "I care more about demand and usage — the bill processing folks care more about the bottom line." Andi Ault (UT Austin): confirmed from the finance side — when asked whether a cross-collaborative tool made sense, her immediate reaction was that the two audiences have completely different questions. Ann Walston (Bon Secours): finance-side focus on cost predictability, not operational efficiency; hedging as the primary lever. Scott Czubkowski (Ascension): described three structurally separated teams — supply side, engineering, and facilities performance — each with different data needs from the same underlying energy spend. Troy Nevells (UT Austin): engineering production lens vs. finance billing lens — same campus, completely different data relationships. John Culver & Scott Sukits (AdventHealth): sustainability reporting vs. operations optimization — John asks "are we on track for our 2030 goals?"; Scott asks "why is this building using 20% more than last month?"

Secondary Corroboration: EnergyCAP user reviews (G2/Capterra, 132 total): system described as not serving both audiences well. No vendor in the competitive landscape offers a single platform with separate views for energy operators and finance users confirmed in secondary competitive research.

Deliverable Link: User Journey Maps; Product Module Recommendations (chat for expert, push dashboard for finance — Dan's architecture is validated by six independent interviews)

Strength: STRONG

Finding 3: Billing Errors Are Real, Material, and Structurally Persistent

Primary Evidence (7 interviews): Roger Goldstein (Panda): paid a neighbor's electricity for an entire year — consistent enough to establish a new baseline, never flagged by NG Insight's threshold system. This is the clearest real-world validation of the "baseline drift" problem. Ann Walston (Bon Secours): over her career she has encountered billing errors totaling $5–6M in a single event, with $1M ultimately recovered — a partial recovery that required months of utility dispute, reached board-level visibility, and illustrated how identified

errors rarely convert fully to realized credits. Texas State (Andee Chamberlain): "We had a weird multiplier happen in one of our bills — they listed 35,000 gallons instead of 3,500, a 10× error in a pattern that should have been obvious." Intermountain Health (Ross Snow): a third party should have flagged a bill that spiked from ~$40K to ~$140K — they didn't; an internal accountant caught it. Scott Czubkowski (Ascension): Peer AI identified a dead meter paying exactly $32/month for months — a charge so consistent it never triggered threshold alerts under the previous system. Across Czubkowski's portfolio, ghost meters and vacated-site billings accumulated to more than $1M/year in payments for locations no longer drawing power — a recurring, systematic error class that threshold-based systems cannot detect. Walt Taylor (Panda): ENGIE Impact flags outliers but cannot catch baseline drift; the management-by-exception approach has a structural blind spot for systematic errors. Don Johnson (Trane): 1 in 5 utility bills contain some form of error — framing consistent with EnergyCAP's own marketing claims.

Secondary Corroboration: FERC audit: $108M mischarged at a single utility (FirstEnergy). Rhode Island PUC: 83% of internally flagged billing exceptions were sent to customers anyway. Harvard Law (2025): structural information asymmetry perpetuates billing errors. EnergyCAP: claims 1 in 5 bills contain errors; 98% of users take monthly action on flagged bills.

Six-Thread Secondary Brief adds three material data points: (1) UtiliSave, a specialist utility bill auditing firm, reports a 94% audit-hit rate — meaning 94% of commercial accounts they audit contain at least one correctable billing error — and has recovered more than $650 million in overcharges for clients since 1991. (2) ENGIE Impact internal estimates put the error rate in commercial utility bills at approximately 17%, meaning roughly 1 in 6 bills has a recoverable error — a more conservative but still significant figure than EnergyCAP's 1-in-5 claim, and from a credible competitor source. (3) Demand charges — the peak-demand component of a commercial electricity bill — represent 30–70% of total commercial electricity cost and are the single most opaque and most frequently miscalculated line item on a commercial utility bill; this is the structural reason billing errors are so persistent and so material.

Note on the "80% overpaying" statistic: This claim cannot be used — it is vendor-generated and fails the claim verification audit. The government-source framing (FERC, RI PUC, Harvard Law) and the UtiliSave/ENGIE Impact competitor data are the credible substitutes.

Deliverable Link: Product Module Recommendations; Engagement Trigger Map; Opportunity Heatmap

Strength: STRONG

Finding 4: Threshold-Based Exception Rules Have a Structural Blind Spot (New — Not in Original Synthesis)

Primary Evidence (4 interviews): Roger Goldstein (Panda): NG Insight's threshold rules work for spikes but miss gradual or systematic errors that establish a new baseline. Panda paid a neighboring tenant's electricity for a year because the charge was consistent, not anomalous. Scott Czubkowski (Ascension): Peer AI's dead meter

detection ($32/month exact, month after month) required pattern-matching across time — not threshold alerting. Walt Taylor (Panda): "management by exception" catches the top 20% of outliers but is blind to systematic errors that blend into the average. Ann Walston (Bon Secours): described meter failure events where the charge was wrong but not obviously wrong — the error was structural, not a spike.

Secondary Corroboration: Rhode Island PUC: 83% of flagged exceptions sent to customers anyway (system-level failure, not just individual errors). No secondary source specifically addresses the baseline drift problem — this is primarily a primary research finding.

Deliverable Link: Product Module Recommendations (anomaly detection must be framed as "pattern deviation detection," not "spike detection")

Strength: STRONG (primary) / EMERGING (secondary)

Finding 5: Regulatory Intelligence Is Confirmed White Space With Documented Business Cases

Primary Evidence (5 interviews): Phil Combs (Trane): "super powerful" — called regulatory intelligence stronger than any other feature discussed; described monitoring PUC filings, utility board meetings, rate hike proposals, and grid operator decisions as a capability no one currently offers. Andi Ault (UT Austin): budget locked in March before City of Austin finalizes rates for the July 1 fiscal year — the exact regulatory timing gap this feature solves. Walt Taylor (Panda): one of the few interviewees with hands-on PSC rate case experience — successfully argued a utility's calculation methodology was inequitable by making a quantitative case before the state commission. Ann Walston (Bon Secours): hedging is the only proactive lever available; forward-looking rate intelligence would give her a second lever. Sean Sevy (Houston Methodist): no current regulatory monitoring capability; described it as a gap he is aware of.

Secondary Corroboration: Ohio PUC mandates 275-day rate case timelines; AES Ohio 2024 rate case resulted in ~9% commercial rate increase with minimal advance notice. PPL Pennsylvania case filed October 2025 with final decision July 1, 2026 — the same day a university's new fiscal year begins. Boston University locks budget in March before final rates are known. Commercial energy CAGR is 5.9% vs. 3% budgeted escalator in most corporate forecasts. Zero competitors (EnergyCAP, Arcadia, Schneider, Gridium, Nectar) offer PUC docket monitoring as a product feature — confirmed white space.

Deep Secondary Research adds four quantified macro headwinds that make regulatory intelligence urgent rather than optional: (1) PJM capacity market surge — PJM capacity prices spiked from approximately $10,000/MW in 2024 to a projected $120,000/MW by the 2027/28 delivery year, a 12× increase that translates to approximately $110,000 of unbudgeted annual capacity cost per megawatt of peak demand for any commercial facility in the PJM footprint (Pennsylvania, Ohio, Maryland, New Jersey, Virginia, and parts of 10 other states). (2) Rate increase breadth — 97.5% of commercial facilities in the U.S. experienced at least one electric rate increase between 2020 and 2025; 71% of those rate increases outpaced general inflation over the same period. (3) ERCOT wholesale volatility — ERCOT wholesale power prices rose approximately 45% from 2025 to 2026, driven by

data center load growth in the DFW and Austin corridors, directly relevant to the Texas-first beachhead strategy. (4) State-level severity — New Jersey commercial rates increased 9.1% in 2024–2025; Rhode Island increased 8.6%; California's commercial rate schedule increased 7.3% — all above the 3% corporate budget escalator, creating a documented budget gap for any customer that locked annual utility budgets before rates were finalized.

Deliverable Link: Product Module Recommendations; Engagement Trigger Map

Strength: STRONG

Finding 6: The IT Security Gate Is a Real and Recurring Enterprise Adoption Barrier

Primary Evidence (3 interviews): Phil Combs (Trane): "You've got guys like me that would love to use this, and then IT says no, no, no." Trane internally restricts AI tools its own employees want to use, despite selling AI to clients. AdventHealth (Scott Sukits): BrainBox AI — delivered via Trane, already contracted — has been stuck in IT/cybersecurity review for 2.5+ years in Central Florida. The AdventHealth IT posture hardened significantly following the Johnson Controls ransomware attack, which hit the healthcare sector broadly and triggered enterprise-wide AI vendor review protocols; this context explains why the delay is measured in years, not months. Not a concern or a slow process: 2.5 years of completed security review with no end date. Scott Czubkowski (Ascension): SOC 2 compliance referenced as the primary threshold for IT sign-off; Peer AI (Constellation Energy) cleared it — making SOC 2 Type II the documented compliance path that converted a healthcare IT security gate into a signed contract.

Secondary Corroboration: Enterprise SaaS literature confirms this is the dominant dynamic across cloud software adoption, not an edge case. 76% of organizations are actively increasing SaaS security budgets (secondary research). Commercial utility data — billing histories, rate structures, PPA terms — is classified as competitively sensitive, making IT's concern rational, not obstructionist. SOC 2 Type II is the primary certification that satisfies IT gatekeepers in the enterprise energy buyer context.

Deep Secondary Research quantifies the security gate with four statistics that should be included in investor and sales materials: (1) Forrester Research: 94% of enterprise IT departments report veto power over SaaS procurement — a single IT department can kill a deal regardless of end-user demand; (2) 45% of enterprise buyers report having abandoned a SaaS deployment mid-cycle specifically because the vendor failed to meet security requirements; (3) vendors with SOC 2 Type II certification win at a 62% higher rate in competitive enterprise deals than vendors without it; (4) SOC 2 certification shortens sales cycles by 3.2 months on average and increases average deal size by $100K+. On the cost side: SOC 2 Type II typically costs $20K–$80K and takes 6–12 months to complete — material for a pre-revenue company but a one-time investment that clears the single largest adoption barrier in the target verticals.

Action: Clear Current should lead with SOC 2 Type II in all materials before it is asked for. The AdventHealth example (2.5-year delay) suggests this is not a checkbox — it needs to be a proactive part of the sales narrative. The Forrester 94% statistic makes it table stakes, not a differentiator.

Deliverable Link: Engagement Trigger Map; User Journey Maps

Strength: STRONG

Finding 7: Excel Is the Real Incumbent for Sophisticated Energy Analysts

Primary Evidence (4 interviews): Phil Combs (Trane): Excel spreadsheets are the primary tool for day-to-day energy analysis — even within Trane's own technology portfolio. Specialized software is only used for guaranteed savings contracts. Roger Goldstein (Panda): NG Insight data extracted and pulled to Excel for scrubbing and year-over-year trend monitoring. Walt Taylor (Panda): described Excel-based exception management workflows as standard across his QSR client base. Troy Nevells (UT Austin): similar Excel-based reconciliation described from his Yale experience.

Secondary Corroboration: Verdantix 2023: all 15 leading EMS vendors incorporating AI because buyers want automation over manual spreadsheet workflows. EnergyCAP user reviews reference spreadsheet-based workarounds as common even among paying customers.

Deliverable Link: Product Module Recommendations; User Journey Maps (the bar for adoption is "faster and more insightful than a well-structured Excel workbook")

Strength: STRONG

Finding 8: EnergyCAP Is the Incumbent in Higher Ed — With Specific, Exploitable Gaps

Primary Evidence (3 interviews): Texas State (Andee Chamberlain): active EnergyCAP users who described specific gaps around district energy metering, blending sub-meter data with utility data, and chargeback complexity. Andi Ault (UT Austin): uses a billing system with limited flexibility; confirmed EnergyCAP limitations from the finance perspective. Troy Nevells (UT Austin): aware of EnergyCAP from Yale experience; noted it is best suited to settings with high external utility invoice volume (which UT Austin's self-generation model minimizes).

Secondary Corroboration: 132 G2/Capterra reviews confirm: "outdated, un-intuitive, glorified spreadsheet"; chargebacks "hard to understand"; report customization refused; $50–75K/year for 1,500 meters via reseller. EnergyCAP's "Watts AI" is nascent, retrofit, and rides on top of a tree-view navigation metaphor — not a conversational interface. No AI-native competitor has published campus logos at CMU/UCSD scale.

GTM Part 4 Research and the EnergyCAP competitive brief add three pricing reference points that anchor Clear Current's own pricing strategy: (1) EnergyCAP's list pricing is $40–$52/meter/year for the core UtilityManagement product, with implementation fees and managed services layered on top; (2) Texas State University (a confirmed EnergyCAP customer from primary research) is paying approximately $50K–$75K/year for 1,500 meters through a reseller, implying a blended rate of $33–$50/meter/year after reseller margin; (3)

University of New Mexico (Lobo Energy) is documented at $9,600/year — a substantially lower contract, suggesting EnergyCAP prices smaller institutions at deeply discounted rates. The practical implication: Clear Current can price at or above EnergyCAP on a per-meter basis in higher ed while offering meaningfully better conversational AI, without triggering sticker shock. On the competitive disruption window: EnergyCAP has been owned by Resurgens Technology Partners since March 2021 — a five-year hold at typical PE holding period. A Resurgens exit or recap in 2026–2027 would create a 6–12 month window during which customers reconsider contracts, integration roadmaps, and vendor relationships — the classic PE-backed SaaS disruption window.

Deliverable Link: Competitive Positioning; Opportunity Heatmap; Product Module Recommendations

Strength: STRONG

Finding 9: Arcadia's Competitive Gap Is Structural — And the Data Access Problem Is Now Confirmed (Updated Finding)

Primary Evidence (3 interviews): Marc Spieler (NVIDIA): named Arcadia as the primary AI-native competitive threat; raised the "why aren't people just signing?" conversion gap question. Walt Taylor (Panda): confirmed the ENGIE Impact / Arcadia relationship — a commercially significant finding about the data access dynamics in the QSR vertical. ENGIE Impact manages all of Panda's utility data; the relationship between ENGIE Impact and Arcadia creates a data-layer dependency that a new entrant must navigate. Adam Zavadsky (ENGIE Impact): ENGIE's moat is the data — 20 years of client utility history — not the analytics. AI is being used internally at ENGIE for bill processing efficiency, not surfaced to clients.

Secondary Corroboration: Arcadia has zero documented higher education customers after 14+ months of enterprise sales. Arcadia's enterprise product is built on a data/API architecture for developers, not a workflow tool for energy managers. Arcadia is simultaneously selling new enterprise accounts while migrating existing customers off the legacy Urjanet/Arc platform — PM job postings confirm this is the primary PM responsibility. No natural language chat interface exists in Arcadia's product. Arcadia enterprise customer list (Cox Enterprises, Arconic, Iron Mountain, Adobe, Intuit) is entirely corporate Fortune 500 — no university, no hospital, no district energy reference. Arcadia's Trustpilot score is 2.6/5 across 1,159 reviews — driven largely by its consumer-facing Perch brand, which is pending spinout and separation from the enterprise product. Arcadia's total disclosed funding is $649M at a last reported $1.5B valuation; the 12–24 month migration window created by the Urjanet/Arc platform transition is the primary competitive window documented in secondary research.

ENGIE Impact OCR coverage is approximately 90% — meaning roughly 10% of commercial utility bills processed through ENGIE's system are not captured electronically. This gap concentrates at the franchisee tier and smaller regional utilities, which is precisely where QSR and multi-site retail operators have the highest location density. ENGIE uses Arcadia's Arc API feeds for electronic bill data, meaning ENGIE's bill intelligence moat is the historical client data custody, not a proprietary data infrastructure.

Trane completed its acquisition of BrainBox AI in January 2025. BrainBox builds AI-based HVAC/BAS optimization using real-time sensors and is not a utility bill intelligence platform. The correct competitive framing: when Clear Current identifies HVAC-driven energy drift, Trane/BrainBox is the resolution pathway — a referral endpoint, not a competitor.

New Implication: The ENGIE Impact / Arcadia data relationship in QSR creates a structural access problem for Clear Current in that vertical. In healthcare and higher education — where ENGIE Impact is less dominant as the primary bill processor — this barrier does not apply, which reinforces the priority of those verticals. The bolt-on analytics framing (Walt Taylor: "just bolt you on") is the right entry strategy in ENGIE-heavy accounts rather than direct displacement at contract renewal.

Deliverable Link: Competitive Positioning; Opportunity Heatmap; Product Module Recommendations

Strength: STRONG

Finding 10: Healthcare Is a Stronger Beachhead Than Initially Documented (New — Not in Original Synthesis)

Primary Evidence (8 interviews across 5 systems): This finding emerges from the depth and consistency of healthcare interviews rather than any single quote. Across five health systems, the research team found: (1) massive scale — 36 to 55 hospitals per system, 1,000–2,000 bills processed daily at AdventHealth; (2) sophisticated energy management teams with dedicated budgets; (3) real billing error experiences with documented dollar amounts; (4) sophisticated use of third-party processors (ENGIE Impact, Get Choice, Peer AI / Constellation) that create either partnership opportunities or competitive positioning; (5) willingness to pay for platforms that reduce budget uncertainty and catch errors; and (6) IT security gate as the primary adoption barrier, not price or product fit.

Scott Czubkowski (Ascension): detailed polar vortex ROI story — one event cost $7.2M in three days; proactive gas nomination alerting is now worth several million annually. This is the clearest documented ROI case for weather-integrated regulatory intelligence in the entire research program.

Ann Walston (Bon Secours): described billing error recovery as rare and difficult — a pain with no current software solution in her stack.

Ross Snow (Intermountain): third party missed a 3.5× bill spike; internal accountant caught it. A sophisticated 14-person team with $14B of BAS data still has a gap in utility invoice anomaly detection.

Secondary Corroboration: Secondary research documented 660+ district energy systems in the U.S. concentrated in universities and hospitals. Healthcare is EnergyCAP's secondary vertical (higher ed is primary) — creating an opening for a challenger. No AI-native competitor has published hospital logos with specific billing intelligence use cases.

Deliverable Link: Opportunity Heatmap; Product Module Recommendations; GTM Beachhead Strategy

Strength: STRONG (primary) / MODERATE (secondary — secondary did not originally cover healthcare depth)

Finding 11: The "Ghost Meter / Address Reconciliation" Problem Is Real in Healthcare M&A (New — Not in Original Synthesis)

Primary Evidence (1 interview, specifically surfaced): AdventHealth (John Culver): described the #1 automation wish as basic data accuracy — consumption occurring at one address while the bill is paid somewhere else. In a 55-hospital system in active M&A, mergers, acquisitions, new builds, and divestments constantly change the portfolio, and keeping billing addresses in sync with physical locations is a continuous, imperfect process. ENGIE Impact sometimes catches this mismatch; sometimes it doesn't, depending on what the utility sends.

Secondary Corroboration: No secondary source specifically addresses the ghost meter / address reconciliation problem in the M&A context — this is a primary-first finding.

Deliverable Link: Product Module Recommendations (meter-to-address reconciliation should be an explicit feature, not an edge case)

Strength: EMERGING (primary only; awaiting secondary corroboration)

Finding 12: The Owner-Operator Structure in Hospitality Changes the Buyer (New — Not in Original Synthesis)

Primary Evidence (1 interview): Randy Dawes (Hyatt): corporate does not touch utility invoices; billing responsibility falls to individual property owners, asset managers, and management companies. Invoice automation (solving late arrivals and late fees) is the most clearly articulated pain. Metrics fragmentation (different units across properties) is the second. The right buyer is not the brand's corporate team — it is the asset manager or management company running multiple properties.

Secondary Corroboration: Secondary research on CRE buyer behavior identified property managers and asset managers as the primary energy decision-maker in multi-tenant or managed property contexts, consistent with Hyatt's owner-operator structure.

Deliverable Link: Opportunity Heatmap (hospitality should be scored against asset manager buyer, not brand corporate buyer)

Strength: MODERATE

Finding 13: Higher Ed Buying Cycle Is Calendar-Driven and Predictable (Reinforced)

Primary Evidence (4 interviews): Andee Chamberlain (Texas State): January–February budget planning; March–May gas purchasing and budget approval; September 1 new fiscal year. Andi Ault (UT Austin): budget submitted in March before City of Austin finalizes rates — the exact regulatory timing gap. Troy Nevells (UT Austin): capital equipment cycle is inverse to operations schedule (cooling replaced in winter; heating replaced in summer). Intermountain Health (Ross Snow): two-layer model — optimization calendar is seasonal; capital replacement calendar is counter-seasonal.

Secondary Corroboration: University of Wisconsin RFP: $500K–$2M contract value issued September with October deadline. Part 4 research: higher ed Fall awareness, Winter budget proposals, Spring RFP, Summer execution.

Deliverable Link: Seasonal Energy Management Calendar; Engagement Trigger Map

Strength: STRONG

Finding 14: ERCOT Deregulation Creates a Defensible Texas Moat (Reinforced)

Primary Evidence (3 interviews): Marc Spieler: deregulated markets create billing complexity that is difficult for competitors to replicate from Boston or DC. Walt Taylor (Panda): PSC rate case experience in a deregulated state — the complexity of managing REP relationships, overlapping billing cycles during supplier transitions, and hedging decisions is directly navigable for a Texas-based team. Don Johnson (Trane): El Paso Electric background; ERCOT's commercial customer dynamics well understood.

Secondary Corroboration: ERCOT: 100+ REPs, 87% commercial switching rate, overlapping billing cycles during supplier transitions. 18 deregulated states mapped in Part 4 — Texas has highest billing error exposure from supplier switching.

Deliverable Link: Opportunity Heatmap; GTM Sequencing

Strength: STRONG

Finding 15: The Three GTM Tracks — Marc Spieler's Framework (New — Not in Original Synthesis)

Primary Evidence (1 interview, strategic framework): Marc Spieler (NVIDIA) identified three parallel GTM strategies that address different buyer types and distribution economics:

Track 1 — Direct to Enterprise: continue founder-led direct sales to large, complex organizations. Current approach. Highest friction, highest margin, right for pre-revenue stage.

Track 2 — Sell to Utilities: offer billing accuracy as a service to the utility itself. Value proposition: catching underbilling (lost revenue) and preventing overbilling (regulatory risk, customer service cost, potential legal exposure). Utilities have whole commercial/industrial customer management teams; C&I customers account for 80% of power consumption.

Track 3 — License to ERP / Billing Software: embed Clear Current's AI audit capability as a module inside existing ERP stacks — Oracle, SAP, Microsoft Dynamics, JD Edwards, QuickBooks. Removes direct sales burden; ERP vendor becomes distribution channel.

Secondary Corroboration: Part 4 research confirmed the land-and-expand motion as the right GTM strategy for a pre-revenue SaaS company in a category where buyers are skeptical. ERP distribution channel not explored in secondary research — this is a primary-first strategic finding.

Deep Secondary adds one material data point for the ERP track: SAP integration projects for mid-to-large commercial organizations typically require 12–24 months from contract to production — meaning any ERP-embedded module strategy has a long lead time, but also creates high switching costs once embedded. This makes Track 3 (ERP licensing) a 24–36 month horizon play, not a near-term revenue source. Marc Spieler's framing of Track 1 (direct enterprise) as the right pre-revenue approach is confirmed by secondary research — ERP licensing only makes sense after the product has proven itself in direct deployments.

Deliverable Link: GTM Strategy; Investor Readiness Narrative

Strength: EMERGING (primary first; secondary adds ERP timeline context)

Finding 16: The Property Tax Protest Pricing Model Is a Better Analogy Than SaaS Subscription (New — Not in Original Synthesis)

Primary Evidence (1 interview, pricing framework): Marc Spieler (NVIDIA): a property tax protest service operates on an evergreen agreement with a 50% savings share. Every year, the firm protests the tax increase and invoices 50% of the savings. If they don't win, the customer pays nothing. The model works because it is automated, recurring, low-friction, and the value is proven every cycle. Marc also described the savings-sharing trap: as soon as recoveries become large, clients push to renegotiate to a fixed fee. The right structure is a low flat monthly fee plus a percentage of identified savings — the fee creates a revenue floor and avoids the renegotiation dynamic.

Secondary Corroboration: Part 4 research confirmed willingness-to-pay dynamics in the enterprise energy buyer market, and GTM Part 4 adds substantial secondary data validating the pricing model and anchoring it with real numbers.

GTM Part 4 quantifies the pricing landscape across four dimensions: (1) **Pilot proof point** — across 12 beta/pilot deployments in comparable utility bill intelligence platforms, the median identified error pool was $10M — meaning a typical pilot produces $10M in identifiable billing errors, of which a percentage is recoverable. This is

the primary investor proof point: the product finds money, which makes the pricing conversation straightforward. (2) **Contingency fee benchmarks** — the contingency fee model is well-established in utility audit services: standard range is 20–50% of recovered overcharges; UtiliSave, the most documented specialist, charges 40%; property tax protest services (Marc Spieler's analogy) typically charge 30–50%. A 20–30% contingency rate positions Clear Current at the lower-cost end of the market while remaining highly credible. (3) **Pricing ladder** — GTM Part 4 documents a three-tier structure benchmarked against competitive platforms: an Essentials tier (basic invoice ingestion and anomaly alerts) at $25–$50/meter/year; an Optimize tier (regulatory intelligence, trend analysis, conversational Q&A) at $75–$100/meter/year; and an Enterprise Platform tier (API access, custom reporting, dedicated CSM) at $120K–$250K/year flat. (4) **ROI benchmarks** — energy management software deployments in the secondary literature produce an average of 7.5% savings on total utility spend, with a median payback period of 2.6 years; at $500K annual utility spend (small healthcare or QSR portfolio), that is $37,500/year in savings, making the willingness-to-pay range of $15K–$40K/year rational and sustainable.

Note on the property tax protest model: The structural insight from Marc Spieler is confirmed by secondary pricing research — the flat-fee-plus-contingency structure (low monthly platform fee + % of identified savings) creates a revenue floor while avoiding the renegotiation dynamic that pure contingency arrangements trigger once savings become large and predictable.

Deliverable Link: Product Module Recommendations; Investor Readiness Narrative

Strength: MODERATE (primary framework strong; secondary adds specific price anchors)

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04 — The Five Findings That Should Shape Every Deliverable

These are the insights that appear in both primary and secondary research across multiple independent interviews with enough consistency and specificity to anchor every deliverable.

FINDING 1 — The Reactive Buyer Is Universal Across Verticals

Eleven of sixteen interviews confirmed that commercial energy buyers operate reactively. The pattern holds across higher education (Texas State, UT Austin), healthcare (Bon Secours, Houston Methodist, AdventHealth), QSR (Panda), and advisory (ENGIE Impact, JLL). The only exception is Intermountain Health, which operates an unusually sophisticated 14-person proactive team — and even they missed a $40K-to-$140K spike via their third-party processor. EnergyCAP's own 2026 survey found 76% of buyers reactive despite knowing costs will rise. The product must surface insights proactively. The push dashboard layer is not optional.

FINDING 2 — The Dual-Audience Problem Is Structural, Documented by Both Verticals, and Unsolved by Any Competitor

Six interviews across higher education and healthcare confirmed the finance/operations split from multiple chairs. Andee Chamberlain described it from operations; Andi Ault confirmed it from finance. Ann Walston (Bon Secours), Scott Czubkowski (Ascension), Troy Nevells, and the AdventHealth team all described versions of the same split. No current tool — EnergyCAP, Arcadia, Peer AI — serves both audiences from a single platform. This is Clear Current's clearest product differentiation opportunity.

FINDING 3 — Billing Errors Are Government-Documented, Not Vendor Folklore, and Threshold Systems Miss the Worst Ones

Seven interviews produced specific, named billing error examples — not anecdotes but specific incidents with dollar amounts. The most important new finding: threshold-based exception systems (NG Insight at Panda, Peer AI's predecessor at Ascension) miss gradual and systematic errors that establish a new baseline. FERC confirmed $108M mischarged at a single utility; Rhode Island's PUC found 83% of flagged exceptions sent to customers anyway. This framing — regulatory enforcement, not vendor marketing — is how Clear Current should position the billing error opportunity in every investor and customer conversation.

FINDING 4 — Regulatory Intelligence Is White Space With a Quantified Business Case Across Two Verticals

Five interviews produced specific, independent evidence for the regulatory intelligence use case: Phil Combs called it the "killer feature nobody has"; Andi Ault's March budget / July rate decision is the archetypal use case; Walt Taylor has actually won a PSC rate case and understands the process deeply; Ann Walston identified rate trend visibility as the second proactive lever her finance team lacks; Sean Sevy acknowledged the gap explicitly. Zero competitors offer PUC docket monitoring. Scott Czubkowski's polar vortex story provides the healthcare-specific version — weather-triggered energy procurement alerting estimated at several million dollars annually for a single health system.

FINDING 5 — The Right Beachhead Is Healthcare First, Higher Ed Second — Not the Reverse

The original synthesis prioritized higher education based on the first five interviews. With eight interviews across five healthcare systems, a stronger picture has emerged. Healthcare systems — 36 to 55 hospitals per organization, $265M annual utility spend at Ascension, 1,000–2,000 daily bills at AdventHealth — represent larger contracts, more sophisticated buyers, and a less fortified incumbent position than higher education. EnergyCAP's higher ed moat (campus chargebacks, district energy) is real and documented. Healthcare is EnergyCAP's secondary vertical — the beachhead is cleaner. The IT security gate applies in both verticals but healthcare IT is better resourced to clear it quickly when the value case is made clearly.

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05 — Market Sizing, Investor Benchmarks & Comparable Transactions

This section compiles secondary data relevant to Clear Current's investor narrative — market size, Series A thresholds, and comparable M&A transactions in the energy intelligence and facility management SaaS space. These data points were not tied to individual primary interviews but are directly relevant to fundraising positioning.

Series A Benchmarks for B2B SaaS (GTM Part 4, Thread 2)

GTM Part 4 secondary research surfaced the following Series A medians for enterprise B2B SaaS companies in adjacent verticals (energy tech, facility management, sustainability software):

Metric	Series A Median	Range
ARR at raise	$1.5M–$2.0M	$800K–$4M
Net Revenue Retention (NRR)	101–104%	95–115%
Burn multiple	1.6×	1.2×–2.5×
Gross margin	68–72%	60–80%
Customer count	15–35 logos	8–60
Lead investor check	$6M–$10M	$4M–$15M

The $10M identified-error proof point from the 12-pilot benchmark is the most investable metric in Clear Current's current story — it directly answers the investor's core question ("does the product find money?") with a specific, secondary-corroborated number.

Comparable M&A Transactions — Energy Intelligence & Facility SaaS

Company	Acquirer	Year	Revenue at Exit	Exit Multiple	Key Insight
Brightly Software (formerly Dude Solutions)	Siemens	2022	~$160M ARR	~9.8× ARR ($1.575B)	Facility management SaaS; highest comp in space
EnerNOC	Enel	2018	~$200M revenue	1.5× revenue	Demand response + analytics; pre-AI
Lucid (BuildingOS)	Verdigris	2019	~$10M ARR	~6× ARR	Energy monitoring SaaS; early AI
Urjanet	Arcadia	2022	Undisclosed	Strategic	Bill data API layer; now inside Arc platform; Arcadia total raised $649M post-acquisition
Wattics	EnergyCAP	2022	~€3M ARR (est.)	Strategic	Interval data analytics; Dublin
Carbon Lighthouse	Hines	2021	~$15M revenue	Strategic	Building efficiency; CRE-focused

The Brightly transaction at 9.8× ARR is the ceiling comp for a well-positioned facility/energy SaaS business with scale. The Arcadia/Urjanet deal is the closest structural analog to Clear Current's data-and-AI positioning — and Arcadia subsequently raised $649M (last valuation $1.5B) and serves 1.35M accounts, establishing the scale ceiling for the category. **Note:** earlier drafts of this synthesis used $566M; the $649M figure reflects the most recent disclosed funding total as used in the D6 Competitive Intelligence Brief and should be treated as current until Arcadia's next public disclosure.

Market Size — Secondary Research Summary

Segment	Market Size (2025)	CAGR	Source
Global utility bill management software	$2.1B	12.4%	MarketsandMarkets 2025
Energy management software (enterprise)	$8.6B	14.2%	Grand View Research 2025
Utility analytics & AI	$4.3B	18.7%	IDC 2025
U.S. commercial electricity spend (annual)	$241B	5.9%	EIA CBECS 2018, projected forward
U.S. commercial energy spend — all fuels (annual)	$380B	5.9%	EIA 2025 (electricity + gas + other)
Addressable error pool (1-in-5 bills × avg commercial spend)	~$11–19B	—	Derived from EIA electricity spend + EnergyCAP/ UtiliSave error rates

TAM note for investor materials: The $241B figure (electricity-only, EIA CBECS 2018 projected at 5.9% CAGR) is the anchor used in the investor narrative — it is the more conservative and auditable number. The $380B figure covers all commercial energy including natural gas and other fuels; both are accurate but serve different framing purposes. Use $241B as the primary investor TAM and clarify the scope if challenged.

BEMS market penetration: Only 4% of U.S. commercial buildings had a Building Energy Management System as of the most recent EIA Commercial Buildings Energy Consumption Survey (CBECS). This penetration figure — from a government source — is the most credible proof of how early the commercial energy software category is, and belongs in the investor narrative's "why now" beat.

Demand charge context (NREL): NREL analysis documents that demand charges represent 30–70% of a typical commercial electricity bill. A practical example from NREL: a commercial customer hitting a 1,000 kW summer peak under an 80% ratchet clause locks approximately $144,000/year in demand charges regardless of subsequent usage — illustrating why ratchet detection and demand charge explanation are high-value product features. This government-sourced math replaces any vendor claim about billing error magnitude in investor and sales materials.

Serviceable Obtainable Market — 24-Month Targets (Aligned to D7 Investor Narrative)

Segment	Target Accounts	ACV Range	24-Month Revenue
Higher education (beachhead)	10–15	$30K–$75K	$300K–$1.1M
Healthcare systems	3–5	$75K–$150K	$225K–$750K
Multi-site commercial / QSR (ERCOT-first)	5–10	$25K–$50K	$125K–$500K
Total SOM (24-month)	**18–30**	—	**$650K–$2.4M**

The SOM range of $650K–$2.4M across 18–30 accounts represents the realistic fundraise narrative for a Series A raise in 2026, consistent with the $1.5–2.0M ARR median documented for comparable B2B SaaS raises in adjacent verticals.

CRE Vertical — Competitive Data

Platform	Scale / Backing	Primary Strength	Relevance to CC
WatchWire (Tango)	500+ CRE customers; Berkshire-backed via 2023 acquisition	GRESB, CDP, GRI reporting; building performance standards	CRE sustainability/reporting lead; limited bill-intelligence depth
Measurabl	Free tier onboarded 11,000+ buildings in 4 weeks (2024)	ESG data collection; GRESB workflow automation	ESG data aggregation at scale; not a tariff-truth or NL bill-Q&A platform
JLL Technologies	Internal AI energy-management and utility-bill intelligence build in active development	Custodied client data across global Fortune 200 portfolio	Most significant CRE competitive signal — enterprise CRE buyer may "build first" not "buy first"

JLL Technologies internal-build flag: Jamare Bates (JLL, primary interview) described JLL Technologies as actively building internal AI energy-management and utility-bill intelligence tooling on data they already hold across their client portfolio. This is the most significant finding for CRE vertical positioning: the enterprise CRE buyer in this segment may already be building the product Clear Current would sell them. CRE beachhead strategy should lean toward partnership or licensing rather than direct SaaS displacement until the JLL Technologies roadmap is clearer. This finding was not captured in original secondary competitive research and is primary-research-first.

GRESB 2025 scale: GRESB engaged 1,002 fund managers and processed 2,382 assessments in 2025 — confirming the ESG reporting compliance demand in CRE is real and growing, even if the primary buyer for Clear Current in that vertical is more complex than initially modeled.

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06 — Research Gaps Identified

The following areas lack sufficient primary or secondary coverage to support strong claims:

Commercial Real Estate Direct Buyer Voice — No primary interview completed with a REIT energy manager, property manager, or CRE operator. Jamare Bates (JLL) provided the advisory perspective. D5 heatmap scores CRE lowest on buyer readiness (2/5) partly for this reason. Additionally, JLL Technologies is actively building internal AI energy-management and utility-bill tooling on custodied client data — a finding from the Jamare Bates primary interview that materially changes the CRE GTM thesis. CRE beachhead strategy should lean toward partnership/licensing until the JLL Technologies internal-build roadmap is clearer.

Hospitality Direct Buyer Voice — Randy Dawes (Hyatt) confirmed the owner-operator structure shifts the real buyer to asset managers and management companies — but no interview was completed with that actual buyer. D5 heatmap scores hospitality at 13/25 (lowest) and explicitly flags "No primary interview captured for this vertical yet." NYC Local Law 97 (annual GHG emissions report due May 1 for buildings over 25,000 sq ft, with fines beginning 2024) is a documented CRE/hospitality compliance trigger with no primary voice of customer behind it. This trigger is catalogued in D1 (REG-04) based on secondary research only.

Sewer/Water Overbilling — Walt Taylor (Panda) flagged that most QSR operators have no meter for how much water is actually leaving the restaurant — creating a structural blind spot for sewer/wastewater billing errors that compound water charges. D6 flags this as "EMERGING — one primary, no secondary." No secondary research on commercial sewer/water billing error rates or detection methods.

Water Billing as a Parallel Use Case — Andee Chamberlain (Texas State) flagged water as increasingly complex and expensive; UT Austin operates its own wells. No secondary research on commercial water billing error rates.

Multi-Family / Apartment Billing — Not covered in primary or secondary research. Billee Technologies ($9.15M seed, July 2025) is active here per secondary research.

Utility-Side Buyer Persona — Marc Spieler identified utilities as a potential second GTM track (D7 Beat 3). No primary interview with a utility commercial/industrial customer management team; no secondary research on utility-side procurement software spend.

ERP Integration Economics — Marc Spieler identified licensing to ERP platforms as a viable third GTM track. SAP integration timelines run 12–24 months making this a 24–36 month horizon play. No secondary research on ERP marketplace economics or billing software integration commission structures.

Willingness-to-Pay Primary Validation — No primary interview produced a named dollar figure a buyer would actually commit to paying for a standalone analytics layer. All pricing validation in Section 05 is secondary benchmark-derived. Two interviews with a healthcare CFO or QSR energy director on actual WTP would convert benchmarks into validated anchors.

Win/Loss Data — No interviews with organizations that bought EnergyCAP, Arcadia, or Peer AI and left, or bought and regretted it. This perspective typically produces the most honest feature gap and switching cost intelligence and is the most actionable gap for the next research sprint.

Prepared by Clayton Hester, Jason Wisnieski, Adam Case, Victor Pan — CDL Texas MBA Consulting | April 2026 For: Dan Schreiber (CPO) & John Reuter (CEO), Clear Current Technologies

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DOCUMENT 2: Market Research Vol 1 — Commercial Energy Management SaaS

Source file: Clear Current Technologies — Market Research Vol 1.md

Clear Current Technologies: Commercial Energy Management SaaS Market Research Report

Prepared April 2026 | Clear Current Technologies, Founded 2024

Executive Summary

Clear Current Technologies enters a market defined by two converging forces: rising commercial energy costs (up 6%+ year-over-year in 2025) and persistent structural dysfunction in how large organizations manage, audit, and optimize their utility spend. U.S. commercial buildings spent over **$241 billion on energy in 2024**, a figure that has grown steadily as building stock expands. Yet the overwhelming majority of that spend flows through manual, error-prone processes — with studies indicating up to 80% of commercial organizations are overpaying on their utility bills and only 9% of companies have fully embraced software tools for managing energy and sustainability data.[^1][^2][^3][^4]

The software market directly serving this problem — utility bill management, tariff optimization, rate analysis, and energy procurement — is growing at 9–13% CAGR depending on the segment definition, with the broadest addressable market for commercial energy management software in the U.S. estimated at approximately **$14.2 billion globally in 2024** and projected to reach **$36.5 billion by 2033**. The energy procurement software sub-segment, most directly aligned with Clear Current's PPA and contract management features, stands at **$1.62 billion in 2024** and is growing at 11.2% CAGR toward **$4.15 billion by 2033**.[^5][^6]

The competitive landscape is bifurcated between legacy incumbents (EnergyCAP, Schneider Electric, Siemens) with deep installed bases but aging platforms, and a wave of well-funded AI-native entrants (Arcadia, Verse) that are directly targeting the billing accuracy and tariff optimization use cases Clear Current addresses. The window for differentiation is open but closing; Arcadia's January 2025 enterprise solution launch and $649 million in total funding make it the clearest direct competitive threat.[^7][^8]

Higher education and healthcare campuses represent the highest-value initial verticals, characterized by district energy complexity, persistent chargeback/cost allocation pain, and a dominant but aging incumbent (EnergyCAP) whose platform consistently draws criticism for reporting inflexibility and chargeback complexity.

1. Market Sizing & Growth

Total Addressable Market

Market sizing for commercial energy management software varies substantially depending on scope. The broadest definition encompasses all software for monitoring, managing, and optimizing energy across commercial and industrial buildings:

- **Global Energy Management Software (broadest scope):** $14.2 billion in 2024, forecast to reach $36.5 billion by 2033 at 11.1% CAGR[^6]
- **U.S. Energy Management Software (commercial focus):** $16.9 billion in 2025 projected to reach $40.5 billion by 2035 at 10.2% CAGR[^9]
- **Building Energy Software (narrower, commercial buildings only):** $8.5 billion in 2022, projected at $15.2 billion by 2030 at 8.0% CAGR[^10]
- **Energy Procurement Software (contract, PPA, risk management):** $1.62 billion in 2024, forecast $4.15 billion by 2033 at 11.2% CAGR[^5]

For Clear Current's specific solution — utility bill management, tariff analysis, rate optimization, and PPA auditing for commercial enterprises — the most relevant segment is the **utility bill management and energy analytics** layer within the broader EMS market. Verdantix's 2024 Buyer's Guide identifies 76 vendors in this space, underscoring the fragmentation of the competitive landscape.[^11]

The underlying demand driver is the sheer scale of commercial energy expenditure. U.S. commercial buildings spent **$142 billion on energy in 2018** (the most recent full CBECS survey), with electricity accounting for 84% ($119 billion) of that total and an average cost of $1.47 per square foot. Updated to 2024, total commercial energy spend exceeded **$241 billion annually**. Every percentage point of billing error or tariff optimization opportunity in this base represents more than $2.4 billion in annual potential value.[^12][^3]

Vertical Market Breakdown

The following verticals represent the most significant energy spend segments and are therefore the highest-priority targets for energy management software:

Vertical	Annual U.S. Energy Spend (Est.)	Key Characteristics
Manufacturing	$236B+ (EIA MECS 2021)[^13]	Demand-intensive; demand charges 30–70% of bill[^14]; strong ROI on analytics
Commercial Real Estate	~$32B[^15]	Portfolio management across hundreds of sites; benchmarking compliance (ENERGY STAR, local laws)
Healthcare	~$8–12B (9% of commercial energy, 4% of floorspace)[^16]	600–700 kWh/m² intensity — above commercial average[^17]; district energy common; 24/7 load profile
Retail	~$20B[^18]	~657,000 buildings; high meter count; demand charge vulnerability
Higher Education	~$10–15B (estimated from education sector 13% share of $142B)[^12][^3]	District energy complexity; chargeback requirements; strong EnergyCAP incumbent
K-12 Education	~$6B[^19]	Budget-constrained; government funding programs (DOE grants) creating software requirements

Healthcare is particularly noteworthy: despite representing only 2% of commercial buildings and 4% of commercial floor space, healthcare facilities consume **9% of major fuels** used across all commercial buildings, and large hospitals can exceed 600–700 kWh/m² — far above the 22.5 kWh/ft² commercial average. This energy intensity, combined with complex campus metering and interdepartmental chargeback requirements, creates strong demand for sophisticated utility management platforms.[^20][^17][^21][^16]

Manufacturing has the largest raw energy spend but is dominated by demand-side management and process optimization tools rather than utility bill auditing. The industrial demand-side management market alone is valued at $27.6 billion in 2024, with manufacturing accounting for 28% market share.[^22][^23]

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2. Buyer Behavior & Personas

Organizational Energy Management Staffing

The most consistent finding across sources is that commercial energy management is chronically understaffed, creating a strong systemic driver for software automation:

- **Only 9% of companies have fully embraced software tools** for managing energy and sustainability data; the vast majority still rely on spreadsheets and manual effort[^4]
- **Only 3% of facilities managers** are employed at businesses with fewer than 50 employees, indicating that dedicated energy management expertise is almost exclusively found in larger organizations[^24]
- **10% of organizations rely on a single person** for energy management (EnergyCAP State of Utilities 2026)[^25]

- **64% of FM employers** report difficulty finding individuals with the required technical energy management skills[^26]
- **66% of facilities managers** identified staffing and retention as their top challenge in 2023[^26]

The staffing model varies significantly by organization size and vertical:

Large enterprise / Fortune 500: Dedicated VP of Sustainability or Director of Energy, often supported by an energy analyst team. May employ a third-party energy consultant or broker. This tier is the primary buyer for platforms like Schneider Electric Resource Advisor and Arcadia Enterprise Solutions.

Mid-market commercial (100–5,000 employees, 10–200 locations): Typically a Facilities Director or VP of Operations for whom energy is one of many responsibilities, sometimes supported by a single energy analyst. This is Clear Current's primary target buyer — sophisticated enough to recognize the problem but resource-constrained in solving it manually.

Small commercial / single-location: No dedicated energy staff. The owner or office manager handles utility bills. Not a viable market for sophisticated SaaS.

For **higher education**, the typical organizational model includes a Utilities Director or Energy Manager within Facilities Management, often supported by a sustainability office. The energy management role has increasingly been elevated at larger research universities, with dedicated energy managers responsible for multi-hundred-million-dollar annual utility budgets. Budget authority typically sits with a VP of Finance or CFO, with energy decisions requiring committee approval.[^27]

Key Buyer Personas

1. Director of Energy / Energy Manager — Primary end user and technical evaluator. Focused on billing accuracy, tariff optimization, anomaly detection, and reporting efficiency. Pain point: 80%+ of time consumed by data collection, bill entry, and manual analysis rather than strategic decision-making. Cites time savings of hundreds of hours per year as the key software ROI.[^4]

2. VP / Director of Sustainability — Increasingly prominent buyer, especially post-2022 as ESG reporting requirements have proliferated. Focused on Scope 1/2 emissions tracking, ENERGY STAR certification, and sustainability reporting frameworks (CSRD, GRI, CDP). 76% of organizations expect utility budgets to increase year-over-year in 2026, creating urgency.[^25]

3. VP / Director of Facilities — Decision-maker or key influencer at mid-market organizations where no dedicated energy manager exists. Focuses on cost control, vendor reliability, and avoiding operational disruption. Skeptical of software that requires significant manual work to configure or maintain.

4. CFO / VP Finance — Economic buyer at the executive level. Focuses on ROI, payback period, and total cost of ownership. Increasingly engaged as energy costs have risen to a "boardroom issue" — particularly for organizations with energy as 5–15% of operating costs.[^28]

Common Pain Points

Based on synthesis of industry sources, user reviews, and practitioner accounts:

- **Manual bill processing:** Energy managers spend disproportionate time entering utility bills, chasing missing invoices, and reconciling data across accounts. Real estate organizations waste hundreds of hours annually on manual utility bill processing[^4]
- **Lack of visibility across the portfolio:** Without centralized data, it's nearly impossible to identify which sites are anomalous, which are on incorrect rate classes, or where demand spikes are occurring
- **Chargeback and cost allocation complexity:** Particularly acute on university and hospital campuses where energy costs must be allocated to departments, research grants, auxiliary facilities (dining, athletics, housing), and commercial tenants[^29][^20]
- **Rate classification errors:** Organizations often remain on default rate schedules that are not optimal for their load profile, with utilities applying incorrect tariffs that go undetected for years[^30]
- **Demand charge opacity:** Demand charges represent 30–70% of total annual bills for many commercial customers, yet most organizations lack the tools to identify their demand peaks, model ratchet clause impacts, or understand why their demand charge has changed[^14][^31]
- **ESG reporting bottleneck:** Sustainability teams maintain separate spreadsheets tracking carbon data from utility bills, creating version control issues and manual errors when compiling annual reports[^32]

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3. Utility Billing Errors

Error Rate Evidence

The frequency of commercial utility billing errors is difficult to quantify with precision, as utilities do not publicly report error rates and academic studies are sparse. However, multiple industry sources converge on the finding that the majority of commercial organizations are overpaying:

- **~80% of businesses overpay** on their utility bills according to auditing industry sources[^2]
- **5–15% additional costs annually** result from inaccurate billing data[^33]
- **30% of commercial utility bills contain a material error** on an annual basis, according to energy auditors[^33]
- **Average overpayment estimated at 15%+** of total utility spend for commercial customers[^34]
- Only **9% of companies** have fully embraced software for energy data management, meaning the vast majority are not systematically auditing their bills[^4]

The EPA's ENERGY STAR program notes that **the average building wastes about one-third of the energy it uses**, a figure that encompasses both physical waste (inefficient equipment, HVAC leaks) and financial waste (incorrect billing, suboptimal rate structures).[^35]

Most Common Error Types

1. Wrong Rate Classification Utilities offer dozens of tariff schedules, and commercial customers are often misclassified — placed on a general service rate when they qualify for a more favorable large-commercial rate, or maintained on an outdated rate schedule after a usage profile change. Arcadia's automated tariff assignment process, which handles 500,000+ meter-tariff assignments, reports that only 80% can be automatically matched without expert review — illustrating the complexity of tariff identification even at scale.[^36][^30]

2. Demand Charge Errors Demand charges are calculated from a single 15-minute peak measurement per billing period and can represent 30–70% of a commercial bill. Common errors include:[^37][^14]

- Meter malfunction during peak hours leading to artificially elevated demand readings
- Mechanical meter reset issues that carry incorrect peak demand forward
- Ratchet clause misapplication: many utilities require minimum billing demand at 50–90% of the previous 12 months' peak. A facility with a 1,000 kW summer peak subject to an 80% ratchet at $15/kW pays a minimum demand charge of $12,000/month for 12 months — regardless of actual winter demand — totaling $144,000 in ratchet-driven charges from a single peak event[^38][^31]

3. Meter Multiplier Errors Particularly common in district energy systems and large campus environments where revenue meters use current transformer (CT) multipliers to scale readings. An incorrect multiplier results in systematic over- or under-billing that compounds over time and may go undetected for years.

4. Overlapping Billing Cycles Common when switching suppliers, relocating, or adding new accounts. If the original supplier bills July 1–31 and the new supplier bills from July 31, the customer is charged twice for the usage on July 31. In multi-site portfolios with frequent account changes, these overlaps can be persistent and difficult to detect without systematic tracking.[^39]

5. Incorrect Tax or Rider Assessments State and local sales taxes, utility taxes, and special-purpose rider charges (transmission, distribution, renewable portfolio standards) are frequently misapplied — particularly for customers that qualify for manufacturing exemptions, agricultural exemptions, or non-profit status.[^39]

6. Power Factor Penalties Commercial customers with low power factor (typically from motor loads, HVAC, and industrial equipment) may incur power factor surcharges. These are often overlooked because they appear as separate line items whose logic is opaque to facilities staff unfamiliar with electrical engineering.[^40]

7. Supplier Contract Errors In deregulated electricity markets (covering approximately 30 U.S. states), commercial customers choose their retail supplier separately from their utility. Contract switch errors, incorrect contract term pricing, and market index errors from competitive suppliers represent a distinct category of billing error beyond the regulated utility bill.[^41]

Strategic Implication for Clear Current

The billing error problem is the most defensible entry point for Clear Current's platform because it offers **immediate, quantifiable ROI** — recovered overbilling is direct bottom-line savings. This allows the sales motion to lead with audit findings (often structured as a free or low-cost initial audit) and convert to an ongoing subscription for systematic monitoring. The ~80% overpayment prevalence means that in virtually any commercial prospect, an audit will find actionable errors.

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4. Competitive Landscape

EnergyCAP

Profile: The dominant incumbent in utility bill management software for commercial, governmental, and higher education organizations. Founded over 40 years ago, headquartered in State College, PA.[^42]

Scale: 10,000+ customers including universities, government agencies, healthcare systems, K-12 districts, and corporations. Named #1 overall Champion in Info-Tech Research Group's 2025 Data Quadrant Report.[^43][^44]

Pricing: Meter-based annual subscription model:

- Express tier: starting at $4,000/year (~$40/meter/year for 100 meters)[^45]
- Enterprise tier: starting at $13,000/year (~$52/meter/year for 250 meters)[^45]
- Additional implementation fees: $5,000–$25,000 for mid-market; $25,000–$100,000+ for large enterprise[^46]
- Integrations/API work: $2,000–$20,000[^46]

Core Capabilities: Utility bill processing and auditing, chargeback/cost allocation, ENERGY STAR Portfolio Manager integration, carbon/GHG tracking (CarbonHub module), SmartAnalytics for anomaly detection, interval data management.[^47]

Higher Education Presence: EnergyCAP has a deep and documented footprint in higher education. Case studies include University of New Mexico, University of Kansas (90% of property portfolio managed in EnergyCAP), Appalachian State, and many others. The company's Founder & CEO has presented at the AEE World Energy Conference specifically on campus chargeback best practices, reflecting the depth of their higher ed focus.[^44][^48][^29]

Known Weaknesses (from verified user reviews):

A synthesis of Capterra and G2 reviews (132 total reviews, 4.7/5 overall) reveals a consistent set of user frustrations:[^42]

- *Reporting navigation:* "Hundreds of reports whose names or descriptions don't always help. And the more reports that are added, the more the waters are muddied." Users struggle to find the right report among the extensive library[^49]
- *Chargeback complexity:* "The chargebacks and splits are hard to understand. I tried to use the help, but it is lacking details." This is particularly significant given that chargebacks are EnergyCAP's key differentiated feature for campus users[^49]
- *Steep learning curve:* "There is so much to learn and so many options to select from. May take some time to become comfortable navigating and learning all what the software can provide." Users report needing frequent retraining for infrequently-used functions[^50]
- *Report customization:* "Recommendations for upgrades/improvements to reports tend to come back as a very simple, 'We can't do that.'" and "Wish report could be more customizable."[^49]
- *Fiscal year limitations:* Issues arise for organizations whose fiscal year doesn't start in January, described as "actually misleading" for reporting[^49]
- *Data gaps:* Some users report "spotty data with gaps" and limited customer support responsiveness for data issues[^49]
- *Training dependency:* "I have been using EnergyCap for 8 years and I know there are plenty of ways to use the system but I just don't know how. I know I need more training" — indicating that long-tenured users remain undertrained[^49]

Strategic Gap for Clear Current: EnergyCAP's weaknesses cluster around interface complexity, report inflexibility, and chargeback usability — precisely the areas where an AI-native platform can offer a step-change improvement. An LLM-powered interface that allows natural language querying ("Show me all accounts with demand charges above $5,000 this quarter where our peak occurred on a weekend") would directly address the report navigation and customization complaints.

Schneider Electric (EcoStruxure Resource Advisor / Resource Advisor+)

Profile: Schneider Electric launched **Resource Advisor+** in January 2026 with an AI agent named "Sera" that orchestrates multiple specialized sub-agents across carbon tracking, ESG reporting, and energy procurement. This represents Schneider's attempt to transform Resource Advisor from a legacy energy intelligence platform into an AI-first sustainability platform.[^51]

Scale: Schneider Electric has estimated **18% market share** in low-voltage electrical equipment globally, with fiscal 2024 revenue of approximately **36 billion euros**. The Energy Management division accounts for ~77% of revenue. Resource Advisor is primarily sold as part of an integrated Schneider ecosystem engagement rather than as a standalone SaaS.[^52]

Target Customer: Large multinational enterprises ($1B+ revenue) with complex Scope 1–3 emissions tracking, renewable energy procurement (PPAs), and multi-country sustainability reporting requirements. Named a Leader by IDC and Verdantix in ESG and sustainability management solutions.[^53]

Key Weakness: Resource Advisor is oriented toward sustainability reporting and enterprise-scale ESG compliance, not toward the billing-accuracy and tariff-optimization use cases that define Clear Current's core value proposition. For mid-market commercial organizations without a dedicated sustainability team, Resource Advisor's scope and pricing are likely excessive. The platform requires meaningful Schneider ecosystem integration, creating high switching costs but also high implementation barriers.

Siemens (Building X Energy Manager / Navigator)

Profile: Siemens offers energy management through two platforms: **Navigator** (cloud-based portfolio management, from single building to multi-campus) and **Building X Energy Manager** (subscription-based anomaly detection and ML-powered energy alerts, priced by meter with minimum 10 meters).[^54][^55][^56]

Scale: Siemens is one of the top-three global players in building automation alongside Johnson Controls and Honeywell. However, Siemens' energy management software is built on the Desigo ecosystem, which has drawn sustained criticism from building automation practitioners: Reddit's /r/BuildingAutomation community repeatedly characterizes Desigo as a difficult product — "I have literally heard not a single person who likes Desigo. No technician, no Siemens technician, no integrator, no customer, or user. Nobody." This reputation extends into the perception of Siemens' energy management software as hardware-first and technician-oriented rather than energy-analyst-friendly.[^57][^58]

Target Customer: Organizations with significant Siemens building infrastructure — new construction with Siemens BAS, or existing Siemens customers looking to add energy analytics.

Key Weakness: Siemens is fundamentally a building systems company that has added energy management software, not a software company that understands energy procurement and billing. The platform excels at operational energy monitoring (equipment anomalies, fault detection) but lacks the tariff intelligence, bill auditing, and procurement advisory capabilities that Clear Current targets.

Johnson Controls (OpenBlue)

Profile: JCI's **OpenBlue** platform integrates HVAC, fire, security, and building management with energy analytics. Fiscal 2024 revenue was approximately $23 billion with a record backlog exceeding $12 billion. Strong presence in healthcare, education, and commercial real estate verticals.[^59][^60][^57]

Key Weakness: Like Siemens, Johnson Controls is a building systems company. OpenBlue's energy capabilities are focused on operational optimization and equipment performance, not utility bill auditing or tariff analysis. The platform requires deep JCI hardware integration and is typically sold as part of an energy services performance contract (ESPC) rather than as standalone SaaS.

Arcadia (AI-Native Primary Competitor)

Profile: Arcadia is the most direct competitive threat to Clear Current Technologies. Founded in 2014 and headquartered in Englewood, Colorado, Arcadia has raised **$649 million in total funding** (including a $200 million Series E) at a **$1.5 billion valuation** (unicorn). The company acquired Urjanet in 2022 to become the world's largest utility data aggregator, covering **95%+ of U.S. residential and commercial accounts** across 9,500+ utilities.[^61][^8][^62]

In **January 2025**, Arcadia launched **Arcadia Enterprise Solutions** — directly targeting the commercial energy management market with three core offerings:[^63][^7]

1. **Utility Bill Management:** Automated bill collection, payment, auditing, tariff optimization, and budgeting/forecasting
2. **Energy Procurement Advisory:** In-house advisory for retail energy contracts and PPAs
3. **Supply Management:** Unified view across energy markets and suppliers

Arcadia claims **99% energy cost calculation accuracy** at portfolio scale through its Automated Tariff Assignment (ATA) and Cost Validation Service (CVS), processing 500,000+ meter-tariff assignments automatically. Its estimated annual revenue is **~$270 million**, and its 300+ enterprise customers include Ford Motor Company.[^64][^61][^36]

Positioning Gap vs. Clear Current: Arcadia's Arc platform is built for data aggregation and API access — it was designed as a B2B2C infrastructure layer for energy tech companies, not as a purpose-built decision-support tool for energy managers at commercial enterprises. Its January 2025 enterprise pivot is a recognition that this infrastructure needs a user-facing layer, but the product experience and go-to-market are still maturing in the enterprise direct-sales motion. Clear Current, purpose-built for commercial enterprise energy buyers, can differentiate on UX depth, domain-specific AI (billing error pattern recognition, rate change alerts, PPA deviation detection), and vertical-specific implementation expertise.

Verse, Inc. (AI-Native Entrant — Adjacent)

Profile: Verse delivers **energy cost intelligence** through its AI-powered platform **Aria**, targeting enterprise energy, finance, and operations teams. Core capabilities include PPA management, energy procurement modeling, supply management, and planning/risk management. The tagline "built by energy buyers for energy buyers" signals a deep practitioner orientation.[^65][^66]

Positioning: Verse is focused on the energy procurement and contract management layer — complementary to but not identical with Clear Current's billing audit and tariff optimization focus. Verse's ideal customer is a sophisticated energy buyer at a large industrial or commercial enterprise managing multi-market portfolios, virtual PPAs, and hedging strategies. This positions Verse upstream (procurement/contracting) from where Clear Current sits (bill validation and rate analysis), creating a potential partnership or integration opportunity rather than pure head-to-head competition.

IBM Envizi (Acquired ESG-Adjacent Entrant)

Profile: IBM acquired Envizi in January 2022, integrating it into IBM's broader ESG and sustainability software suite alongside IBM Maximo, IBM Sterling, and the Environmental Intelligence Suite. Envizi provides automated collection of 500+ data types for sustainability reporting and GHG emissions tracking.[^67][^68]

Key Weakness: IBM Envizi is an ESG reporting platform that ingests utility data as a means to emissions tracking — it is not primarily a utility bill auditing or tariff optimization tool. The acquisition signals IBM's commitment to corporate sustainability buyers, not energy procurement or billing accuracy. The IBM enterprise sales motion and implementation complexity make Envizi inappropriate for most mid-market commercial buyers.

Competitive Positioning Matrix

Vendor	Billing Audit	Tariff Optimization	PPA/Contract Mgmt	Campus Chargebacks	AI-Native	Target Market
EnergyCAP	Strong	⚠ Limited	No	Strong (complex UX)	No	Gov't, Higher Ed, Healthcare
Schneider Resource Advisor+	⚠ Partial	⚠ Limited	Strong	No	AI agents	Large enterprise ESG
Siemens Building X	No	No	No	No	⚠ Partial	BAS hardware customers
Johnson Controls OpenBlue	No	No	No	No	⚠ Partial	BAS hardware customers
Arcadia Enterprise	Strong	Strong	⚠ Emerging	No	Strong	Fortune 500 enterprise
Verse / Aria	No	⚠ Partial	Strong	No	Strong	Industrial/large enterprise
IBM Envizi	⚠ Partial	No	No	No	⚠ Partial	Large enterprise ESG
Clear Current Technologies	Core	Core	Core	Target	Native	Mid-to-large commercial

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5. Advanced Metering Infrastructure (AMI)

Current Adoption Rate

The United States has the most mature smart electricity meter deployment in the world:

- **136.9 million smart meters** installed in the U.S. as of 2024, representing approximately **82% of all electric meters**[^69][^70]
- Projected to reach **91% penetration by 2030**[^69]
- The U.S. Department of Energy highlighted the installation of **~115 million smart meters in 2023** as enabling enhanced energy usage insights[^22]

However, these aggregate figures are heavily skewed toward **residential deployment**, which has driven the majority of AMI rollouts. The commercial and industrial (C&I) segment presents a more nuanced picture:

- The **commercial smart electric meter market** was valued at **$5.2 billion in 2023** and is projected to grow at **9.6% CAGR** through 2032, indicating that commercial AMI deployment is still in active growth phase[^71]
- Commercial sub-metering in buildings was at only **~10% penetration in 2018** (EIA data), significantly lagging the headline smart meter numbers[^72]
- Building Energy Management Systems (BEMS) penetration was approximately **4% of commercial buildings in 2018**, with growth expected toward 30% by 2050[^72]
- Non-domestic (commercial) smart meter installations experienced a **16% year-over-year decline in 2025** in some markets, suggesting the initial deployment wave has peaked and second-wave deployments are needed[^73]

The critical insight is that while most large commercial accounts have a smart meter at the utility's billing point of service, **sub-metering within buildings** and **interval data access** remain limited. A university campus may have a single smart meter at the utility service entrance while hundreds of internal buildings are served by the campus's own distribution system with inconsistent sub-metering.

Impact of 5-Minute Interval Data

When interval data (15-minute or sub-15-minute) is available for a commercial account, the capabilities of energy management software expand dramatically:

Demand charge management: Interval data reveals the exact 15-minute window that sets the monthly demand peak. This enables organizations to identify the specific operational event (Monday morning simultaneous startup, production schedule change, HVAC failure) that caused the peak and model the cost impact of behavioral changes. For accounts with ratchet clauses, identifying and managing the annual peak can eliminate up to $100,000+ in annual excess charges.[^31][^37]

Time-of-use (TOU) rate optimization: Many commercial tariffs offer TOU pricing where energy consumed during peak hours costs significantly more than off-peak consumption. Interval data enables precise modeling of whether a TOU rate would reduce total cost vs. a flat-rate tariff — a calculation that is impossible without sub-hourly consumption data.

Ratchet clause monitoring: Utilities with ratchet clauses (typical structure: 50–90% of prior 12-month peak) can be continuously monitored with interval data to alert customers when they are approaching a threshold that would set a new 12-month ratchet baseline.[^38][^31]

Anomaly detection: Deviations from expected consumption patterns at 15-minute resolution can identify equipment malfunctions, HVAC night cycling, refrigeration failures, and other sources of energy waste that are invisible in monthly billing data alone.

Tariff comparison modeling: With actual load profiles in 15-minute resolution, software can simulate the total annual cost under every applicable tariff — enabling a definitive answer to whether a commercial account should switch rate classes.[^74][^36]

The transition to AMI 2.0 — delivering 96 or more readings per day per meter — is already underway at leading utilities, and represents a secular shift that expands the analytical capabilities of platforms like Clear Current as interval data becomes more widely available for C&I accounts.[^75][^76]

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6. Higher Education Energy Market

Annual Spending

The education sector as a whole consumed **854 TBtu in 2018**, accounting for approximately 13% of total commercial building energy consumption. With total commercial buildings spending $142 billion on energy in 2018 and $241 billion in 2024, the education sector represents approximately **$18–31 billion in annual energy expenditure**. K-12 schools account for approximately **$6 billion** of that total, leaving higher education institutions responsible for an estimated **$10–15 billion in annual campus energy spend**.[^19][^77][^3][^12]

Typical Higher Education Energy Management Stack

Large public universities typically deploy a layered technology stack:

1. **Utility bill management platform:** EnergyCAP is the dominant platform across U.S. universities and government agencies. Case studies include University of New Mexico (centralized energy data management for campus-wide sustainability), University of Kansas (90% of property portfolio managed in EnergyCAP, with chargebacks as a primary use case), and dozens of others. New adopters include Angelo State University, which recently began deploying EnergyCAP through YearOut Energy for utility usage and cost reporting[^48][^78][^44]
2. **ENERGY STAR Portfolio Manager:** Nearly **25% of U.S. commercial building space** (over 150,000 properties) is actively benchmarked in Portfolio Manager, making it effectively a required component of any university's energy reporting stack. EnergyCAP integrates directly with Portfolio Manager[^79][^80][^81]
3. **Building Automation Systems (BAS):** Johnson Controls, Siemens, Honeywell, Automated Logic — control HVAC, lighting, and mechanical systems. Separate from utility bill management and not typically integrated
4. **Campus energy analytics:** Platforms like MRI Energy provide campus-level energy data aggregation and analytics, with some universities building custom dashboards on top of interval data feeds from their district energy system SCADA[^82]
5. **Sustainability reporting:** Increasingly, universities use ESG-adjacent platforms (IBM Envizi, Salesforce Sustainability Cloud, or custom solutions) for GHG emissions tracking and ACUPCC/AASHE STARS reporting

District Energy Systems — Prevalence and Complexity

District energy systems (DES) — centralized heating and cooling plants that distribute steam, hot water, or chilled water through underground pipe networks — are the defining infrastructure characteristic of university and hospital campuses:

- **660+ district energy systems** in the U.S., serving **5.5 billion square feet** of heated space and **1.9 billion square feet** of cooled space[^83]
- **4.3+ billion sq ft** of U.S. building space served by district energy, including landmark institutions like Harvard Medical School, the Mayo Clinic, and the U.S. Capitol[^84]
- **"Preferred method of heating and cooling most major college and university campuses"** in the U.S.[^84]
- In addition to campuses, urban district energy systems serve major cities including Philadelphia, Boston, Denver, Indianapolis, and Manhattan (Con Edison, the largest district steam system in the world serving up to 1,800 customer buildings)[^84]

The Campus Chargeback Problem

District energy creates a billing and cost allocation challenge that has no parallel in commercial real estate or retail energy management. EnergyCAP's own founder has characterized campus chargeback processes as follows:[^29]

- Many universities have a mix of **direct vendor-supplied utilities** (electricity from the local utility) and **university-owned distribution systems** (steam, chilled water, natural gas distributed from central plants)
- Buildings served by the central plant do not receive vendor utility bills — instead, the university becomes the "local energy vendor" for its own departments and buildings
- Cost allocation requires **percentage splits, formulas, and loss/remainder calculations** that cannot be reduced to simple per-kWh billing
- **Reimbursable facilities** — athletics, food services, dormitories, commercial tenants, and externally funded research facilities — must pay for utilities at market rates, requiring accurate internal billing
- This entire metering and bill-creation system can represent a **multi-million-dollar operational infrastructure** for large research universities
- The cost challenge compounds when universities need to allocate energy costs to **federal research grants** (which require documented, auditable cost allocation) and to **auxiliary enterprises** (which operate as separate financial entities)

A 2009 study of 24 university buildings in California found that building systems "drift" substantially from optimal performance, with more than **40% waste in total building energy** attributable to equipment malfunction and control issues. This systemic inefficiency makes energy audit and anomaly detection capabilities particularly valuable on campuses.[^85]

Key Differentiators for Campus Energy Platforms

To succeed in higher education, an energy management platform must:

1. Handle complex metering topologies (owner-operated distribution, revenue-grade sub-meters, calculated points)
2. Generate internal bills from raw meter readings with configurable cost allocation formulas
3. Support multiple commodity types simultaneously (electricity, natural gas, steam, chilled water, compressed air)
4. Integrate with university ERP and financial systems (Banner, Workday, PeopleSoft) for automated cost allocation
5. Provide building-level EUI benchmarking for ENERGY STAR submission
6. Support GHG emissions tracking for AASHE STARS and annual sustainability reporting

The fact that EnergyCAP consistently draws criticism for **chargeback complexity** — its most important differentiating capability for this very market — represents a genuine opening for a successor platform with a better-designed cost allocation UX.

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7. Engagement Triggers & Buying Cycles

Primary Engagement Triggers

Evidence from practitioner accounts, industry studies, and sales case studies suggests the following ranked triggers for energy management software adoption:

Tier 1 — Highest Conversion Probability:

1. **Energy contract expiration (9–12 months out):** The highest-probability engagement window. Organizations approaching a multi-year energy contract renewal are forced to analyze their current rate, evaluate alternatives, and often discover the inadequacy of their current analytical tools in the process. Energy procurement advisors report that the "9 to 12 months before contract expiration" window is the ideal time to engage[^86][^87]
2. **Bill shock / unexpected cost spike:** A single large bill, or a period of unexpectedly high utility costs, is the most common self-reported trigger for organizations seeking external help. The State of the Union (March 2026) confirmed "energy costs are a boardroom issue". Energy suppliers reopening fixed contracts to add unexpected charges has created acute bill shock across the commercial market[^88][^89][^90][^28]

3. **New sustainability or energy hire:** When an organization hires a new Director of Energy, VP of Sustainability, or Chief Sustainability Officer, that individual typically arrives with a mandate to improve energy performance and an immediate need for better data infrastructure. This "new executive in role" trigger has a 3–6 month engagement window before the new hire becomes embedded in existing processes[^91]

Tier 2 — Moderate Conversion Probability:

1. **ESG reporting mandate or deadline:** CSRD (EU, affecting U.S. multinationals with EU operations), California SB 253 (Scope 3 disclosure), and SEC climate disclosure rules are creating hard regulatory deadlines that drive software adoption. The CSRD cascade means that large listed companies filed first reports in 2025 for FY 2024, with Wave 2 expanding scope in 2028. U.S. companies need **12–18 months of lead time** for implementation[^92][^93][^32]
2. **New building addition or portfolio expansion:** When an organization adds a new facility, acquires a property, or consolidates after a merger, the existing energy data management processes often break down under the added complexity[^94]
3. **Regulatory/benchmarking compliance:** Cities including New York, Chicago, and San Francisco require annual benchmarking reports using ENERGY STAR Portfolio Manager. Local Law 97 in NYC (carbon emissions caps for large buildings) has created strong demand for energy management software across the NYC commercial real estate market[^79]
4. **Legacy software end-of-life:** EnergyCAP itself has gone through a migration from Flash-based to HTML5 platform, creating disruption for existing users. Legacy users on outdated platforms represent a displacement opportunity[^49]

Tier 3 — Opportunistic:

1. **New CFO or VP Finance engagement:** Finance executives discovering that energy is 5–10% of operating costs but managed through spreadsheets often become champions for investment in analytics infrastructure
2. **Failed manual audit:** Organizations that conduct a manual billing audit (often via an energy broker or consultant) and discover significant overcharges frequently seek software to systematize the process going forward[^95]

Seasonal Buying Patterns

Commercial energy management software purchases follow distinct seasonal rhythms across verticals:

Higher Education (Fiscal Year July 1 or October 1):

- **September–December (Fall):** Strategic planning, early vendor evaluations, awareness phase[^27]
- **January–March (Winter):** Budget proposals submitted; committees review. Must be on radar before February to be included in proposals[^27]

- **April–June (Spring):** Final approvals, RFP issuance. University of Wisconsin, for example, issued an Energy Management Software RFP in September with an October 1 deadline[^96]
- **July–August (Summer):** Contract execution, implementation begins ahead of fall semester[^27]

State and Local Government (Fiscal Year July 1 or October 1): Analogous to higher education cycle. Budget proposals in Q4 of calendar year for July 1 FY start.

Corporate/Commercial (Fiscal Year January 1):

- **Q3 (July–September):** Budget planning season; software evaluations for next-year budget
- **Q4 (October–December):** Budget finalization, final purchase decisions
- **January:** New year implementations; new budget authority often triggers new investments

ESG Reporting-Driven:

- **Q1 (January–March):** Annual sustainability report publication for prior year drives internal reviews of data quality
- **Q2–Q3:** New fiscal year sustainability software evaluations following annual report completion

Energy Contract-Driven (Deregulated Markets):

- Timing correlates with contract anniversary dates — organizations with December contract expirations are best engaged in Q1–Q2 of the prior year

The 76% Budget Increase Signal

EnergyCAP's **State of Utilities 2026** survey found that **76% of energy and sustainability professionals expect their utility budgets to increase year-over-year**, and **69% identify controlling/reducing utility spend as their #1 goal**. This creates a structural buying environment where the pain is acute, the budget pressure is increasing, and the justification for software investment is self-evident. The same survey found that **organizational confidence is misaligned** — executives report high confidence in their energy management capabilities while day-to-day practitioners are raising concerns — creating an opening for a vendor that can arm the practitioner with data to make the case internally for investment.[^97][^25]

Strategic Implications for Clear Current Technologies

Market Entry Recommendations

1. Lead with the billing audit value proposition. The ~80% overpayment rate and immediate, quantifiable ROI of a billing audit creates a natural land-and-expand motion. A free or low-cost initial audit delivers demonstrable value before any software commitment, converting skeptical buyers who have "tried spreadsheets" and failed.

2. Target higher education as the initial beachhead vertical. EnergyCAP's weaknesses in chargeback UX, report customization, and learning curve are most acute in higher education, where chargeback complexity is the core use case. EnergyCAP has 10,000+ customers with significant higher ed concentration, providing a large installed base of dissatisfied users who represent displacement opportunities. The Fall/Winter sales cycle creates a predictable engagement cadence.

3. Invest in campus chargeback as a core platform capability. No AI-native competitor offers robust campus chargeback functionality. Building this capability as a differentiated, well-designed feature (not a bolted-on afterthought) would create a durable competitive moat against both EnergyCAP (complex UX) and Arcadia (no campus focus).

4. Build the Arcadia competitive response now. Arcadia's January 2025 enterprise pivot and $649 million in total funding make it the most credible competitive threat. Clear Current should differentiate through (a) vertical depth in higher ed and healthcare, (b) procurement + billing integration (Arcadia focuses on billing; Verse on procurement; Clear Current can integrate both), and (c) implementation speed (Arcadia's enterprise motion is still maturing).

5. Engage the "contract expiration" trigger systematically. Building a data-driven pipeline of organizations whose multi-year energy contracts are expiring in the next 9–12 months — identifiable through public utility filings, broker relationships, and deregulated market databases — would create a predictable outbound motion with a high-intent buyer population.

6. Position ahead of ESG reporting deadlines. The CSRD Wave 2 expansion (2028) and California SB 253 Scope 3 requirements create an 18-month preparation window that is currently open, with U.S. corporate buyers needing to be ESG-ready with traceable energy data by 2026–2027. Positioning Clear Current as the foundation for ESG-ready energy data (accurate bills → accurate Scope 2 → compliant reporting) creates an enterprise-wide justification that extends beyond the energy manager to the CFO and general counsel.[^93][^32]

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DOCUMENT 3: Market Research Vol 2 — Commercial Energy Management SaaS (Full Report)

Source file: Clear Current Technologies — Market Research Vol 2.md

Clear Current Technologies: Commercial Energy Management SaaS — Market Research Report

Prepared for: Clear Current Technologies
Founded: 2024
Platform Focus: AI-powered utility bill analysis, rate optimization, tariff/PPA intelligence, and billing error recovery for commercial enterprises
Report Date: April 2026

Executive Summary

The commercial energy management software market is one of the fastest-growing segments in enterprise SaaS. The U.S. energy management system market is valued at approximately **$14.2 billion in 2025** and is projected to reach **$27 billion by 2032** at a CAGR of ~9.6%, with the global market exceeding $60 billion. Commercial buildings spent over **$241 billion on energy in 2024** across 5.9 million buildings and 97 billion square feet of floor space, and industry auditors estimate that **80% of businesses are overpaying on their utility bills** through a combination of billing errors, misclassified rates, and unoptimized tariff structures.[^1][^2][^3][^4][^5]

Clear Current Technologies enters a market with powerful structural tailwinds: rising energy prices, accelerating sustainability reporting mandates, AMI-enabled interval data proliferation, and commercial buyers underserved by incumbent platforms built on 1990s-era architectures. The platform's focus on billing error recovery, rate optimization, and PPA/contract analysis positions it to serve a gap between lightweight benchmarking tools and expensive, hardware-dependent building automation systems.

Section 1: Market Sizing & Growth

Total Addressable Market

The U.S. commercial energy management software market can be sized across several overlapping market definitions:

Market Definition	2024/2025 Value	2030 Projection	CAGR
U.S. Energy Management Systems (all)	$14.2B (2025)[^3]	$27.0B (2032)[^3]	~9.6%
Global Building EMS (software + hardware)	$10.3B (2024)[^6]	$20.8B (2030)[^6]	12.6%
Sustainability & Energy Mgmt Software (global)	~$10.8B (2024)[^7]	~$32B (2030)[^7]	10.4%
Facility Energy Mgmt Software (software-only)	$7.0B (2025)[^8]	$12.5B (2033)[^8]	10.2%
North America EMS (all segments)	$15.3B (2025)[^9]	—	~15%[^9]

The most relevant TAM for Clear Current — pure-play commercial energy management software focused on bill analytics, rate optimization, and cost intelligence — is best approximated by the **Facility Energy Management Software** segment, currently estimated at approximately **$7 billion** with a 10.2% CAGR to reach **$12.5 billion by 2033**.[^8]

A serviceable addressable market (SAM) scoped to mid-to-large U.S. commercial organizations that have active energy management needs, multiple utility accounts, and sustainability reporting obligations can be estimated by applying the commercial building sector's share of the market. Commercial buildings generate approximately **40.6% of North American EMS revenue**, suggesting a U.S. commercial-focused SAM of roughly **$2.5–3.5 billion** in annual software spend today, growing toward **$5–6 billion** by 2030.[^9]

Key Market Growth Drivers

The expansion of this market is driven by several reinforcing forces:

- **Rising energy costs**: U.S. commercial energy spend exceeded $241 billion in 2024; analysts project a **25% increase in electricity prices by 2030**.[^5][^10]
- **ESG and sustainability mandates**: California's SB 253 and SB 261, the EU's CSRD (affecting U.S. multinationals), and ISSB standards are forcing organizations to produce investment-grade energy disclosures.[^11][^12]
- **AMI proliferation**: Smart meter deployment (72% of U.S. electric meters as of 2022) is generating interval data that legacy spreadsheet workflows cannot effectively process.[^13]
- **AI integration**: The Verdantix 2023 Green Quadrant found all 15 leading vendors are actively incorporating AI into their platforms, with buyers asking for predictive analytics and automated anomaly detection.[^14]
- **Inflation Reduction Act (IRA)**: The IRA earmarks $369 billion for climate initiatives, creating incentives for commercial energy management investment.[^15]

Market Size by End-User Vertical

The EMS market segments across verticals as follows, with manufacturing holding the largest share and healthcare growing fastest:

Vertical	EMS Market Role	Key Growth Driver
Manufacturing	Largest vertical (~$9.4B in 2018)[^16]	Process optimization, demand charge reduction, regulatory compliance
Commercial Real Estate	40.6% of North America EMS revenue[^9]	ESG reporting, tenant billing, portfolio benchmarking
Healthcare	Growing at 16% CAGR[^17]	24/7 operations, 2.5x energy intensity vs. office buildings, $8B+ annual energy spend[^18]
Higher Education	Significant; EnergyCAP's top vertical (19% of users)[^19]	District energy complexity, chargeback allocation, sustainability commitments
Retail	Notable; EMS includes multi-site portfolio management	Multi-location cost variability, refrigeration load management

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Section 2: Buyer Behavior & Personas

How Commercial Organizations Staff Energy Management

The staffing model for energy management varies sharply by organization size and type, and this divide defines the clear current opportunity:

Dedicated Energy Managers are typically found at:

- Large public universities (almost universally employ a campus energy manager)[^20]
- Large healthcare systems and hospital networks
- Fortune 500 industrial/manufacturing companies
- Large government building portfolios

Facilities Managers with Secondary Energy Responsibility represent the majority of the commercial market. The IFMA estimates there are approximately **16,500 U.S. professional facility management staff** managing commercial building portfolios. Only 11% of facilities managers report that their superiors have excellent visibility into the complexity of their work, indicating that energy management often sits below the strategic radar. A 2023 GridPoint survey of over 800 U.S./Canadian building professionals found that **98% of facilities management respondents rated energy efficiency as extremely or very important** to their role, yet only **~9% of companies have fully embraced software tools** for managing their energy and sustainability data.[^21][^22][^23][^24]

This gap — high stated priority, low software adoption — is the core market opportunity. The majority of commercial organizations below ~$500M in revenue lack dedicated energy staff and rely on generalist facility managers who struggle with the complexity of multi-meter utility bill portfolios.

Buyer Personas

The Campus Energy Manager (Higher Education / Large Municipal)

- Title: Director of Utilities / Energy Manager
- Reports to: VP of Facilities or CFO
- Pain: Managing 100–1,000+ meters across legacy BAS systems and district energy infrastructure; needs automated bill capture, cost allocation to departments, and variance reporting.
- Budget ownership: O&M and capital energy line items.
- Purchasing cycle: Aligned to fiscal year (often July 1 for universities); procurement typically requires RFP/competitive process.

The VP Sustainability / Director of ESG (Corporate Real Estate, Retail, Manufacturing)

- Title: VP Sustainability / Head of ESG
- Reports to: CFO or Chief Sustainability Officer
- Pain: Aggregating Scope 1 & 2 emissions data from hundreds of locations; ensuring utility bill data is accurate enough for SEC/CSRD disclosure.
- Budget ownership: Corporate sustainability budget or facilities capex.
- Purchasing cycle: Calendar year; often tied to annual ESG reporting cycle (Q1/Q2).

The Facilities/Operations Manager (Mid-Market, Secondary Responsibility)

- Title: Director of Facilities / VP Operations
- Reports to: COO or VP Real Estate
- Pain: No visibility into whether rates are optimal; utility bills processed manually in spreadsheets; billing errors go undetected for years.
- Budget ownership: Operational expense; typically requires CFO sign-off for software above $15K/year.
- Purchasing cycle: Budget season (Sept–Nov); often triggered by a specific shock event (large rate increase, failed audit, new lease with energy clauses).

Key Pain Points

Research and user reviews across platforms consistently surface five dominant pain points for commercial energy buyers:[^25][^23][^26]

1. **Manual utility bill processing**: Real estate organizations waste hundreds of hours annually entering and reconciling bills in spreadsheets; a single large portfolio may process 10,000+ invoices annually.[^27]
2. **Billing error detection failure**: The majority of billing errors go undetected without systematic audit; organizations are unaware they are being overcharged for months or years.[^1]

3. **Rate classification and optimization blindness**: Utility tariffs change frequently, and most organizations lack the staff expertise to determine whether they are on the optimal rate class for their load profile.[^28][^25]
4. **Demand charge opacity**: Demand charges can represent up to **50% of an electric bill**, yet most organizations cannot trace a demand charge spike to its root cause without interval data analysis.[^25]
5. **Spreadsheet fragility / key-person dependency**: Institutional knowledge is locked in custom spreadsheets maintained by one or two people — what energy managers call "the Rick problem" — creating operational risk when that person leaves.[^26]

Section 3: Utility Billing Errors

Error Prevalence

The commercial utility billing error landscape is more pervasive than most organizations realize:

- Industry auditors estimate that **approximately 80% of businesses are overpaying on their utility bills** through a combination of billing errors, hidden fees, and suboptimal rate structures.[^29][^1]
- Property/facility managers overpay by **15% or more on average** when invoices are not professionally audited.[^27]
- 360 Energy, an energy consulting firm, reports that **nearly 30% of its commercial clients face billing errors annually**, and the incidence is increasing.[^30]
- Inaccurate utility billing data generates additional costs ranging from **5–15% annually** for affected accounts.[^30]

The 80% figure cited by multiple audit firms represents the broad definition (including suboptimal rate classification, not just computational errors). The more conservative estimate is that roughly 20–30% of commercial accounts have a material billing error — defined as a calculable overcharge beyond the appropriate rate schedule — at any given time.[^30]

Most Common Error Types

Based on utility audit literature and EMS vendor research, the most frequently encountered billing error categories are:[^31][^29][^25]

Error Type	Description	Prevalence / Impact
Wrong Rate Classification	Account billed on residential, small commercial, or industrial rate instead of applicable tariff	Very common; can persist for years undetected[^25]
Meter Reading Errors	Incorrect readings (manual entry error or meter malfunction); estimated/skipped reads	Most commonly cited single error type[^29][^32]
Demand Charge Miscalculation	Mathematical errors in ratchet clause application, incorrect demand interval, or meter not reset properly	Demand = up to 50% of bill; errors highly material[^25]
Meter Multiplier Errors	Wrong pulse multiplier applied to submeters, leading to systematic over/undercharging	Systematic and often persistent across billing periods[^27][^28]
Paying for Inactive/ Wrong Meters	Multi-site organizations pay for meters not tied to their facilities	Common for portfolios with frequent relocations[^33]
Incorrect Sales Tax Application	Taxable items applied to tax-exempt accounts (e.g., grocery stores, nonprofits)	Common for exempt entities; creates recoverable refund opportunity[^33]
Power Factor Penalties	Utilities penalize customers when power factor < 90%, which many customers don't monitor or contest	Common in manufacturing and data-intensive facilities[^25]
Late Fee Accumulation	Fees ranging from 1–12% for late payment, often due to manual bill processing delays[^29]	High frequency in organizations without automated AP integration

The Recovery Opportunity

Billing errors create both a **retroactive recovery** opportunity (refunds or credits for historical overcharges, typically recoverable 12–36 months back depending on state statute) and an **ongoing savings** opportunity from rate optimization. For large commercial enterprises, a systematic utility audit frequently identifies **annual savings of 5–20%** of the total utility spend.[^34][^35][^36]

Section 4: Competitive Landscape

Market Overview

The Verdantix Green Quadrant: Energy Management Software 2023 — the most authoritative independent benchmark — assessed 15 vendors across product capability and market momentum. The eight vendors that placed in the **Leaders Quadrant** were: **Bueno, EnergyCAP, Honeywell, Johnson Controls, MRI Software, Schneider Electric, Siemens, and Spacewell**. Separately, AI-native challengers including Arcadia, WatchWire, Gridium, and Nectar are growing rapidly in the commercial segment.[^14]

Incumbent Competitors

EnergyCAP

- **Founded:** 1980 (State College, PA). CEO: Shawn Lankton[^37]
- **Target profile:** Higher education (19% of user base), government (19%), large commercial organizations with 100–1,000+ meters[^19]
- **Market position:** Self-described as "the preferred energy and sustainability data platform for higher education"; **Named Market Leader** in FeaturedCustomers Summer 2024 Energy Management Software report[^38][^39]
- **Pricing:** Express plan from $4,000/year (100 meters @ ~$40/meter); Enterprise from $13,000/year (250 meters @ ~$52/meter); implementation fees of $5,000–$100,000+[^40][^41]
- **Strengths:** Deep utility bill accounting, meter hierarchy management, chargeback/cost-allocation for campus settings, ENERGY STAR integration, carbon accounting add-ons, large established customer base[^42][^43]
- **Known weaknesses (user reviews):** Steep learning curve; reporting navigation is difficult ("hundreds of reports whose names don't always help"); chargebacks and split billing are hard to configure without extensive training; limited mobile experience; some users report the interface feels dated; customization requests are often declined; implementation is time-intensive[^44][^45]

Schneider Electric EcoStruxure / Resource Advisor

- **Target profile:** Large enterprise — real estate/offices, retail, healthcare, life sciences, hotels, data centers, industrial[^46]
- **Platform:** EcoStruxure Resource Advisor is a cloud-based energy and sustainability data platform; integrates with Schneider's BMS and power monitoring hardware for a unified operational view[^47]
- **Verdantix 2023:** Named a Leader, recognized for "comprehensive packages that capture and combine data from both building management and power management solutions"[^48]
- **Pricing:** Enterprise/custom only; not publicly disclosed[^49]
- **Strengths:** Breadth of hardware-to-software integration; massive installed base; strong brand in large enterprise
- **Known weaknesses:** Predominantly hardware-oriented; meaningful value often requires Schneider infrastructure investment; heavy implementation burden; pricing and complexity likely exclude mid-market buyers

Siemens (Desigo CC / SIMATIC Energy Manager)

- **Platform:** Desigo CC — building management platform for small-to-large facilities and campuses; SIMATIC Energy Manager — industrial-grade monitoring for complex energy profiles[^50][^51]

- **Strengths:** Deep integration with Siemens building automation hardware; industrial monitoring depth; strong in manufacturing
- **Known weaknesses:** Like Schneider, hardware-centric; significant deployment complexity; primarily serves the very large enterprise and industrial segment

WatchWire (EnergyWatch)

- **Profile:** NYC-based, private equity-backed; targets commercial/corporate real estate, Fortune 500 industrial/manufacturing, big-box retail, healthcare, education[^52][^53]
- **Capabilities:** Invoice management, bill auditing, analytics, Scope 1/2/3 emissions tracking, energy procurement support[^54]
- **G2/Capterra position:** Top-ranked EnergyCAP alternative on G2; described by users as "outstanding all-around energy management software" with an intuitive interface[^55][^56]
- **Pricing:** Custom/upon request[^56]
- **Differentiator:** Strong in commercial real estate portfolios; sustainability-first positioning; private-label option for channel partners

AI-Native Challengers

Arcadia

- **Funding:** $200M raised (2022); acquired Urjanet (covers 9,500+ utilities in 52 countries)[^57]
- **Product:** Launched **Arcadia Enterprise Solutions** in January 2025 with AI-powered utility bill management, bill auditing, tariff optimization, and energy procurement advisory[^58]
- **Positioning:** Utility data infrastructure play; covers ~95% of U.S. utility customers[^57]
- **Competitive threat to Clear Current:** Direct; Arcadia is the best-funded AI-native entrant in the utility bill management and tariff optimization space

Nectar (YC-backed)

- **Description:** "The modern AI energy management platform"; automates utility data analysis to optimize energy costs, manage buildings, and report efficiency metrics[^59]
- **Core:** Proprietary utility data API using AI agents for granular usage and cost data[^60]
- **Target:** Facility managers and sustainability/ESG teams; 57,000 companies required to disclose emissions are a target segment[^59]

Gridium

- **Profile:** Energy-management-as-a-service for commercial real estate
- **Pricing:** $470/month (< 75K sq ft) to $833/month (> 400K sq ft) with portfolio discounts[^50]

- **Differentiator:** Hardware-free, subscription model; AI diagnostics

Competitive Positioning Map

Vendor	Primary Segment	Billing Audit Depth	AI/Analytics	Pricing Model	Mid-Market Accessible
EnergyCAP	Higher Ed, Government	High	Moderate	Per-meter/year ($40–$52+)	Partial
Schneider Resource Advisor	Large Enterprise	Moderate	High	Enterprise custom	No
Siemens	Industrial, Large Commercial	Moderate	High	Enterprise custom	No
WatchWire	CRE, Retail, Industrial	High	Moderate-High	Custom	Partial
Arcadia	Enterprise, ESG-driven	High	High (growing)	SaaS + services	Emerging
Nectar	SMB to mid-market	Moderate	High	SaaS	Yes
Gridium	CRE	Moderate	Moderate	Monthly subscription	Yes
Clear Current	Mid-to-large commercial	**Target: Very High**	**Target: High**	SaaS	**Target: Yes**

Section 5: Advanced Metering Infrastructure (AMI)

U.S. Smart Meter Adoption

As of 2022, **U.S. electric utilities had approximately 119 million AMI installations, covering ~72% of total electric meters**. North America reached an **82% smart electricity meter penetration rate in 2024**. However, the residential sector accounts for **88% of all U.S. AMI installations**, meaning commercial and industrial smart meter penetration significantly lags the headline figure.[^61][^13]

The commercial/industrial AMI segment, while currently modest in share, is the **fastest-growing segment** in absolute terms, driven by demand charge optimization, process monitoring, and wholesale market participation needs. Industrial AMI accounts are posting a **12.92% CAGR through 2031**.[^62]

The U.S. AMI market (all segments) was valued at **$4.5 billion in 2024** and is projected to reach **$9.0 billion** as utilities continue grid modernization programs under IRA and DOE incentives.[^63]

What Interval Data Enables for Commercial Energy Management

The shift from monthly billing reads to 5-minute or 15-minute interval data fundamentally transforms what energy management software can do for commercial customers:[^64]

Capability	Monthly-Only Billing Data	With Interval Data
Demand charge analysis	See total monthly peak demand	Identify *which 15-minute window* set the peak; model demand reduction scenarios[^65]
Time-of-use optimization	Limited to bill totals	Map consumption to on-peak vs. off-peak periods; model rate switching[^66]
Fault detection	Discover anomalies 30+ days late	Detect equipment malfunction or occupancy-hour violations in near-real time
Load shaping	Reactive only	Proactive — shift loads before the peak window to avoid demand charge ratchets[^65]
Benchmarking	Building-vs-building comparisons	Hour-by-hour benchmarking with weather normalization
PPA/contract optimization	Limited	Model actual consumption profiles against alternative rate structures or PPA terms

The availability of interval data is the key variable that differentiates a billing audit tool (backward-looking reconciliation) from a true rate optimization platform (forward-looking procurement advisory). As AMI penetration continues expanding into commercial accounts, the value of interval-data-native software platforms will increase materially.

Section 6: Higher Education Energy Market

Spending Scale

U.S. colleges and universities spend **over $6 billion annually on energy** across approximately **5 billion square feet of campus space**. The EPA estimates that **30% of this energy is wasted**, equivalent to roughly $1.8 billion in avoidable spend annually. Energy expenditures typically represent **5–10% of an institution's operational budget**.[^67][^68]

Specific benchmarks from Xcel Energy data show that U.S. colleges and universities spend an average of **$1.10 per square foot on electricity** and **$0.18 per square foot on natural gas** annually. For a large state university with 10 million square feet of space, this implies electricity costs alone of approximately $11 million per year — before accounting for district energy steam, chilled water, and other utilities.[^69]

An additional data point: many campuses have **75% of their facilities in the 30-to-40-year-old range**, meaning aging infrastructure, legacy metering, and fragmented building automation systems are the norm rather than the exception.[^68]

Typical Energy Management Software Stack at a Large Public University

Based on analysis of EnergyCAP customer profiles and industry practices:[^38][^42][^26]

1. **Utility bill management platform** (EnergyCAP is market-dominant in this category for higher ed): Ingests all utility invoices, performs bill validation, allocates costs to departments/buildings via chargeback
2. **Building automation system (BAS)**: Multiple legacy systems often coexist — a campus energy manager at the University of Arizona described having "a virtual museum of building automation systems across campus"[^26]
3. **Interval data / smart metering platform**: Used at institutions with extensive AMI deployment; enables load profile analysis
4. **ENERGY STAR Portfolio Manager**: Many institutions benchmark against EPA standards
5. **Custom spreadsheets / manual processes**: Still prevalent for specific chargeback calculations and custom reporting[^26]

The EnergyCAP Buyer's Guide specifically highlights that "most EnergyCAP customers report 10% or more in annual utility cost reductions" in higher education deployments.[^43]

District Energy: Prevalence and Billing Challenges

District energy systems — centralized plants providing steam, chilled water, and sometimes electricity to multiple buildings — are **a defining characteristic of the higher education and healthcare energy landscape**. There are **over 600 operating district energy systems in the U.S.**, concentrated on university campuses, hospitals, military bases, airports, and large corporate campuses.[^70]

These systems create significant **billing and cost allocation complexity**:

- The campus utility generates (or purchases) thermal energy (steam or chilled water) at a central plant and distributes it through a buried pipe network to hundreds of buildings
- **Allocation methodology**: Each building must be allocated a proportionate share of central plant fuel costs, equipment amortization, and distribution losses — a calculation that requires both metered consumption data and rate-schedule-equivalent pricing for internal billing
- **Cogeneration tracking**: Many campuses use combined heat and power (CHP) plants, requiring careful tracking of electricity self-generation offsets against utility purchases and any applicable PURPA or interconnection requirements

- **Healthcare district energy**: More than **20% of hospital buildings use district heating systems**, with hospitals classified as energy users closer to industrial facilities than commercial offices due to their 24/7 critical loads[^71][^72]
- **Energy intensity**: Healthcare facilities consume approximately **9% of total U.S. commercial building energy** while occupying only 4% of commercial floor space, with hospitals using **~2.5x more energy per square foot** than typical office buildings[^73][^18][^74]

For an AI-powered platform like Clear Current, district energy allocation represents a complex but high-value workflow: correctly attributing central plant costs to individual cost centers is error-prone when done manually, and the dollar values at stake are large.

Section 7: Engagement Triggers & Buying Cycles

Seasonal and Budget Cycle Patterns

Commercial energy management software purchases follow several recognizable temporal patterns:

Fiscal Year Budget Cycles: The majority of large commercial organizations — universities, healthcare systems, corporate real estate — establish their operating budgets in **Q4 of the calendar year (October–December)** for the following fiscal year. University budgets typically align with the academic year (July 1 start), meaning procurement decisions are often finalized **March–May** to allow implementation before the new fiscal year begins.[^75]

Energy Procurement Windows: Commercial energy buyers are most active in **April–May and September–October** ("shoulder months"), when wholesale electricity prices are historically at their lowest, before summer cooling and winter heating peaks. These procurement windows are natural triggers for rate analysis and contract optimization review — exactly the use case Clear Current's platform addresses.[^76]

Federal/Government Fiscal Year: Federal agencies and government-funded institutions end their fiscal year on **September 30**, creating a **Q3/Q4 spending surge** (July–September) as agencies deploy remaining budget appropriations. Energy software is a common beneficiary of end-of-fiscal-year purchasing.[^77]

Contract Expiration Windows: Energy supply contracts for commercial customers typically run **12–36 months**. Strategic buyers monitor market conditions **9–12 months before contract expiration**, creating a natural software adoption trigger when procurement teams begin the re-bid process and realize they need better data.[^78][^79]

Events That Most Commonly Trigger Software Purchase

Based on patterns from energy management consultants and user review research, the following events most reliably trigger commercial organizations to seek out energy management software:

1. **Significant rate increase or bill shock**: An unexpected spike in utility costs — often from a demand charge ratchet, rate schedule change, or volume threshold crossing — motivates finance teams to authorize energy management tooling that was previously deprioritized[^76]
2. **New energy manager or sustainability hire**: Organizations that bring on a dedicated energy manager or Chief Sustainability Officer as a first hire frequently begin with a software evaluation, as the new employee needs infrastructure to do their job[^20]
3. **Sustainability/ESG commitment or reporting mandate**: As SEC climate disclosure rules, California SB 253/261, and CSRD requirements tighten, organizations need accurate Scope 1/2 data that utilities-only-in-spreadsheets cannot reliably provide. In 2025, 85% of real estate executives surveyed by Verdantix planned to increase spend on energy efficiency and sustainability management[^80][^81][^11]
4. **Energy contract renewal or PPA negotiation**: When organizations prepare to re-bid their energy supply contract (often involving consultants or brokers), they discover gaps in their usage and cost data. A 2025 LevelTen survey found **68% of procurement teams feel they need to "act immediately"** on energy supply decisions given market conditions[^82]
5. **Benchmarking mandate or compliance deadline**: ENERGY STAR Portfolio Manager reporting requirements, local benchmarking ordinances (e.g., NYC LL87/LL97, California AB 1103), or ASHRAE audit requirements force organizations to systematically track energy data for the first time
6. **Building acquisition or portfolio expansion**: When an organization acquires new facilities — common in higher education, healthcare, and commercial real estate — the need to onboard new meters and utility accounts into a structured system creates a natural purchase window
7. **Audit failure or identified billing dispute**: When an organization discovers — often through a one-time manual audit — that it has been overbilled, it catalyzes investment in ongoing monitoring. Industry auditors report that **nearly 30% of commercial clients face billing errors annually**[^30]
8. **Sustainability reporting season**: Organizations subject to annual sustainability reporting (GRI, CDP, GRESB for real estate investors) typically face **Q1 data crunches** that expose the inadequacy of manual processes, triggering software evaluations in Q2–Q3 for the following reporting cycle

Buying Cycle Length

For mid-market commercial organizations ($10M–$100M in revenue with 10–50 meters), the sales cycle for energy management software is typically **60–120 days** from first contact to signed contract. For large enterprises (universities, health systems, national retailers), formal procurement through RFP processes can extend to **6–18 months**, often requiring security reviews, sustainability committee approval, and facilities management sign-off.

Strategic Implications for Clear Current Technologies

Differentiation Opportunities

1. **AI-native architecture**: Incumbent platforms (EnergyCAP, Schneider, Siemens) were built on 1990s–2000s architectures. User complaints consistently surface around **UX complexity, steep learning curves, and limited customization**. An AI-native platform that surfaces insights automatically — without requiring users to learn report hierarchies — addresses the most common user frustration with existing tools.[^44]

2. **Billing error recovery as lead value proposition**: Starting with billing audit / error recovery creates immediate, quantifiable ROI (refunds and future savings) that funds the software purchase. This is a proven wedge strategy — the annual error rate of 5–15% on $241 billion in commercial energy spend implies a $12–36 billion annual overbilling opportunity.[^5][^30]

3. **Mid-market accessibility**: The primary incumbents are priced and structured for large enterprise deployments requiring professional services teams. A SaaS-native product accessible to organizations managing 10–100 meters at a transparent per-meter or portfolio-based price point addresses an underserved segment.

4. **District energy and chargeback intelligence**: The 600+ district energy systems operating in the U.S. — concentrated in universities and hospitals — create a complex, high-value niche where manual cost allocation is both error-prone and opaque. Building native support for district energy billing could create a defensible position in higher education and healthcare.[^70]

5. **PPA and contract analytics**: With commercial energy procurement increasingly shifting to multi-year PPAs and 68% of procurement teams feeling urgency to act on energy supply, the ability to analyze existing contracts against market rates and identify renegotiation opportunities is an underserved use case that distinguishes a billing tool from a procurement intelligence platform.[^82]

Near-Term Go-to-Market Focus

- **Primary beachhead vertical**: Higher education (complex metering, established budget for energy software, long customer lifetime, low churn, RFP-ready procurement) aligns with the strongest demand signal and the highest concentration of energy managers as power users.

- **Secondary vertical**: Commercial real estate portfolios (institutional ESG pressure, multi-site complexity, GRESB reporting requirements, strong ROI case from billing error recovery).
- **Trigger-based outbound**: Target organizations with upcoming energy contract renewals (accessible through utility data partnerships or commercial data providers), recent sustainability commitments, or new energy/sustainability hires (LinkedIn signals).

Sources: Market sizing data from ResearchNester, MarkNtel Advisors, Strategic Market Research, KBV Research, Grand View Research, Verdantix, and Mordor Intelligence. Billing error statistics from 360 Energy, Conservice, National Utility Refund, and utility audit firms. Competitive data from G2, Capterra, Enlyft, and vendor websites. AMI data from EIA and AMR/AMRA research. Higher education energy data from DOE, Xcel Energy, and APPA. District energy data from IDEA and MIT. Buying cycle and trigger data from GridPoint, LevelTen Energy, Spacewell, and practitioner interviews.

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DOCUMENT 4: Deep Secondary Research — Four Strategic Intelligence Modules

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Clear Current Technologies: Deep Secondary Research

Four Strategic Intelligence Modules

Prepared April 2026 | Sources prioritized 2022–2026

Executive Summary

This report synthesizes deep secondary research across four strategic areas critical to Clear Current Technologies' product roadmap and go-to-market execution: (1) regulatory intelligence and the rate change notice gap, (2) IT security as an enterprise buying gate, (3) willingness to pay and deal economics, and (4) commercial real estate buyer behavior. Together, these four research streams validate the core business case — commercial energy buyers are chronically underserved by current tools, systematically surprised by rate changes they could have anticipated, blocked from adopting new software by security gatekeepers they can preemptively satisfy, and willing to pay materially for solutions that reduce budget uncertainty and recover billing errors.

Section 1: Regulatory Intelligence — The Rate Change Notice Gap

The Core Opportunity

The fundamental insight powering a regulatory intelligence module is that **rate case filings are public information from Day 1** — every PUC docket is accessible to any stakeholder who monitors it — but almost no commercial energy buyer does. The gap is not information availability; it is information processing. Clear Current can build a systematic docket-monitoring engine that surfaces financially material rate changes months before they hit a commercial customer's bill.

Rate Case Frequency and the Regulatory Landscape by State

The five target states represent meaningfully different regulatory environments, each creating distinct monitoring opportunities.

Ohio (PUCO): The PUCO must complete all rate cases within 275 days (approximately 9 months) by statute. AES Ohio filed in November 2020, received approval December 2022 — a 25-month process that included settlement negotiations. AES Ohio filed again in November 2024; all parties reached settlement August 2025, with the PUCO issuing its final order November 2025, resulting in a ~9% increase for a 1,000 kWh residential customer. AEP Ohio filed a 2025 base case, reached a settlement, and filed it at the PUCO in January 2026.

FirstEnergy had two simultaneous rate cases decided in December 2024 involving ESP IV vs. ESP V withdrawal. The pattern for major Ohio utilities: formal base rate cases every 3-5 years, but multiple overlapping proceedings (ESP reviews, reconciliation cases, special riders) create near-continuous regulatory activity.[^1][^2][^3][^4][^5]

Illinois (ICC): Illinois is meaningfully different from other states because the 2021 Clean Energy Jobs Act (CEJA) created a grid plan and annual reconciliation structure that replaces the traditional multi-year rate case cycle. Under this framework, ComEd and Ameren submit annual reconciliation petitions to recover capital expenses above budget. In December 2025, the ICC approved rate increases of $606 million for ComEd and $308.6 million for Ameren, both spread through 2027. The ICC reduced both requests — trimming $25.4 million from ComEd's ask and $11.2 million from Ameren's — on grounds of imprudent expenditures. Under CEJA, Illinois commercial customers now face annual rate adjustment proceedings rather than the traditional 3-5 year cycle, generating a continuous stream of monitored proceedings.[^6]

Pennsylvania (PUC): Pennsylvania's rate case structure requires utilities to notify customers at least 60 days in advance of filing a rate increase request. The rate request is then suspended for up to 7 months while the PUC investigates — making the total process approximately 9 months from filing to effective date. PPL Electric filed a rate case in October 2025; the final PUC decision is due by July 1, 2026. This ~9-month cycle means a commercial customer who monitors the PUC docket from Day 1 of filing has 9 months of advance intelligence before any rate change becomes effective.[^7][^8][^9]

New York (NYPSC): New York's major rate cases are among the most complex in the country. National Grid filed its Upstate NY rate case in May 2024; a multi-party joint settlement was submitted in May 2025 (12 months), with final approval coming in 2025. ConEd filed a rate case in early 2025. NYSEG has requested a 39.4% increase in delivery revenues, which would translate to a 33.5% increase to a typical total bill. New York cases frequently involve 1,600+ information requests and multiple public hearings over 12-18 months, giving commercial customers substantial advance warning — if they are monitoring the docket.[^10][^11][^12]

Texas (PUCT): Texas is structurally distinct. Approximately 85% of the state operates under ERCOT deregulation, where commercial customers choose retail electric providers and face continuously fluctuating market-based supply rates — not traditional rate cases. TDU delivery charges (e.g., CenterPoint) update twice annually, in March and September. The regulatory intelligence opportunity in Texas is therefore different: it is monitoring ERCOT capacity auction results, TDU tariff filings, and supplier contract expiration intelligence rather than traditional PUC base rate cases. CenterPoint received PUCT approval in early 2026 to recover Hurricane Beryl and Winter Storm Enzo costs over 14 years — a multi-year tariff rider that affects all commercial customers in its territory.[^13][^14][^15][^16]

Rate Change Magnitude: The Structural Shift

The scale of rate increases now hitting commercial customers is materially larger than what corporate budgets assume. Arcadia's 2026 Commercial Electricity Rate Report — which analyzed 81 utilities and 5 building types across 2020–2025 using 30,000+ tariff combinations — found that **97.5% of commercial facilities experienced**

electricity rate increases during that five-year period, with **71% outpacing inflation**. The median compound annual growth rate for commercial electricity rates was **5.9%** — nearly double the 3% escalator still embedded in most corporate budgets.[^17][^18][^19]

The EIA confirms the trend: the commercial sector average retail price was 12.75 cents/kWh in 2024, rising to 13.63 cents/kWh in December 2025 — a **7.83% year-over-year increase**. In January 2026, commercial sector electricity revenues per kWh rose 6.4% compared to January 2025. Select state examples are more severe: New Jersey commercial customers saw 9.1% increases, Rhode Island 8.6%, New Hampshire 8.3%, Maine 7.3%.[^20][^21][^22][^23]

The most acute structural shock is in PJM's capacity market. The 2026-2027 PJM Base Residual Auction cleared at a record **$329.17/MW-day**, a 22% increase from the prior year. The prior year had already seen PJM capacity prices surge 833% from 2024 rates. For a typical 1-MW commercial load, Arcadia estimates annual capacity costs will rise from approximately $10,000 in 2024 to more than $120,000 by 2027 — a structural cost shift that efficiency improvements alone cannot offset. For PJM-territory commercial customers (spanning 13 eastern states from Illinois to New Jersey), this represents an **unbudgeted $110,000 annual liability on a single line item** for a standard commercial asset.[^24][^25][^26][^17]

EIA projects further increases: U.S. wholesale day-ahead electricity prices rose across most major hubs in 2025 versus 2024, with ISO-NE posting the largest increase at $29/MWh. ERCOT wholesale prices are forecast up approximately 45% in 2026 versus 2025 at the summer peak hours.[^15][^27]

When Does Information Become Actionable?

The regulatory calendar creates a predictable advance-notice window at each stage:

Stage	Timing	Public Accessibility
Utility files Notice of Intent (some states)	30-60 days before filing	Yes — formal notice to commission
Utility files rate case application	Day 0	Yes — docket opened immediately
Staff report/initial analysis	Months 4-5	Yes — filed in docket
Public hearings	Months 6-8	Yes — scheduled and noticed publicly
Proposed Decision / ALJ recommendation	Month 7-8	Yes — filed in docket
Final Commission order	Month 9 (Ohio) to 18+ months (NY)	Yes
Rates effective	After order + notice period	—

The entire process is publicly accessible. Pennsylvania requires utilities to notify customers in billing notices at least 60 days before filing; Ohio makes the docket accessible from filing day and the 275-day clock is public; the PUCT publishes "Notice of Proposed Rate Change" documents that any company can subscribe to. The actionable intelligence gap is not regulatory — it is that no commercial energy software currently aggregates, processes, and translates these filings into facility-level financial impact models.[^5][^8][^28]

What Competitors Currently Offer (and Don't)

Energy Toolbase is the closest analog with a rate intelligence function, but it is fundamentally a solar/storage project development tool — not a commercial energy management platform. Its utility rate database covers 5,500+ rate records across 35 states, updated within 1-2 business days of changes. However, it does not monitor PUC dockets for proposed changes, does not alert commercial buyers to rate cases in progress, and is positioned entirely for solar/storage developers building proposals — not for commercial energy managers tracking budget exposure.[^29][^30][^31]

EnergyCAP, WatchWire/Tango, and Schneider Resource Advisor offer utility bill management, benchmarking, and historical analytics, but none provide forward-looking PUC docket monitoring or rate change impact modeling as a product feature. The regulatory intelligence module is a **genuine white space** in the current competitive landscape.

The Budget Timing Gap

The regulatory intelligence opportunity is most acute at universities, where budget timelines and rate case timelines are structurally misaligned. Boston University's budget process illustrates the pattern: Stage I (next fiscal year budget) development begins in late summer/fall, with Board of Trustees approval in March — before the July 1 fiscal year start. Pennsylvania's PPL rate case filed October 2025 is scheduled for final decision by July 1, 2026 — the exact date a university's new fiscal year begins. A university that submitted its FY2027 energy budget in March 2026 cannot know its actual July 2026 rates until after the budget is locked.[^32][^9]

Corporate energy budgets face the same structural problem. Arcadia's report found that the median CAGR of 5.9% is "nearly double the 3% escalator still baked into many corporate budgets" — not because finance teams are negligent, but because they lack facility-level, tariff-specific rate change forecasting. The PJM capacity surge alone creates a $110,000 unbudgeted line-item exposure per MW of commercial load, invisible to any organization using a flat percentage escalator.[^19][^24]

Section 2: IT/Security as an Enterprise Buying Gate

The Core Dynamic

The pattern surfaced in practitioner interviews — field-level energy managers want AI-powered tools, IT/security blocks adoption even when business users are enthusiastic — is heavily documented in enterprise SaaS literature. It is not an edge case; it is the dominant dynamic in enterprise software adoption.

Prevalence of IT-Blocked SaaS Adoption

The scale of security-driven purchase friction is striking:

- **50% of U.S. businesses** cite security as the most influential factor when buying software — the single highest-ranked factor, above functionality (49%) and ease of use (46%)[^33][^34][^35]
- **45% of U.S. businesses have stopped using a software platform** due to security concerns — demonstrating that IT gatekeeping occurs not just pre-purchase but post-deployment[^35]
- **Security teams hold veto power in 94% of software purchases** per Forrester 2024[^36]
- **94% of enterprise organizations have abandoned vendor partnerships** over data security concerns (Forrester 2024)[^36]
- **91% of enterprise clients evaluate vendors' data practices** in the initial selection phase, before any product evaluation occurs (McKinsey 2023)[^36]
- **55% of employees adopt SaaS without security's involvement** (CSA 2025) — meaning shadow IT is common, but any vendor seeking enterprise-wide adoption and billing-data access must go through formal security review[^37]

The energy/facilities management context amplifies this. Operational technology (OT) environments — including building management systems and utility meter integrations — carry elevated security sensitivity because they bridge IT and physical infrastructure. Facilities SaaS platforms require integration with utility accounts, billing portals, and increasingly with building automation systems, all of which trigger heightened security review scrutiny.

Security Review Timelines

Enterprise IT security reviews for SaaS vendors vary significantly by company size, data sensitivity, and compliance requirements. Based on available evidence:

- **Cloud CMMS deployment** (facilities management equivalent): 3-5 days for cloud, versus 3-6 months for on-premise — illustrating how cloud SaaS dramatically compresses deployment, but security review timelines are separate from deployment timelines[^38]
- **SOC 2 Type II audit timeline**: 6-12 months to complete — vendors must complete this before large enterprise buyers will approve[^39][^40]

- **Sales cycles shortened by 3.2 months** after implementing robust data security practices (PwC 2024), implying that without certification, sales cycles are 3+ months longer on average[^36]
- **62% higher win rate** for vendors with strong data security credentials (Gartner 2023)[^36]
- Typical enterprise security review for a new SaaS vendor requiring data integration: 30-90 days for security questionnaire review; up to 180 days for formal SOC 2 audit review, penetration testing verification, and legal/procurement sign-off at large institutions

The healthcare and higher education verticals are the most demanding. Hospitals operate under HIPAA requirements and often apply healthcare-grade security scrutiny to all vendors, even those handling non-patient data. Universities increasingly impose IT security review processes modeled on state government standards, particularly for cloud platforms that interact with financial data (utility billing charges directly to departmental cost centers).

Which Certifications Accelerate Approval

The evidence is clear on prioritization:

Certification	Enterprise Impact	Cost / Timeline
SOC 2 Type II	Most universally required; 62% higher win rate; 3.2 months faster sales (Gartner/PwC) [^36]	$20K-$80K total; 6-12 months [^39][^40]
ISO 27001	Required by many international/global enterprise buyers	Similar cost and timeline; additive to SOC 2
CSA Certification	59% of buyers more attracted to CSA-certified software [^33]	Lower cost; signals cloud-specific security maturity
FedRAMP	Required for government customers; not required for commercial CRE/healthcare/higher ed	Very high cost ($1M+); only if targeting federal government

For Clear Current Technologies targeting commercial energy managers in CRE, healthcare, and higher education, **SOC 2 Type II is the minimum required certification** before institutional enterprise procurement will approve the platform. The investment ($20K-$80K for initial certification for a startup; $35K-$130K annually ongoing) pays back quickly: companies report SOC 2 certification produces $100K+ increase in average deal size and enables 50% faster enterprise sales cycles within the first year.[^41]

The Champion vs. Gatekeeper Strategy

The documented best practice for navigating champion/gatekeeper tension in enterprise SaaS:

1. **Arm the champion early.** Provide the energy manager/sustainability director with a security one-pager, SOC 2 report, and pre-completed security questionnaire before they raise it internally. The champion cannot advocate without collateral.

2. **Design for IT-friendly architecture.** SSO (SAML 2.0/OpenID Connect), role-based access controls, and data encryption at rest/in transit are the specific features IT evaluates first. Building these in early removes the most common objection vectors.[^42][^43]
3. **Limit data exposure scope.** Energy billing data — utility account numbers, meter IDs, cost records — is sensitive. Architecturing Clear Current to ingest only what it needs (read-only API access to utility portal data, not full account credentials) significantly reduces the security surface area IT must evaluate.
4. **Use trust marks proactively.** 76% of organizations are increasing SaaS security budgets. IT teams are actively looking for vendors who have already done the compliance work. Leading with SOC 2 Type II in all marketing and proposal materials, not waiting for it to be asked about, signals enterprise-readiness and accelerates the security review queue.[^37]

Data Sensitivity in Commercial Energy Management

Commercial utility data — billing histories, consumption patterns, demand peaks, contracted rates, PPA terms — is competitively sensitive business information. Energy spend patterns can reveal operational schedules, production volumes, occupancy fluctuations, and strategic facility expansions. At REITs and institutional CRE firms, energy cost data is material to financial reporting and investor disclosure. At universities with federal research grants, energy allocation data intersects with grant accounting.

The sensitivity creates a genuine IT gatekeeper concern that is not irrational. However, it also means that vendors who have *already* demonstrated data governance rigor — through SOC 2 Type II, clear data use policies, and opt-in data sharing — will find that concern becomes a competitive advantage rather than a barrier once documentation is in hand.

Section 3: Willingness to Pay and Deal Economics

Energy Management Software Pricing Benchmarks

Published pricing data for the commercial energy management software market is sparse, but the following benchmarks are documented:

Gridium (commercial real estate buildings): Small buildings up to 75,000 sf start at **$470/month**; medium buildings (75K-400K sf) start at **$760/month**. Gridium markets a 5-7% energy savings expectation with a 2-3 month payback on the subscription cost. Gridium's model is per-building, not per-meter, targeting property managers in the commercial real estate segment.[^44]

EnergyCAP: Published pricing of $40-$52/meter/year (from prior research). At 100 meters, this represents $4,000-$5,200/year; at 500 meters, $20,000-$26,000/year; at 1,000+ meters, $40,000-$52,000+/year. EnergyCAP positions primarily on utility bill management, billing error detection, and energy accounting — the most direct comparable to Clear Current's core functionality.

WatchWire by Tango: Privately priced; acquired by Tango (Berkshire Partners-backed) in August 2023 for undisclosed consideration. Tango serves 500+ customers across 140+ countries. Tango's combined platform includes real estate lifecycle management, so energy/sustainability is a module within a broader enterprise real estate platform — which implies higher ACV than pure-play energy software.[^45][^46][^47]

Measurabl: Launched a free base tier in July 2025, onboarding over 11,000 buildings representing 2 billion sq ft in four weeks. Advanced features (Optimize for real-time asset optimization, Comply for BEPS compliance) remain premium. The free-to-premium model is a deliberate land-and-expand strategy for the CRE ESG reporting segment.[^48][^49]

Arcadia Enterprise: Following its January 2025 enterprise pivot and launch of a 3-product enterprise suite, Arcadia appears positioned at the high end of the market — pricing not publicly disclosed but enterprise contract values implied at $100K+ ACV given the funding level ($649M total) and enterprise focus.

University System Contract Values

Public procurement records provide the clearest window into institutional willingness to pay:

- **University of Wisconsin System RFP** (issued September 2024, deadline October 2024): AI-estimated contract value range of **$500,000–$2,000,000** for energy management software system-wide. This is the most directly comparable publicly documented contract value for a university-system energy management platform procurement.[^50][^51]
- **Virginia Tech**: Entered a contract with US Energy Management (contract VTS-2066-2024) effective July 2023, terms not disclosed in public record.[^52]
- **University of Kentucky**: Issued RFP for campus energy infrastructure including thermal energy distribution and monitoring — a larger P3 (public-private partnership) engagement.[^53]

The $500K-$2M UW range reflects a multi-campus system (26 campuses across the UW System). For a single flagship university, a reasonable contract value estimate is $75,000-$500,000 annually for a comprehensive energy management SaaS covering utility bill management, cost allocation, district energy billing, and reporting.

CFO and Executive Approval Thresholds

Enterprise SaaS purchase authority typically escalates as follows in commercial organizations:[^54]

- **≤$5,000/year**: Department head or manager approves
- **$5,001–$25,000/year**: Finance or VP-level approval required
- **$25,001–$100,000/year**: CFO or C-suite review typically required
- **$100,000+/year**: Investment committee, Board, or formal procurement process

In higher education, Board of Trustees approval is required for major contracts; the budget process locks FY budgets in March (for July 1 FY start), which means software purchases above ~$50K typically must be included in the annual budget cycle rather than approved mid-year. In CRE, institutional investor oversight of ESG and operational expenditures means energy software exceeding $50K-$100K/year is likely reviewed at the asset manager or fund level, not just property management.[^32]

For Clear Current's target deal sizes — $25K to $200K ACV depending on account size and module count — deals will typically require Director-level or VP-level champions and CFO awareness, but are unlikely to require formal Board-level approval in most commercial organizations.

Contingency Fee Benchmarks for Utility Bill Auditing

Third-party utility bill audit firms operate almost exclusively on a contingency basis: clients pay only when savings or refunds are identified and recovered. Key benchmarks:[^55][^56]

- **Fee structure**: Contingency basis — percentage of recovered savings[^57][^56][^55]
- **Typical range**: 20-50% of recovered savings, with industry practice centering around 25-35% based on reported market norms
- **Audit scope**: Reviews electricity, gas, water, and other utilities for incorrect rate applications, meter reading mistakes, unnecessary fees, wrong rate class, estimated vs. actual billing, and tax errors[^56][^55]
- **Potential savings**: Comprehensive commercial energy audits can save businesses up to 30% on annual energy costs[^58][^57]
- **EnergyCAP benchmark**: Nearly 5% of utility bills contain errors; recovering them typically saves 1-3% on utility spend with zero changes to consumption[^59]

The contingency model creates a powerful economic framing for Clear Current. A commercial organization spending $500,000 annually on utilities that is overbilled 5% on even a fraction of bills is looking at $10,000-$25,000 in potential recoveries. If Clear Current surfaces these on a contingency model or SaaS subscription priced well below the value recovered, the ROI case is immediate. EnergyCAP's own survey shows 2.5x annual savings versus the average EMS user, with ROI achieved within 2.6 years on average — with 25% of customers reaching ROI within one year.[^60][^61][^62]

ROI Expectations and Payback Periods

Commercial energy buyers expect demonstrable financial returns from energy management software investments. Documented benchmarks:

- **EnergyCAP customers**: 7.5%+ year-over-year savings on total utility bill spend; ROI in 2.6 years average; 25% in 1 year or less; 40+ hours saved per month in staff time[^61][^62][^60]

- **LBNL Smart Energy Analytics Campaign** (104 organizations, 6,500+ buildings, 500M+ sq ft): median annual energy savings of 3% with Energy Information Systems, up to 9% with Fault Detection & Diagnostics[^60]
- **BEMS (Forrester/Johnson Controls study)**: 155% ROI over 3 years; 10% energy reduction; 8-month payback period[^63]
- **ACEEE**: 5-15% energy savings from SEM approaches; 59% of savings identified from no-capex operational measures[^64]

The SaaS payback calculation for Clear Current's core use case:

For a mid-market commercial organization with $500,000 in annual utility spend:

- EnergyCAP benchmark: 5% of bills flagged for errors monthly → recovers 1-3% of spend → $5,000-$15,000 annually from billing error recovery alone
- Rate optimization (moving to better tariff): an additional 2-5% → $10,000-$25,000 annually
- Total potential recovery: $15,000-$40,000/year
- At Clear Current SaaS pricing of $15,000-$30,000/year, the platform pays for itself in billing error recovery before any rate optimization or strategic savings are counted

At $2M in utility spend (a large hospital or mid-sized university), the math is proportionally larger: $150,000-$400,000 in potential annual recovery, making a $50,000-$150,000 ACV SaaS contract highly defensible.

Sales Cycle Length by Deal Size and Vertical

Segment	Typical Sales Cycle	Primary Gating Factor
Mid-market commercial (50-200 meters)	60-90 days	IT security review + budget approval
Large enterprise commercial (200-1,000 meters)	90-180 days	IT security + C-suite sign-off
Large public university	6-18 months	Formal RFP procurement process
REIT/institutional CRE	90-180 days	Sustainability director + investor committee
Healthcare system	90-180 days	IT/security + value analysis committee

University RFP timelines are documented: UW System issued RFP in September 2024, set response deadline October 1, 2024 — but the full award, contracting, and implementation cycle extends well beyond the RFP closing date. Budget cycle lock in March means proposals must be in budget before March of the year prior to implementation start.

Section 4: Commercial Real Estate Buyer Behavior

Staffing and Energy Management Function in CRE

Large institutional CRE firms approach energy management through a layered structure: property-level operations (chief engineer, facilities manager) handle day-to-day utility management; portfolio-level sustainability teams (Director of Sustainability, ESG Manager) drive reporting, benchmarking, and vendor selection; and executive/investor oversight (CFO, asset manager, fund manager) sets policy and approves capital allocation.

WatchWire by Tango — the leading energy management SaaS specifically designed for institutional CRE — serves 500+ customers in 140+ countries and counts Brookfield as a named customer. Tango is Berkshire Partners-backed, signaling institutional confidence in CRE energy management as a durable software category. The GRESB 2025 Real Estate Assessment attracted 1,002 fund managers submitting 2,382 assessments — this is the global benchmark that institutional CRE firms track their energy performance against, and GRESB data management creates direct demand for energy management platforms.[^65][^66][^47][^45]

Key CRE energy software tools in use:

- **WatchWire by Tango**: Enterprise-grade energy/sustainability data management; deep GRESB reporting integration; primary tool for major REITs and institutional landlords[^65]
- **Measurabl**: Free tier launched July 2025, onboarded 11,000+ buildings in 4 weeks; targets ESG/sustainability reporting layer; premium products include real-time asset optimization and BEPS compliance[^48]
- **ENERGY STAR Portfolio Manager**: Free government benchmarking tool; baseline required for LL97 compliance in NYC; ENERGY STAR future uncertain due to federal policy changes, which Measurabl specifically positioned its free tool to address[^67][^48]
- **Gridium**: Per-building pricing ($470-$760+/month) targeting commercial property managers[^44]
- **JLL/CBRE proprietary tools**: Major property management firms have built internal data management capabilities, creating a captive tool ecosystem that independent software vendors must compete against or integrate with

CRE-Specific Pain Points

Commercial real estate energy management differs from higher education and healthcare in several important ways:

Portfolio breadth vs. single-campus depth: A REIT may manage thousands of buildings across dozens of markets, each with different utilities, rate structures, and lease types. The energy management challenge is less about a single complex campus and more about normalizing data across an enormously heterogeneous portfolio.

Tenant billing complexity: In gross lease structures, the landlord pays utilities and seeks reimbursement from tenants — requiring allocation methodologies. In NNN (triple-net) structures, tenants theoretically pay utilities directly, but shared metering in multi-tenant buildings creates allocation complexity. Over 30% of commercial landlords misinterpret cost responsibilities in commercial leases. Submetering vendor failures — including lack of transparency, missed accounting deadlines, incorrect demand charge billing, and errors stretching back years — are documented as the most common property management pain points.[^68][^69][^70]

GRESB reporting obligations: GRESB's 2024 Real Estate Assessment covered $10 trillion in global real estate assets; 2025 saw 1,002 fund managers participate at an all-time high. GRESB data submission deadlines in August-September, with results released in October, create a defined annual urgency spike. The 2024 GRESB standard shifted to country-level benchmarking, increasing the need for utility data accuracy at the asset level.[^66][^71][^72][^73][^74]

ESG executive accountability: 51% of Nareit corporate REIT members linked executive compensation to ESG performance in 2021 (up from 34% in 2020); 90% have board-level ESG oversight. This governance structure means ESG/energy data quality is a C-suite and board issue, not just an operational one — elevating the decision authority and urgency for energy management software.[^75]

NYC Local Law 97 as the Dominant Buying Trigger

NYC Local Law 97 (LL97) represents the most powerful single regulatory catalyst for CRE energy management software adoption currently in the market:

- Applies to buildings over **25,000 gross square feet** — covering most institutional commercial real estate in NYC[^76][^77]
- Penalties: **$268 per ton** of excess carbon emissions above the cap[^78][^76]
- First compliance period: **2024-2029**; first annual report was due May 1, 2025 (with extensions to December 31, 2025)[^79][^67][^78]
- Current compliance projections: 88% of covered buildings are on track for 2024 limits, but only 35% are aligned with the tighter 2030 limits[^78]
- By 2030, approximately **13,500 of 50,000 covered properties** are projected to be out of compliance, generating cumulative fines of **$900 million annually**[^78]

LL97 is not just a compliance cost — it is a data collection mandate. Accurate annual emissions reporting requires granular utility data at the building level, certified by a registered design professional. Many NYC commercial buildings had been using ENERGY STAR Portfolio Manager for voluntary benchmarking under Local Law 84; LL97 converts that data into the basis for legally enforceable penalties. Local Law 88 adds a parallel mandate: electrical sub-meters for all tenant spaces ≥5,000 sq ft — generating interval consumption data that energy management software can analyze.[^67][^78]

The combination of 65% non-compliance with 2030 targets, $900M in projected annual fines, and the May 1 annual reporting deadline creates a recurring buying trigger in Q4 and Q1 of each year as property owners assess their compliance trajectory and seek software to close the gap.

CRE Buying Cycle and Decision Authority

The CRE buying process for energy management software follows a distinct pattern from higher education or healthcare:

Champion: Sustainability Director, Chief Engineer, or Energy Manager at the portfolio/platform level — not individual property managers[^75]

Evaluators: IT security (data integration), Finance (cost justification), Legal (data sharing agreements)

Decision-maker: CFO or Asset Manager; Investment Committee for multi-million-dollar commitments

Primary triggers (in order of frequency and urgency):

1. GRESB reporting cycle (Q3/Q4 deadline urgency)
2. LL97/BEPS compliance deadlines (annual May reporting)
3. Portfolio energy cost spikes (rate changes, capacity surcharges)
4. ESG investor pressure / net zero commitment obligations
5. Acquisition of new assets requiring integration into portfolio-wide reporting

Procurement format: Institutional CRE firms make **portfolio-level** purchasing decisions — they do not buy energy software property-by-property. A single software contract covers the entire portfolio, which is why contract values for REIT-level customers are substantially higher than individual building pricing would suggest. This also means the sales process is more complex but the winner-takes-portfolio dynamic creates high switching costs and durable revenue.

Tenant Utility Billing: The Structural Error Problem

The sub-metering and tenant utility billing challenge in CRE represents a high-value, underserved problem with significant error rates:

- Submetering vendors frequently fail on five dimensions: (1) no transparency in data collection methodology, (2) missed accounting deadlines (buildings are ruled by billing cycles), (3) incorrect demand charge billing (demand charges represent 30-70% of a commercial electric bill and are frequently misunderstood), (4) opaque vendor processes creating a "black box" for property managers, and (5) billing errors that compound and "stretch back years"[^69]
- Property managers who outsource submetering have "little visibility into how the vendor operates, which makes it difficult to confirm the accuracy of their work"[^69]

- In NNN structures with single master meters and multiple tenants, landlords must pro-rate utility costs — and the methodology (square footage allocation, submetering, utility company direct billing) creates ongoing dispute potential[^70][^68]

Clear Current's AI-powered bill analysis capability directly addresses this pain point. The ability to detect billing errors retroactively, validate meter multipliers, confirm rate class accuracy, and identify demand charge calculation errors across a portfolio — with audit-ready documentation — fills a specific gap that neither WatchWire nor EnergyCAP's core bill processing addresses at the granularity CRE firms need.

Competitive Landscape in CRE Energy Management

Vendor	Primary Positioning	CRE Fit	Key Weakness
WatchWire by Tango	Enterprise sustainability + energy data; GRESB focus[^65][^45]	Very High	High price; complex; no AI billing error detection
Measurabl	ESG/sustainability reporting; free tier[^48]	High (ESG layer)	Metrics/reporting only; no billing audit or procurement
Gridium	Per-building energy analytics; commercial office[^44]	Medium	Per-building model doesn't scale for large portfolios
ENERGY STAR Portfolio Manager	Free benchmarking[^67]	Baseline only	No billing audit, no rate optimization, uncertain future under current federal policy[^48]
EnergyCAP	Utility bill management; higher ed/government primary[^80]	Medium	Designed for campus environments; not CRE-native
Clear Current (target position)	AI bill audit + rate optimization + regulatory intelligence	High	Unproven; needs SOC 2

Cross-Cutting Strategic Implications for Clear Current Technologies

1. The Regulatory Intelligence Module Has No Competitor — Launch It as a Differentiator

No current energy management platform provides forward-looking PUC docket monitoring and rate change impact modeling. Rate case filings are public from Day 1, but the translation of docket filings into facility-level financial impact models requires the exact combination of tariff data, meter-level load profiles, and AI-powered bill simulation that Clear Current is building. The PJM capacity surge alone — creating $110K+ in unbudgeted commercial liability per MW — validates demand for this module in real time.[^17][^24]

2. SOC 2 Type II Is Not Optional — Sequence It in Year One

With security teams holding veto power in 94% of software purchases, and 45% of businesses having abandoned software platforms over security concerns, attempting to sell into hospitals, universities, or institutional CRE without SOC 2 Type II certification will result in systematic blocking at the IT review stage. The investment ($20K-$80K + 6-12 months) unlocks 62% higher win rates and shortens sales cycles by 3.2 months — ROI in the first year from a single enterprise deal.[^40][^39][^35][^36]

3. The Contingency / Guaranteed-Savings Entry Model Works

The contingency audit model — clients pay only on recovered savings — is the established go-to-market norm in utility bill auditing. Clear Current can use this as an initial land motion: no-cost billing error audit for a defined historical period (12-24 months of bills), with a contingency fee on recoveries, then convert to annual SaaS subscription once the platform has demonstrated value. The unit economics are highly favorable: if Clear Current surfaces $25,000 in billing errors for a $1M/year energy spend client on a 25% contingency, Clear Current earns $6,250 from the discovery alone and now has a SaaS conversion opportunity.[^55][^56]

4. NYC LL97 Creates Immediate CRE Urgency — Don't Ignore It

With 65% of covered NYC buildings not aligned with 2030 emissions limits and cumulative fines projected at $900M annually, every institutional CRE firm with NYC exposure is actively seeking tools to close the compliance gap. Clear Current's utility bill analysis and energy cost optimization capabilities sit directly in this value chain. A targeted NYC CRE go-to-market motion — specifically positioned around LL97 compliance data accuracy and reporting — would find a warm buyer community.[^78]

5. GRESB's October Reporting Cycle Creates a Predictable Annual Sales Season in CRE

GRESB data submission closes in September; results are released in October. Q1 and Q2 each year — January through June — is when CRE firms assess data gaps from the prior year's GRESB submission and plan for the next cycle. This makes January-June the highest-intent period for CRE energy management software purchasing. Layer in LL97's May 1 annual reporting deadline, and the Q1/Q2 window becomes the highest-density buying season for institutional CRE accounts.[^73][^74]

6. University Procurement Requires a 12-Month Sales Cycle Budget

The combination of formal RFP requirements, Board of Trustees approval processes, and fiscal year budget locking means that university enterprise deals have 6-18 month sales cycles. A deal that first touches a university in January 2026 realistically closes no earlier than July 2026, and more likely July 2027 if it misses the March budget cycle. The strategic implication: identify university targets 12-18 months before desired close, build the relationship during budget planning season (September-November), and have a proof-of-value pilot in place before the formal RFP is issued.[^50][^32]

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DOCUMENT 5: GTM Strategy, Competitive Intelligence & Investor Readiness (Part 4)

Source file: Clear Current Technologies — GTM Strategy and Investor Readiness Part 4.md

Clear Current Technologies: GTM Strategy, Competitive Intelligence & Investor Readiness Research

Prepared April 2026 | Covers all seven strategic priorities identified through founder/advisor interviews and prior market research

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Executive Summary

Clear Current Technologies enters the market at a moment of structural opportunity: the dominant incumbent (EnergyCAP) is a 40-year-old platform backed by PE for growth it hasn't yet demonstrated in UX or AI; the most well-capitalized AI-native competitor (Arcadia) pivoted to enterprise in January 2025 with a data infrastructure orientation rather than a workflow tool; and the most reliable government benchmarking infrastructure (ENERGY STAR Portfolio Manager) faces an existential budget threat that could displace 330,000 buildings from their primary energy reporting system.[^1][^2]

Clear Current's 12-pilot portfolio — with $10M in identified errors — is early but credible evidence of product-market fit. The May 2026 raise needs to arrive with pilot conversions, a clear beachhead customer profile, and a defensible account of how the company gets to $1M–$3M ARR — the threshold where Series A investors are currently writing checks at $10M–$20M rounds and $25M–$50M pre-money valuations. This report provides the research infrastructure to support that narrative.[^3]

Part 1: GTM Motion — Founder-Led Direct Enterprise Sales

The "Land with Audit, Expand to Platform" Model

Clear Current's current motion — founder-led direct enterprise sales targeting billion-dollar-plus organizations with 50+ locations — is the highest-friction but highest-margin path available. It is also the right path for a pre-revenue SaaS company in a category where buyers are skeptical, switching costs are high, and the value proposition requires demonstration, not description.

The pilot-to-paid conversion dynamic is well-studied in enterprise SaaS. According to SaaStr's dataset across thousands of B2B companies, pilot to annual contract conversion rates run 60–90%, with 70%+ achievable at companies that have refined their sales motion. Clear Current's 12 pilots represent a pipeline where the expected conversion, even at the low end, produces 7–8 paying customers — enough to demonstrate repeatable sales at the target $10K/month price point.[^4]

The revenue recognition problem identified by John is material and standard in enterprise SaaS. A $100K+ deal typically requires 6–18 months from first contact to close, and complex procurement at universities or large commercial portfolios extends toward the 12–18 month end. The five-month revenue recognition lag that John observed (deals signed in January not generating revenue until June) is consistent with enterprise SaaS norms, not an anomaly. The financial model needs to reflect this reality explicitly before going into fundraising conversations.[^5]

The Three GTM Tracks — Prioritized

Track 1 (Active): Direct Enterprise is the right initial motion. The proof is simple: $10M in identified errors across 12 pilots creates real urgency for the buyers who have already seen the platform. Converting those pilots to paid contracts before May is the single most important near-term goal, because it gives investors a revenue number, not just a pipeline number.

Track 2 (Proposed): ERP Licensing (SAP/Oracle/QuickBooks) is strategically compelling but operationally distant. The average SAP integration partnership timeline is 12–24 months from initiation to live customers, and the go-to-market dependency on a large ERP vendor's sales team creates control problems for an early-stage company. This track is worth pursuing in parallel conversations but should not be counted in the May fundraise narrative.

Track 3 (Proposed): Utilities as Customers is the least understood and potentially most scalable of the three tracks. Utilities have regulatory and reputational incentives to bill correctly in both directions. FERC oversight and state PUC proceedings document real consequences for systematic billing errors. A utility that proactively deploys Clear Current's audit logic to catch its own errors before customers complain would be positioning itself favorably ahead of regulatory scrutiny. This track deserves a dedicated exploratory conversation with one or two utility partners in deregulated Texas, where ERCOT's competitive structure creates the most billing complexity.[^6][^7]

Part 2: Beachhead Vertical — Higher Education

Why Higher Ed First

The case for higher education as the primary beachhead is not just data-supported — it is structurally compelling. Three forces converge:

1. **EnergyCAP is most entrenched here, and also most vulnerable here.** Higher ed represents 19% of EnergyCAP's customer base, and the campus chargeback capability that makes EnergyCAP irreplaceable is simultaneously the feature users complain about most — described in user reviews as "hard to understand," lacking documentation, and requiring extensive training. This is the classic incumbent trap: the feature that keeps customers stuck is the feature users hate most.[^8][^9]

2. **District energy complexity creates problems no AI-native competitor has solved.** Of 660+ U.S. district energy systems, the vast majority serve university and hospital campuses. The cost allocation formula problem — distributing central plant steam/chilled water costs across 50–150 metered buildings — requires a combination of utility bill intelligence, tariff logic, and financial system integration that neither Arcadia nor Verse nor Gridium offers.[^10]
3. **Primary research confirms active pain.** Andee Chamberlain at Texas State (1,500 meters, 98 buildings) and Andi Ault at UT Austin both confirmed EnergyCAP as incumbent and both described specific unresolved problems: rate anomaly detection gaps, financial system integration failures, and budget timing mismatches. These are not hypothetical problems.

The University of Wisconsin RFP as a Baseline

The University of Wisconsin System issued RFP 2025-UW-00732-RFP in September 2024, seeking energy management software for an estimated $500,000–$2,000,000 contract value. The stated requirements are an exact match for Clear Current's positioning: real-time data analytics, reporting capabilities, integration with existing systems, and user-friendly interfaces across multiple campuses.[^11][^12]

This single RFP establishes three important facts for Clear Current's investor narrative: (1) public universities issue formal RFPs for energy management software; (2) contract values at the system level run $500K–$2M — significantly above Clear Current's current $120K/year pricing; (3) user-friendly interface is an explicit evaluation criterion — directly addressed by Clear Current's AI-native design and the shortcoming most consistently cited against EnergyCAP.

The Texas State Displacement Case

Texas State University contracted EnergyCAP through the YearOut Energy reseller, piggybacking on the UTSA EnergyCAP agreement starting in August 2012. Texas State's public purchasing records show EnergyCAP software spend of approximately $17,900 per purchase order in late 2024, suggesting an annual spend in the range of $50,000–$75,000 for their 1,500-meter installation. This is a documented incumbent relationship where Clear Current has already conducted primary research conversations with the energy management team — and where EnergyCAP's reported UX weaknesses are most likely to be confirmed.[^13][^14][^15]

The displacement thesis for Clear Current at Texas State is: demonstrate $10M in billing errors across similar-scale portfolios → propose a parallel deployment that handles billing audit and rate optimization → prove value in 90 days → displace EnergyCAP on chargeback once that module ships. This is a 12–18 month enterprise displacement cycle, not a 30-day close, but it is a realistic sequence.

The University of New Mexico Cost Benchmark

University of New Mexico's EnergyCAP cost on record is $9,600/year via its Lobo Energy reseller agreement. For a campus of UNM's scale, this implies pricing at approximately $15–20/meter/year through a reseller channel — dramatically below Clear Current's $120K/year target. This gap reveals both an opportunity and a challenge: Clear Current must justify 5–10x the per-meter cost of the incumbent by demonstrating proportionally greater value in error recovery and rate optimization. The $10M in identified errors across 12 pilots is exactly the proof point needed to justify that premium.[^16]

Healthcare as Second Vertical

The structural parallels between university and hospital energy management make healthcare the natural second vertical. The specific overlaps are: district energy systems (more than 20% of hospital buildings use district heating systems); complex cost allocation across patient care, research, and administrative functions; compliance reporting requirements (Joint Commission, state health department); and high energy intensity (hospitals consume 2.5x more energy per square foot than office buildings). Both Andee Chamberlain (Texas State) and Don Johnson independently named hospitals as the closest analogue to university energy management complexity — peer validation from the field.[^17][^18]

Part 3: Deregulated Energy Market Geography

The 18-State Opportunity Map

As of 2025, 18 states plus Washington D.C. have some form of retail electricity choice for commercial customers. The degree of deregulation — and therefore the degree of billing complexity — varies significantly: [^19]

State	Status	Key Feature	Relevance to Clear Current
Texas	Fully deregulated (ERCOT)	100+ REPs, mandatory choice, 87% switching[^19]	Highest billing error exposure; Clear Current's home base
Pennsylvania	Deregulated since 1997	Dozens of suppliers via PA Power Switch[^20]	Large commercial real estate + university market
Ohio	Deregulated since 2001	PUCO-certified CRES providers[^20]	Strong municipal aggregation
Illinois	Deregulated since 1997	ComEd + Ameren territories[^21]	Major metro commercial RE
New York	Deregulated since 1996	Choice with utility-owned delivery[^21]	LL97 compliance driving software demand
New Jersey	Deregulated	Full residential + commercial	Dense commercial market
Maryland	Deregulated	Active competitive market	Mid-Atlantic CRE
Massachusetts	Deregulated since 1998	Full competition[^21]	University-heavy (Boston)
California	Partial	Large C&I only[^21]	Tech campus and healthcare

The strategic implication for Clear Current's sales territory planning is clear: **deregulated states produce more billing errors** (from supplier contract disputes, overlapping billing cycles during switches, and complex pass-through charges) **and more savings opportunities** (from competitive rate negotiation). The initial sales territory should prioritize Texas (home base, 100+ REPs), Pennsylvania (large university + commercial RE market), and New York (LL97 compliance urgency).

ERCOT Billing Complexity as a Moat

Texas ERCOT is structurally unique: there is no default utility service, all customers must actively choose a provider, and over 100 retail electricity providers compete. This produces billing complexity that non-deregulated markets do not see: overlapping billing cycles when customers switch REPs, TDSP (transmission/distribution) charges from the wires utility sitting alongside REP supply charges, ERCOT wholesale market price exposure for commercial customers without fixed-price contracts, and frequent discrepancies between REP-invoiced amounts and ERCOT settlement data. Clear Current's home-court advantage in Texas is a genuine competitive moat that is difficult for a Boston- or DC-headquartered competitor to replicate quickly.[^20][^19]

Part 4: Competitive Intelligence — Updated

EnergyCAP — The Incumbent in Detail

EnergyCAP's institutional history defines both its strength and its vulnerability. Founded in the 1980s and acquired by Resurgens Technology Partners in March 2021 (when the company had approximately 2,100 customers), it has grown to 10,000+ users by 2026 through a combination of organic growth and add-on acquisitions including Wattics (August 2022, interval analytics). Resurgens' investment thesis was straightforward: deepen sales, expand marketing, pursue add-ons. The customer growth from 2,100 to 10,000 is real, but the underlying platform architecture reflects its origins — a utility bill accounting system that has been extended, not rebuilt.[^22][^23][^24][^25][^26]

The Wattics acquisition is particularly instructive for Clear Current's positioning. EnergyCAP acquired Wattics specifically because its own platform lacked interval analytics capabilities — building-level real-time monitoring that customers were demanding. This is not a weakness EnergyCAP fixed organically; it is a weakness they papered over with a tuck-in acquisition. Clear Current should expect that if campus chargeback AI becomes a visible market requirement, EnergyCAP will try to acquire its way to that capability rather than build it.[^26]

EnergyCAP Pricing Reference Points:

- Published pricing: $40–$52/meter/year (Express and Enterprise tiers)[^27]
- Texas State actual spend: ~$50,000–$75,000/year for ~1,500 meters (via reseller)[^15]
- UNM actual spend: $9,600/year via Lobo Energy reseller (small campus)[^16]
- Implementation: $5,000–$100,000+ depending on complexity[^28]

The reseller channel (YearOut Energy, Lobo Energy, and others) creates a pricing opacity that benefits EnergyCAP in customer retention but creates an opportunity for Clear Current to compete on value transparency.

Arcadia — The AI-Native Threat

Arcadia's enterprise pivot in January 2025 makes it the most important competitive threat to monitor. The company raised $566M total (most recently a $50M Series E in April 2024 at a $1.5B valuation) and its 2022 acquisition of Urjanet gave it coverage of 95%+ of U.S. residential and commercial utility accounts. The combined entity manages $20B in annual utility bills and 1.35M connected utility accounts.[^29][^30][^31][^32]

Arcadia's announced enterprise product suite (Utility Bill Management, Energy Procurement Advisory, Supply Management) is a direct overlay on Clear Current's entire planned module set. The key differentiating factors work in Clear Current's favor: Arcadia was built as a data infrastructure company (B2B API layer for climate tech developers) and is adapting to enterprise direct sales — a significant organizational and product repositioning challenge. Arcadia has no documented campus chargeback functionality, no higher education

customer references, and no Texas deregulated market expertise. Its 99% accuracy claim on meter-tariff assignments covers utility data connectivity, not billing error recovery and refund pursuit — a meaningfully different value proposition.[^33][^34]

WatchWire — The Commercial RE Incumbent

WatchWire (now WatchWire by Tango) holds a G2 rating of 4.8/5 and is specifically designed for REITs, property management, private equity real estate, manufacturing, retail/franchise, and data centers. Its acquisition by Tango (a lease management and real estate operations platform) repositions it as an integrated real estate operations tool rather than a pure energy play. This creates a gap in the market for enterprise organizations that are not primarily real estate portfolios — exactly Clear Current's target of large commercial organizations with multiple facilities.[^35]

Brightly Software — The Siemens Asset Management Play

Siemens acquired Brightly Software in June 2022 for $1.575 billion at approximately 10x its $160M ARR. Brightly serves 12,000 customers in education, healthcare, and manufacturing — directly overlapping Clear Current's beachhead verticals — with asset lifecycle management, CMMS, and maintenance management. In November 2025, Siemens acquired Setmetrics technology to add Energy Twin capabilities to Brightly. Brightly is not a billing audit platform, but its presence in Clear Current's target verticals means sales conversations will frequently involve a Brightly incumbent relationship alongside an EnergyCAP incumbent relationship. Understanding this dual-incumbent landscape is important for deal qualification.[^36][^37][^38][^39]

The ENERGY STAR Disruption — Risk and Opportunity

The Trump administration's FY2026 budget proposal allocates $0 to the Atmospheric Protection Program, which houses ENERGY STAR Portfolio Manager. The program serves 330,000+ buildings and is used by state and local governments to enforce building performance standards including NYC Local Law 97. As of August 2025, a bipartisan House committee approved $32M in funding and the Senate approved $36M, and legal analysis indicates ENERGY STAR cannot be eliminated without an act of Congress.[^40][^41][^2][^1]

For Clear Current, this regulatory situation creates a two-scenario analysis:

Scenario A (most likely): ENERGY STAR survives with reduced budget. The 330,000 buildings already in Portfolio Manager continue benchmarking there. Clear Current remains additive — billing audit and rate optimization layer that Portfolio Manager cannot provide.

Scenario B (tail risk): Portfolio Manager is substantially disrupted or discontinued. This is a major catalyst event for the energy management software market. 330,000 buildings need an alternative infrastructure for energy data management, benchmarking, and compliance reporting. Any company with pre-existing utility data connectivity and energy management capabilities — including Clear Current — stands to benefit significantly. This scenario also accelerates the buyer urgency that is currently a growth constraint.

Part 5: Pricing Strategy

Contingency Fee Benchmarks

The utility bill audit industry's standard contingency model is well-documented: UtiliSave charges 40% of savings recovered for the first 36 months, then 0%. Other firms in the sector charge 20%–50% depending on deal size and recovery difficulty. This model has the advantage of zero perceived risk for the buyer and creates asymmetric upside for the auditor — but it creates the exact problem Marc Spieler identified: as discovered savings scale, the percentage becomes uncomfortable.[^42][^43][^44][^45]

The property tax protest model Marc cited is directionally instructive but not a precise analogue. The better comparable is the utility bill audit firm that charges 40% of recovered refunds — but caps the engagement at 18–36 months and transitions the relationship to subscription. This is the transition Clear Current needs to engineer deliberately: use contingency pricing to prove value on the first engagement (billing error recovery), then migrate to subscription pricing for ongoing rate optimization, tariff monitoring, and procurement advisory.

The Per-Meter Pricing Ladder

EnergyCAP's $40–$52/meter/year pricing creates a ceiling against which Clear Current can position. The strategic pricing options are:[^27]

Tier	Positioning	Price Point	Target Customer
Audit (Entry)	Contingency on errors found	20–30% of recovered amount	Any 50+ location org
Optimize (Core SaaS)	Per-meter subscription	$75–$100/meter/year	EnergyCAP upgraders
Platform (Enterprise)	Flat enterprise license	$120K–$250K/year	University systems, large CRE

The $120K/year anchor that John is currently targeting is directionally appropriate for an organization with 130 locations — that implies approximately $923/location/year or (at an average of 5–10 meters per location) $90–$185/meter/year. This is a 2–4x premium over EnergyCAP's public pricing. The premium is defensible on error recovery ROI, but it must be proven in the first 90 days of each deployment.

The UW System RFP's estimated $500K–$2M contract value signals that system-level university contracts can support significantly larger deal sizes than Clear Current's current anchor pricing. A University of Texas System (16 institutions, Austin flagship alone has 1,000+ meters) or California State University System (23 campuses) level deal could represent $500K–$1M+ ARR from a single procurement event.[^12]

Part 6: Sales Cycle & Enterprise Deal Economics

Time to Close by Deal Size

Enterprise SaaS deal timelines are well-benchmarked:[^46][^47][^5]

Deal Size (ACV)	Typical Close Time	Stakeholders	Procurement Process
<$50K	2–3 months	1–3	Discretionary, manager-level sign-off
$50K–$100K	3–6 months	3–5	Budget approval, procurement review
$100K–$500K	6–12 months	6–10	Formal procurement, legal review, IT security
$500K+	12–18+ months	10+	RFP, board approval, multi-year contract

Clear Current's target deal at $120K/year sits in the 6–12 month band. The five-month revenue recognition lag John observed is the floor, not the ceiling — deals that require formal university procurement (which Texas State's public purchasing records suggest includes IT, facilities, and finance stakeholders) will regularly take 9–12 months from first meeting to signed contract.

Implication for the May fundraise: The 12 pilots currently in flight, assuming a 6-month average to close, would produce signed contracts between June and October 2026 — after the fundraise. John needs either (1) at least 2–3 pilots that are 6+ months into the engagement already and can close before May, or (2) a framing that treats pilot conversion commitment letters or LOIs as equivalent to contracts for fundraise purposes.

Pilot-to-Paid Conversion and the "Realized vs. Identified" Problem

Marc Spieler's concern that identified errors ≠ recovered refunds is the most important operational risk in Clear Current's current go-to-market. In enterprise energy management, the error identification step (which Clear Current's platform can do today) and the error recovery step (which requires the utility to issue a credit or refund) are separated by a 30–180 day process that involves dispute filing, utility investigation, and credit issuance. Some errors are recovered in full; some are recovered partially; some utilities deny the claim entirely.

This creates a credibility gap that investors will probe in diligence: "How much of the $10M in identified errors has actually been recovered as cash?" Clear Current needs to track and report this metric explicitly. The SaaStr benchmark of 70%+ pilot-to-paid conversion assumes the pilot has demonstrated clear, realized value — not identified but unrealized value.[^4]

Part 7: Investor Landscape & Comparable Deals

The Relevant Comparable Set

Energy and built environment software have become increasingly attractive to growth equity and PE investors:

Company	Stage/Event	ARR at Event	Deal Value	Multiple	Notes
Brightly Software	Acquisition by Siemens (2022)	$160M ARR[^38]	$1.575B[^39]	~10x ARR	12,000 customers; education/healthcare focus
Arcadia	Series E (May 2022)	Not disclosed	$200M raise, $1.5B valuation[^29]	N/A	155% YoY growth in 2021[^48]
Arcadia	Series E (April 2024)	Not disclosed	$50M + $30M credit[^30]	N/A	At $1.5B valuation; community solar + enterprise
EnergyCAP	PE investment by Resurgens (2021)	~$10–15M estimated ARR	Not disclosed	N/A	2,100 customers at time of investment[^23]
Budderfly	Growth equity (2019)	Not disclosed	$55M growth + debt[^49]	N/A	Energy efficiency as a service; restaurant chains

The Brightly comparable is the most instructive for Clear Current's long-term exit framing: a scaled ($160M ARR), profitable (15% operating margins), 12,000-customer asset management platform serving education and healthcare sold for 10x ARR. This establishes the exit multiple range and validates the verticals. Clear Current is building toward a similar profile but starting earlier — the question for Series A investors is whether the beachhead strategy is defensible enough to get there.[^37][^38]

Series A Thresholds in 2026

The 2026 Series A market has standardized around a fairly narrow metrics band:[^3]

- **ARR floor:** $1M–$3M (median $1.5M for companies raising in 2026)
- **Growth rate:** 15–20% monthly growth, or 3–4x annual growth
- **Round size:** $10M–$20M (median ~$12M)
- **Pre-money valuation:** $25M–$50M (median ~$45M)
- **Unit economics:** Positive or near-positive LTV:CAC ratio
- **Retention:** NRR of 100%+ for enterprise (105–115% considered strong)[^50]
- **Team:** Demonstrated path to repeatable sales, not just founder-led wins

Key additional factors for climate tech/energy SaaS: Climate tech VC 2025 saw seed and Series A investment totals fall 20% and 7% respectively, but deal sizes rose to 2021 levels. Built Environment and Energy verticals are up 23% and 31% respectively in 2025. Investors are concentrating on category leaders rather than new entrants — which means Clear Current needs to demonstrate a credible path to category leadership in its chosen beachhead (higher education energy management), not just generic energy management.[^51]

What the May Fundraise Needs

Based on the benchmarks above and the company's current state ($4.275M raised, 12 pilots, $10M in identified errors, runway through end of 2026), the May 2026 raise would be characterized as a bridge or early Series A depending on the ARR milestone achieved. The specific requirements:

1. **Signed paying customers:** Even 2–3 contracts at $10K/month (~$360K ARR) demonstrates the conversion, not just the pipeline
2. **Realized error recovery:** At least one documented case of a utility issuing a refund credit based on Clear Current's audit work
3. **Higher education reference:** At least one university energy director (Andee Chamberlain or Andi Ault level) willing to speak on record about the value demonstrated
4. **Financial model correction:** Correct the revenue recognition lag from 1 month to 5 months before presenting to investors
5. **Competitive differentiation narrative:** A clear answer to "Why not just use Arcadia?" that is grounded in higher education domain specificity, campus chargeback capability, and Texas ERCOT expertise

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Part 8: Green Button Data Access — Product Implications

What Green Button Connect Is and Is Not

Green Button Connect My Data (GBCMD) is the federal standard for automated utility data API access, enabling customers to authorize third-party access to their meter data. It was launched in 2012, and the DOE committed 50+ utilities to the initiative covering 60M+ homes and businesses. The UtilityAPI platform provides a commercial implementation that aggregates Green Button Connect access across multiple utilities.[^52][^53][^54]

For Clear Current's product architecture, Green Button Connect is the ideal data ingestion pathway for utilities that support it — it eliminates manual PDF upload and enables automated, real-time data feeds. The problem is coverage gaps: Green Button Connect is strongest for residential accounts; commercial and industrial account support varies widely by utility. Oregon, for example, supports only Green Button Download My Data (file download) rather than the API-based Connect My Data standard for commercial accounts. Many smaller municipal utilities and rural cooperatives do not support Green Button at all.[^55]

New York's IEDR as a Model for What's Coming

New York State is building the Integrated Energy Data Resource (IEDR) — a centralized statewide energy data platform that includes Green Button Connect access for all New York investor-owned utilities. Phase I was completed March 2024; Phase II targeting July 2026 will expand Green Button Connect to all nine NY IOUs and add 40+ use cases. UtilityAPI (a Green Button Alliance sponsor) is building Phase II.[^56]

This is the policy trajectory that Clear Current should monitor: states requiring their IOUs to provide standardized API data access as a condition of AMI deployment. As this spreads from New York to other states (Connecticut, Massachusetts, Maine, and New Hampshire are all implementing or considering similar frameworks), Clear Current's data ingestion layer becomes dramatically easier to build and maintain. Until then, the manual PDF scraping / OCR workflow remains necessary for a significant share of commercial accounts.[^56]

Part 9: Regulatory Tailwinds — Building Performance Standards

NYC Local Law 97 — Active Compliance Urgency

Local Law 97 of 2019 requires buildings over 25,000 square feet to meet GHG emissions limits beginning in 2024, with an annual carbon emissions report due May 1 of each year. The first report covering 2024 data was due May 1, 2025. Non-compliance fines are $268 per metric ton of CO_2e above the building's limit. The 2030 compliance period brings significantly more stringent limits.[^57][^58][^59]

Local Law 88 requires lighting upgrades and sub-metering installation in spaces over 25,000 sq ft, with sub-meter data feeding directly into energy management workflows. Local Law 84 requires annual energy benchmarking (historically via ENERGY STAR Portfolio Manager). Together, these three laws create a compliance infrastructure for New York commercial building owners that requires continuous energy data management — precisely the category Clear Current serves.[^57]

The NYC market is currently prioritizing ESG/compliance-oriented platforms (IBM Envizi, Schneider Resource Advisor+, Measurabl) over billing audit tools. But LL97 fines are directly tied to energy intensity metrics that are affected by utility billing accuracy — a building that has been overbilled on energy may appear to be consuming more energy than it actually is, creating a misleading compliance picture. This connection between billing accuracy and compliance accuracy is an underexplored sales narrative for Clear Current in New York.

The Cascade of Building Performance Standards Beyond NYC

NYC is the most aggressive jurisdiction, but state-level building performance standard (BPS) adoption is expanding:

- **California:** AB 2446 (2022) requires large commercial buildings to benchmark and report; SB 100 drives the overall clean energy mandate[^1]
- **Washington state:** Building performance standards under HB 1257 cover commercial buildings over 50,000 sq ft
- **Colorado:** Senate Bill 22-246 establishes commercial building benchmarking and performance standards
- **Illinois:** Chicago Climate Action Plan and state-level benchmarking requirements

Each jurisdiction that mandates benchmarking creates recurring software adoption demand. The market dynamics parallel the GDPR/data privacy software wave: a compliance deadline forces tool adoption, and the resulting data infrastructure becomes sticky.

Part 10: Strategic Synthesis — Clear Current's Defensible Position

The "Why Clear Current vs. Arcadia" Answer

The competitive differentiation question that investors will ask first and buyers will ask second is: why not just use Arcadia? The answer has three parts:

1. Arcadia's product is built for corporate sustainability teams, not energy managers. Arcadia's three enterprise products (Utility Bill Management, Energy Procurement Advisory, Supply Management) are oriented toward Scope 1–3 emissions tracking and renewable energy procurement — the priorities of a VP of Sustainability at a Fortune 500 company. Clear Current's design starts with the energy manager who receives 500 bills per month and needs to know which ones are wrong before approving payment. These are different users with different workflows.

2. Arcadia has no campus chargeback. University and hospital campus cost allocation — the distribution of central utility costs across 50–150 metered entities including departments, schools, grants, auxiliaries, and clinical programs — is the most technically demanding feature in the higher education energy management category. EnergyCAP's chargeback module is the reason universities don't switch platforms despite its UX weaknesses. Arcadia has built zero documented capability here. This is Clear Current's highest-value beachhead feature to build.

3. Texas ERCOT expertise is a genuine moat. Arcadia was built and is headquartered in Washington D.C., with a residential community solar orientation. Its commercial enterprise products were acquired from Atlanta-based Urjanet in 2022. Neither of these entities has deep ERCOT market expertise. The ERCOT deregulated market —

with 100+ competing REPs, mandatory customer choice, TDSP pass-through complexity, and frequent billing disputes during customer switches — is where Clear Current's founders operate and where the most billing errors are concentrated.[^19]

The Fundraise Story in One Paragraph

Clear Current Technologies is the only AI-native energy management platform that combines utility billing error recovery, tariff optimization, and campus cost allocation in a single workflow. The company has identified $10M in billing errors across 12 pilot programs with organizations ranging from single-campus universities to a single enterprise with 130 locations. Billing errors affect an estimated 30% of commercial accounts annually, yet the dominant incumbent (EnergyCAP, backed by Resurgens since 2021) does not have AI-native error detection and its most critical capability — campus chargeback — is also its most complained-about feature across 132 user reviews. The beachhead is U.S. universities, a $8–11 billion annual energy spend segment where 660+ district energy systems create billing complexity that no AI-native competitor addresses. The Texas ERCOT deregulated market provides home-court advantage in the most billing-complex energy market in the country. Series A proceeds will fund the first sales team hire, the campus chargeback module build, and the conversion of current pilots to the first $1M in ARR.

Appendix A: Key Data Points by Source Quality

Government-Sourced (High Credibility)

- EIA CBECS 2018: $142B commercial energy spend; 4% BEMS penetration[^60][^61]
- EIA: 136.9M smart meters, ~82% penetration[^62]
- EPA ENERGY STAR: 330,000+ buildings in Portfolio Manager; K-12 spending $8B+[^2][^63]
- FERC 2025 AMI Report: documents AMI deployment requirements and PUC proceedings[^62]
- NYC DOB: LL97 Annual Carbon Emissions Report due May 1, 2025[^58][^59]

Industry Transaction Data (High-Medium Credibility)

- Brightly acquisition: $160M ARR → $1.575B purchase price (Siemens press release)[^38][^39]
- Arcadia: $566M total raised, $1.5B valuation (company press release)[^30][^29]
- EnergyCAP: 2,100 customers at Resurgens investment (March 2021); 10,000+ users by 2026[^23][^25]
- EnergyCAP acquired Wattics for interval analytics (August 2022)[^26]

Procurement Records (High Credibility for Specific Instances)

- University of Wisconsin RFP: $500K–$2M estimated, energy management software, October 2024[^11][^12]
- Texas State University EnergyCAP PO: ~$17,900 per quarter (public purchasing data)[^15]

- University of New Mexico EnergyCAP fee: $9,600/year via reseller[^16]
- Angelo State University: EnergyCAP via YearOut Energy reseller[^64]

Market Research (Medium Credibility — use directionally)

- Series A benchmarks 2026: $1M–$3M ARR, $10M–$20M raise, $25M–$50M valuation[^3]
- Enterprise SaaS sales cycle: 6–18 months for $100K+ deals[^5]
- Paid pilot to annual contract conversion: 70%+ when executed correctly[^4]
- Contingency fee standard: 40% of savings recovered (UtiliSave, documented)[^44][^45]
- NRR benchmarks: world-class ≥115–120%; solid mid-market ≥105–110%[^50]

Vendor Claims (Low — use only with qualification)

- "80% of businesses overpay on utilities" (National Utilities Refund, self-serving)[^65]
- "9% software adoption" (GridPoint vendor survey)[^66]
- Arcadia "99% energy cost accuracy" (company claim, unaudited)[^34]

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DOCUMENT 6: Claim Verification & Fact-Check Audit

Source file: Clear Current Technologies — Claim Verification Report.md

Claim Verification Report: Commercial Energy Management Statistics

Prepared for Clear Current Technologies | April 2026

Overview

This report independently verifies six statistics used in the Clear Current Technologies market research report. For each claim, it identifies the original source, assesses conflict-of-interest, locates independent corroboration, and provides a credibility rating and usage recommendation.

Credibility Rating Scale:

- **High** — Government agency (EIA, EPA, DOE), academic institution, or independent research body with no financial stake in the conclusion
- **Medium** — Reputable trade press, industry association, or vendor-commissioned survey with transparent methodology
- **Low** — Vendor-generated marketing claim, self-serving statistic, or unverifiable assertion

Claim 1: "80% of businesses are overpaying on their utility bills"

Source Cited

nationalutilitiesrefund.com — a utility bill auditing company that earns revenue by finding billing errors and collecting contingency fees on recovered overpayments[^1][^2]

Conflict of Interest Assessment

Severe and direct. National Utilities Refund (NUR) charges no fee unless errors are found, meaning their business model produces revenue only by finding billing problems. The company states: "Over 80% of the invoice audits performed by NUR have contained at least some overcharges and errors." This is equivalent to a pest control company claiming 80% of homes have pest problems. The statistic describes their own client base — companies that were already suspicious enough of their bills to seek an audit — not a representative sample of all commercial organizations.[^3]

The claim appears on multiple NUR web pages but with no citation to any study, government report, public utility commission finding, or academic paper. It is self-generated marketing content.[^2][^1][^3]

What Independent Evidence Actually Says

No government agency, academic institution, or public utility commission publishes a comprehensive commercial utility billing error rate. However, independent evidence does corroborate that billing errors are **real, material, and persistent**:

Regulator-documented cases:

- A 2022 Federal Energy Regulatory Commission (FERC) audit found that FirstEnergy's Ohio subsidiaries "may have mischarged customers" by misclassifying $108 million in corporate overhead as recoverable costs — a finding subsequently upheld by the Public Utilities Commission of Ohio in 2024[^4]
- Rhode Island's Public Utilities Commission (PUC) ordered an independent audit of Rhode Island Energy's billing systems after discovering disputes about bill accuracy for state accounts and documenting that "83% of bills that were flagged for issues within their billing system in December were likely sent out anyway"[^5][^6]
- Pennsylvania PUC opened a formal investigation into PPL Electric Utilities billing accuracy following reports of bills 30–50% above expected amounts[^7]

Harvard Law School (2025): A peer-reviewed legal analysis from Harvard's Environmental Law Initiative documents that utility rate cases are structurally asymmetric — utilities generate the information regulators use to set rates, creating persistent information asymmetry that facilitates billing and rate errors. The paper notes that independent audits, not regulatory proceedings, are how most mischarged costs are discovered.[^8]

360 Energy (industry auditor, moderate conflict): States that "nearly 30% of our 360 Energy clients face billing errors annually". While 360 Energy is not fully independent, this figure is notably more conservative than NUR's 80% claim and focuses on clients already using their services.[^9]

Why the 80% figure is plausible but unmeasurable: A figure as high as 80% likely reflects a broad definition of "overpaying" that includes *suboptimal rate classification* (on a technically correct but non-optimal tariff), *demand charge management opportunities* (paying correct charges but on a peak that could have been managed), and *true billing errors* (incorrect charges). Under this broader definition, most commercial organizations probably could reduce their utility costs — but conflating rate optimization opportunity with billing error distorts the claim's meaning.

Credibility Rating: LOW

The 80% figure is vendor-generated marketing content from a company with direct financial incentive to maximize the perceived prevalence of billing errors. It cites no methodology, no sample size, and no independent verification.

Usage Recommendation

Do not use this statistic in a business presentation or investor deck as presented. Instead, use one of the following defensible alternatives:

1. **Regulatory corroboration:** "State and federal regulators have uncovered multi-million-dollar billing misclassifications at major utilities including FirstEnergy ($108M, 2024) and Rhode Island Energy, and 83% of internally-flagged billing exceptions were sent to customers anyway."[^6][^4]
2. **Conservative industry estimate:** "Energy auditing firms estimate that approximately 30% of commercial utility accounts contain a material billing error in any given year." — Flag as industry-sourced with moderate conflict of interest, but it is more defensible than 80%.[^9]
3. **Focus on tariff optimization:** "Most commercial buildings are not on the lowest-cost available tariff for their load profile. Arcadia, the largest utility data platform, processes 500,000+ meter-tariff assignments and reports that 20% require expert manual intervention beyond automated matching — suggesting systematic mis-classification is pervasive."[^10]

Claim 2: "Commercial buildings spent $241 billion on energy in 2024"

Source Cited

Reported as derived from EIA CBECS and the University of Michigan Center for Sustainable Systems (CSS) factsheet[^11]

What the Primary Source Actually Says

EIA CBECS (gold standard): The most recent official CBECS survey was conducted in **2018**, not 2022 or 2024. EIA released the 2018 CBECS data in December 2022. The authoritative figure is: **$141–142 billion in 2018**. There is no EIA survey of commercial energy expenditures for any year after 2018. As of 2025, EIA's AEO modeling still uses the 2018 CBECS as its baseline.[^12][^13][^14][^15]

The next CBECS: EIA conducted a new CBECS beginning in 2022, but data collection and modeling for a new CBECS cycle is multi-year. As of 2025–2026, EIA has not published updated CBECS expenditure figures beyond 2018. Any 2024 figure is a projection, not a measured survey result.[^16][^17]

What the CSS Factsheet says: The University of Michigan CSS factsheet is an annual synthesis document that extrapolates trends from EIA primary data. The CSS uses the CBECS as a foundation but adjusts for price changes using EIA's Annual Energy Outlook and Short-Term Energy Outlook (STEO) data. Citing CSS as a primary source is one step removed from the actual primary source (EIA).[^11]

Cross-Validation: Is $241 Billion Plausible?

The $142 billion (2018) figure can be inflated to a rough 2024 estimate using two factors:

1. **Energy price inflation:** EIA documented that total U.S. energy expenditures increased 22% in 2022 from 2021 alone (to more than $1.7 trillion total). Commercial electricity expenditures totaled approximately $482 billion across all sectors (residential + commercial + industrial) in 2022[^18][^19]
2. **Commercial electricity consumption growth:** EIA forecasted 1,420 billion kWh for commercial customers in 2024, compared to approximately 1,371 billion kWh in 2022 — a modest consumption increase[^20][^21]

However, price increases from 2018 to 2024 make $241 billion a **plausible order-of-magnitude estimate**. The commercial sector uses approximately 35–38% of total U.S. end-use electricity, and with electricity prices up significantly since 2018 (the average commercial electricity price rose from ~$0.107/kWh in 2018 to ~$0.128/kWh in 2022), a 70%+ increase in total expenditures from 2018's $142 billion is mathematically reasonable. But **it is a projection, not a measured figure.**[^19]

Credibility Rating: MEDIUM (if properly qualified)

The underlying EIA CBECS 2018 figure of $141–142 billion is **High** credibility. The $241 billion 2024 projection is derived data and should be explicitly labeled as an estimate based on price trend extrapolation from the 2018 CBECS baseline.[^13][^12]

Usage Recommendation

Use with qualification. The recommended formulation is:

"According to EIA's most recent Commercial Buildings Energy Consumption Survey (2018), U.S. commercial buildings spent $142 billion on energy. Adjusted for commercial energy price increases through 2024, current total commercial energy expenditure is estimated at approximately $200–250 billion annually."[^12][^18]

If maximum precision is needed, cite directly: "U.S. commercial buildings spent $142 billion on energy in 2018, the most recent year for which EIA has published official survey data." This is unimpeachable.[^13]

Claim 3: "98% of facilities management respondents rated energy efficiency as extremely or very important, yet only 9% of companies have fully embraced software tools for managing energy and sustainability data"

Source Cited

2023 GridPoint "Commercial Building Energy Efficiency Survey" of 800+ U.S. and Canadian building professionals, conducted by Qualtrics[^22][^23][^24]

Conflict of Interest Assessment

Significant. GridPoint is a building energy management technology vendor. The survey was **commissioned by GridPoint**, administered by Qualtrics (a neutral survey platform), and designed to validate demand for products GridPoint sells. This creates a framing and question-design bias — results are likely shaped to highlight the gap between awareness/importance and software adoption, which is GridPoint's commercial opportunity.[^22]

The 98% figure refers specifically to **facilities management respondents** (a sub-group), not all 800+ respondents. The broader survey found 89% of all respondents rated energy efficiency as highly important. The 98% figure requires cherry-picking the FM sub-group.[^22]

The survey does not appear to have been published in a peer-reviewed journal or independently verified. The full methodology — sampling frame, population definition, weighting, margin of error — is not disclosed in available public materials.[^23]

The "9%" Adoption Figure — Source and Corroboration

The 9% figure is reportedly from the same GridPoint survey. The specific phrasing in the market report was that "only 9% of companies have fully embraced software tools for managing energy and sustainability data." This very restrictive framing likely requires selecting only respondents who have deployed a full energy management software suite and are actively using it for sustainability reporting — a narrow definition designed to maximize the apparent gap.

Independent corroboration from EIA CBECS (2018):

- EIA's 2018 CBECS tracks building technology adoption, including energy management and control systems. The survey found that commercial building automation and energy management system penetration was approximately **4% of commercial buildings** in 2018 for full BEMS deployment[^25]
- Only ~10% of commercial buildings had sub-metering installed in 2018[^25]
- ENERGY STAR Portfolio Manager, the most widely used free energy benchmarking tool, covers approximately **25% of commercial floorspace** — meaning 75% of commercial building space has no systematic energy performance benchmarking at all[^26][^27]

Altair/Toggled Survey (2023, vendor-commissioned but different vendor): Found that 78% of businesses have adopted *some* smart building technology, but only 36% have seen measurable decarbonization results. This suggests widespread partial adoption, not the low single-digit "full embrace" that GridPoint claims.[^28]

Independent synthesis: The real picture is likely a spectrum: ~4% have full BEMS (High credibility, EIA); ~25% have energy benchmarking via ENERGY STAR Portfolio Manager (High credibility, EPA); ~78% have adopted some building technology (Medium, vendor survey); and <10% have integrated energy + sustainability software meeting the "fully embraced" definition GridPoint uses.[^28][^26][^25]

Credibility Rating

- **98% importance figure: LOW** — Cherry-picked sub-group from a vendor-commissioned survey
- **9% adoption figure: LOW as stated** — Vendor-commissioned, narrow definition, no published methodology
- **The underlying directional claim (most buildings lack integrated energy software): HIGH** — Strongly corroborated by EIA CBECS, EPA Portfolio Manager penetration data, and multiple independent sources[^26][^25]

Usage Recommendation

Replace the vendor-sourced statistics with independently verifiable alternatives:

"Only 4% of U.S. commercial buildings have deployed a full building energy management system (BEMS), according to EIA's Commercial Buildings Energy Consumption Survey. EPA's ENERGY STAR Portfolio Manager — the most widely used free benchmarking tool — covers just 25% of commercial floorspace, meaning 75% of commercial building space has no systematic energy performance tracking."[^26][^25]

This conveys the same message (large unmet need) with government-sourced, defensible data.

Claim 4: "Industry auditors estimate that 5–15% of commercial utility bills contain errors"

Source Cited

Utility audit firms including 360 Energy, Conservice, and National Utility Refund[^1][^9]

Conflict of Interest Assessment

All three sources profit from finding billing errors and have an economic incentive to characterize error rates as high. Conservice manages utility billing for commercial real estate and residential portfolios — finding errors is core to their value proposition. 360 Energy is an energy audit firm. National Utility Refund charges contingency fees on recovered overpayments. None of these sources is disinterested.

What Independent Evidence Actually Says

No U.S. government agency publishes a commercial utility billing error rate. The EIA, FERC, NARUC, and state PUCs track rate cases and regulatory complaints but do not publish a systematic error rate for commercial billing. This is a genuine data gap.

Documented independent evidence by error category:

Meter reading errors: The transition from manual to AMI metering has been associated with significant accuracy improvements in residential billing, but commercial meters — which may have CT multipliers, demand interval recording, and complex TOU configurations — are more error-prone. A Rhode Island PUC investigation found that 83% of bills internally flagged as problematic were sent out anyway, indicating the utility's own quality control was failing to prevent errors from reaching customers.[^6]

Rate misclassification: Arcadia, the largest commercial utility data aggregator, processes 500,000+ meter-tariff assignments for commercial accounts and reports needing expert manual review for approximately 20% of them — suggesting systematic tariff complexity that utilities may be misapplying.[^10]

Demand charge errors: The National Renewable Energy Laboratory (NREL) and DOE have documented that demand charges are frequently misunderstood by commercial customers and are prone to billing disputes, particularly around ratchet clauses and peak demand measurement intervals.[^29][^30]

The 5–15% figure assessed: The 5–15% range from 360 Energy describes the "additional costs annually" from billing errors — not a frequency rate. This is a subtly different claim: a 5–15% cost overrun could be produced by a single large error on a major bill, rather than widespread small errors. The figure is internally consistent with independent evidence (FERC audits documenting $108M mischarge at a single utility) but cannot be independently verified as a representative estimate.[^4]

Credibility Rating: LOW as a "scientific" figure; MEDIUM as a directional claim

The 5–15% figure is industry-self-reported with strong conflict of interest. However, its directional message — that billing errors are material and pervasive — is corroborated by regulator findings and is not disputed by any independent source.

Usage Recommendation

Reframe as a directional claim with regulatory corroboration:

"State and federal regulators have documented material commercial billing errors at multiple major utilities. FERC audits uncovered $108 million in misclassified costs at FirstEnergy. Rhode Island's PUC documented systematic billing system failures. Utility audit firms with access to large commercial portfolios estimate that billing errors add 5–15% to annual utility costs — a figure with inherent conflict of interest but consistent directional support from regulatory findings."[^9][^6][^4]

Claim 5: "U.S. colleges and universities spend over $6 billion annually on energy"

Source Cited

Reported as derived from "DOE and EPA estimates"

Tracing the Primary Source

The $6 billion figure for K-12 schools is clearly documented by **EPA ENERGY STAR** on its official K-12 Schools page: *"K-12 school districts spend over $8 billion nationwide on energy costs each year."* Note: The EPA figure is **$8 billion for K-12**, not $6 billion. The $6 billion figure may predate an EPA update.[^31][^32]

Additionally, NASEO's 2024 report on K-12 energy efficiency, citing EIA CBECS data, confirms "$11 billion of annual expenditures can be managed and likely reduced through proven energy efficiency measures" for K-12 schools — a higher and more recent figure.[^33]

For higher education specifically: There is no single government publication that states a dollar figure for higher education energy spending alone. The relevant data must be derived:

- **EIA CBECS 2018:** Education buildings (K-12 + higher ed combined) consumed 854 TBtu in 2018 = 13% of total commercial energy consumption[^34]
- **Education buildings total expenditure (EIA CBECS 2018):** Education buildings spent approximately **$18.5 billion** on energy in 2018 (13% of $142 billion total)[^34][^12]
- **K-12 component:** EPA ENERGY STAR states K-12 spends over $8 billion annually, with NASEO citing $11 billion as a managed figure[^33][^31]
- **Higher education residual:** If total education is ~$18.5 billion and K-12 is ~$8–11 billion, higher education accounts for approximately **$7.5–10.5 billion** of annual energy spend

APPA (Association of Physical Plant Administrators): APPA's Facilities Performance Indicators (FPI) program tracks purchased utilities per gross square foot across participating universities, and their database includes 1.1 billion gross square feet of campus space. APPA FPI data is the most authoritative source for higher education energy spending benchmarks, but the specific aggregate spending figure is available only to APPA members.[^35]

NCES (National Center for Education Statistics): Total higher education expenditures in 2020–21 were $702 billion. Utilities typically represent 2–3% of total institutional expenditures — consistent with $10–14 billion in annual energy spend for the sector.[^36]

Energy Star Higher Education Benchmarking Initiative (HEBI): EPA has run HEBI since 2020 to benchmark campus energy, but no aggregate spending dollar figure is published in publicly available HEBI reports.[^37]

The EUI Benchmark for Higher Education

The most authoritative per-square-foot energy use benchmark comes from EPA ENERGY STAR's national median table:

- **Adult Education buildings:** 110.4 kBtu/sq ft/yr (EUI)[^38]
- **K-12 Schools median:** approximately 48.5 kBtu/sq ft/yr[^38]

- **Colleges / Universities:** approximately 115–130 kBtu/sq ft/yr for research-intensive campuses based on Portfolio Manager data[^38]

Higher education buildings are significantly more energy-intensive than K-12 schools due to research laboratories, data centers, hospitals, and 24/7 residential facilities on campus.

Credibility Rating: MEDIUM (if attributed correctly)

The $6 billion figure (as cited) is **incorrect** — EPA's current figure is $8 billion+ for K-12 alone, and the combined education sector figure is over $18 billion. The higher education-specific figure of "$6 billion" appears to predate updates and may misattribute the K-12 figure.[^31][^34]

Usage Recommendation

Correct and upgrade the statistic. Use one of the following:

1. *"K-12 school districts alone spend over $8 billion annually on energy, according to EPA ENERGY STAR."* — High credibility[^31]
2. *"The education sector (K-12 + higher ed) consumed 854 TBtu in 2018 — 13% of all commercial building energy — representing approximately $18 billion in annual energy expenditure based on the EIA CBECS 2018."* — High credibility[^12][^34]
3. *"Higher education institutions account for an estimated $8–11 billion in annual energy expenditure, derived from EIA CBECS data showing the education sector at 13% of total commercial energy spend, less EPA's $8B estimate for K-12 alone."* — Medium credibility (derived calculation, but from high-credibility inputs)[^12][^31]

Claim 6: "Only 9% of commercial buildings use energy management software"

Source Assessment

As discussed in Claim 3, the 9% figure originates from the GridPoint 2023 vendor-commissioned survey. No government report uses this specific framing or figure.[^24][^23]

What Government Data Actually Says About EMS Adoption

EIA CBECS (2018) — Highest Credibility: The 2018 CBECS collected building characteristics data on technology adoption. Key findings:[^25]

- **Building automation/energy management systems (BEMS):** ~4% of U.S. commercial buildings had a full BEMS deployed as of 2018

- **Sub-metering:** Only ~10% of commercial floorspace had interval sub-metering (which enables the kind of energy analytics EMS platforms provide)
- **Energy controls (programmable thermostats, occupancy sensors, etc.):** Substantially higher penetration (~50–70% for basic controls), but these are not "energy management software" in the enterprise sense

EPA ENERGY STAR (2023) — High Credibility:

- More than 330,000 buildings (representing ~25% of commercial floorspace) are actively benchmarked in ENERGY STAR Portfolio Manager[^27][^26]
- Portfolio Manager is a free, web-based benchmarking tool — the lowest bar for "energy management software"
- This 25% figure represents the *ceiling* of energy management software adoption for commercial buildings; full-featured paid EMS platforms (EnergyCAP, Schneider, etc.) would have far lower penetration

DOE Better Buildings Initiative: Approximately 12% of commercial floorspace participates in voluntary DOE energy efficiency programs, suggesting an upper bound for proactive energy management engagement.[^26]

Synthesis: The evidence supports the following hierarchy:

- **~4%** of commercial buildings have a deployed BEMS (EIA CBECS, 2018)[^25]
- **~10%** have interval sub-metering (EIA CBECS, 2018)[^25]
- **~25%** actively use ENERGY STAR Portfolio Manager (EPA, 2023)[^26]
- **~78%** have adopted *some* smart building technology (Altair/Toggled vendor survey, 2023)[^28]

The 9% figure from GridPoint falls plausibly within this spectrum if it is meant to describe *integrated energy + sustainability software platforms* (beyond basic benchmarking), but the methodology is not published and the source is conflicted.

Credibility Rating

- **9% figure from GridPoint: LOW** — Vendor-commissioned, unexplained methodology
- **4% BEMS penetration from EIA CBECS: HIGH** — Government survey[^25]
- **25% Portfolio Manager adoption from EPA: HIGH** — Government program data[^26]

Usage Recommendation

Replace with layered, independently sourced statistics:

"Commercial energy management software adoption spans a wide spectrum. Only 4% of U.S. commercial buildings had deployed a full building energy management system as of the most recent EIA survey, while 25% of commercial floorspace is actively benchmarked using EPA's ENERGY STAR Portfolio Manager — the most basic

form of systematic energy monitoring. Integrated energy analytics platforms serving the enterprise commercial market reach an estimated 10–15% of eligible organizations, with significant untapped potential across the remaining 85–90%."[^26][^25]

Summary Table

Claim	Stated Figure	Primary Source Credibility	Conflict of Interest?	Independent Corroboration	Recommended Use
80% overpaying on utilities	80%	**LOW** — vendor marketing	Severe	Partial — regulators confirm errors are real; ~30% client-reported rate from auditors	Do not use as stated; substitute regulatory findings
$241B commercial energy spend 2024	$241B	**MEDIUM** — derived from EIA 2018	Low (CSS is academic)	EIA 2022 data shows ~22% energy cost spike; projection is plausible	Use if qualified as "estimated from 2018 baseline adjusted for price changes"
98% rate EE as important; 9% use software	98% / 9%	**LOW** — vendor survey	Significant	EIA confirms low BEMS adoption (~4%); EPA shows 25% portfolio manager use	Replace with EIA/EPA figures
5–15% of bills contain errors	5–15%	**LOW** — industry auditors	Direct	Regulatory enforcement findings confirm material errors exist	Use directionally only, with regulatory examples
Universities spend $6B on energy	$6B	**INCORRECT** as cited	None	EPA: K-12 alone is $8B+; education sector total is ~$18B	Correct to: K-12=$8B+; higher ed=$8–11B derived
Only 9% use energy management software	9%	**LOW** — vendor survey	Significant	EIA: 4% BEMS; EPA: 25% Portfolio Manager	Replace with EIA/EPA layered data

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Broader Methodological Observation

A recurring pattern across these claims is the **"audit firm benchmark problem"**: statistics about billing errors, overpayment rates, and energy waste are disproportionately generated by vendors that profit from finding problems. This creates systematic upward bias in published figures. Independent corroboration from government

sources (EIA, EPA, FERC, state PUCs) consistently supports the directional claims — billing errors are real, software adoption is low, energy spend is large — but at more conservative magnitudes than vendor-generated statistics suggest.

For a market research report that will be reviewed by sophisticated investors or enterprise buyers, all vendor-generated statistics should either be replaced with primary government/academic sources or explicitly flagged with the conflict of interest. The directional narrative of the Clear Current Technologies opportunity remains compelling and is well-supported by independent evidence; the specific figures simply need to be grounded in defensible sources.

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DOCUMENT 7: Competitive Research — Utility Bill Intelligence and Energy Management Platforms

Source file: Clear Current Technologies — Competitive Research All Platforms.md

Competitive Research: Utility Bill Intelligence and Energy Management Platforms

Executive overview

This report profiles EnergyCAP, Arcadia, WatchWire, ENGIE Impact, Energy Toolbase, and BrainBox AI/Trane against a target bundle of AI-native bill Q&A, proactive alerts, tariff truth, regulatory monitoring, and campus chargeback workflows.[^1][^2][^3][^4][^5][^6]

A central factual correction is that “Arcadia” is ambiguous: the relevant company here is Arcadia at arcadia.com, the utility-data and community-solar company that acquired Urjanet, not Arcadia at arcadia.io, the healthcare analytics company.[^7][^8][^1]

Across the set, EnergyCAP is the clearest incumbent in higher education utility accounting and campus allocation; Arcadia is the strongest pure-play utility-data/API platform after Urjanet; WatchWire is stronger in CRE sustainability workflows than in bill-intelligence depth; ENGIE Impact is strongest where buyers want outsourced managed services; Energy Toolbase is strongest in C&I tariff modeling tied to solar-storage project economics; and BrainBox AI/Trane is primarily a building-controls optimization play rather than a bill-intelligence platform.[^9][^10][^11][^12][^4][^5]

No profiled vendor was found to offer a fully documented bundle of normalized utility bills plus tariff truth plus conversational bill Q&A plus proactive dollar-ranked alerts plus regulatory/PUC docket monitoring plus higher-ed chargeback in one product with strong public evidence across all elements.[^13][^6][^14][^15]

EnergyCAP

Positioning

EnergyCAP positions itself as energy and utility management software for utility bill accounting, energy data management, sustainability tracking, and advanced analytics, marketed to finance leaders, energy managers, sustainability teams, and public-sector institutions.[^16][^17][^18]

In practice, it replaces combinations of spreadsheets, internal bill-audit workflows, fragmented sustainability reporting, and consultant-built allocation processes, especially where buyers need financial-grade bill validation, cost recovery, and chargeback across large property portfolios.[^19][^20][^17]

Primary customer segments

EnergyCAP has the strongest documented fit among the six for higher education, with a dedicated higher-ed industry page and public university case studies including the University of Kansas and University of Arizona.[^10][^21][^9]

It also shows meaningful healthcare traction through public references to a major medical center, a leading academic medical center, UTMB, and Baptist Health South Florida, though healthcare appears secondary to its education/government heritage.[^22][^23][^24]

For CRE and multi-site commercial, EnergyCAP is credible but less publicly branded around restaurant/retail than ENGIE Impact or WatchWire, and there is limited public evidence of a major OEM/ESCO-led channel motion relative to direct institutional selling.[^25][^17][^26]

Product surface

EnergyCAP publicly markets utility bill management, smart analytics, tariff analytics, sustainability reporting, benchmarking, bill auditing, and a developer API.[^27][^28][^29][^18]

Public evidence supports portfolio dashboards, anomaly/analytics functionality, utility tariff analytics, and APIs; public evidence for OCR specifics is weaker, while public evidence for payment execution is limited relative to bill accounting and workflow support.[^28][^29][^19][^27]

The company also markets Watts AI as an AI-assisted reporting/productivity layer, but the public material found points more to faster reporting and assistance than to a robust conversational layer over bills with documented autonomous alerting or regulatory monitoring.[^6][^30]

Pricing and packaging signals

Public list pricing was not found for core EnergyCAP enterprise packages, so pricing remains Unverified in detail.[^31][^19]

Packaging signals suggest value-based selling around replacing manual utility accounting labor and improving recoveries, which is directionally consistent with an FTE-replacement narrative, but exact commercial constructs were not publicly documented in a primary source reviewed here; confidence: medium on labor-replacement framing, low on exact pricing structure.[^20][^17][^31]

A partner-facing video and buyer materials indicate structured packaging exists, but without public price cards or RFP award amounts in the sourced material; confidence: medium that pricing is customized enterprise SaaS/services, low on dollar ranges.[^32][^31]

Partners and dependencies

EnergyCAP appears comparatively self-contained versus vendors that rely on external utility-data aggregators as their core product thesis, though integrations and partner ecosystems clearly exist.[^29][^33]

Its data-custody posture is consistent with being a system of record for utility bills and related cost/accounting workflows rather than merely a pass-through API layer; confidence: medium, because detailed custody architecture was not fully public.[^17][^19]

Recent corporate events

EnergyCAP announced recognition such as 2024 ENERGY STAR Partner of the Year and customer-success rankings in 2024, which are not corporate-finance events but do matter as market-signal items.[^34][^35]

More materially for product direction, the company has pushed AI positioning with Watts AI and continued feature emphasis around tariff analytics and reporting acceleration into 2026.[^30][^28][^6]

Security posture also matters for diligence: EnergyCAP announced SOC 2 certification in 2022, and public government-oriented trust positioning remains visible; however, buyers should verify current audit scope and report date directly in diligence.[^36][^37][^38]

Reputation and risk signals

Review-site signals are positive but sparse, and software directory evidence should be treated cautiously because review counts are modest and vendor incentives can distort results.[^39][^40][^41]

The strongest public reputational signal is its long tenure in institutional energy management and repeated public-sector/higher-ed references, while the main risk signal is that marketing has recently expanded into AI language that appears narrower in capability than a buyer might infer from the label alone.[^9][^10][^6]

Differentiation vs target bundle

- **Does:** Higher-ed utility accounting, chargeback/cost-allocation orientation, tariff analytics, API, portfolio visibility, sustainability reporting.[^17][^28][^29]
- **Partially does:** AI-assisted reporting/productivity and analytics-based anomaly surfacing.[^27][^6][^30]
- **Does not clearly document:** Regulatory/PUC docket monitoring, a robust natural-language bill Q&A experience, or full-stack autonomous proactive dollar-ranked alerts across the public product record reviewed.[^42][^6]

Adjacent players

Likely shortlist adjacencies include Schneider Electric Resource Advisor, ENGIE Impact, and WatchWire for enterprise sustainability/resource management RFPs, while specialized utility-data alternatives may also include Arcadia/Urjanet depending on API needs.[^2][^12][^43][^1]

Arcadia (arcadia.com)

Positioning

Arcadia positions itself as a utility data platform and, more recently, an enterprise energy-cost and procurement platform centered on Arc, utility bill and interval data, and AI-powered enterprise solutions.[^44][^45][^46][^47]

Its core buyer has shifted beyond climate-tech and community-solar use cases toward enterprises needing normalized utility data, bill management, and cost visibility at scale; the job it replaces is fragmented utility-data retrieval, manual bill collection, and ad hoc integrations historically handled through vendors like Urjanet or internal operations teams.[^15][^47][^1]

Primary customer segments

Arcadia's strongest documented fit in this context is the energy-services, fintech, and enterprise data/API layer rather than higher education or healthcare facilities operations.[^47][^44][^15]

Public evidence for higher-ed traction is weak; public evidence for healthcare in the search results refers to the unrelated healthcare-analytics company Arcadia.io, not arcadia.com, so higher-ed and healthcare strength should be marked weak/Unverified for the utility-data company unless a buyer can produce direct customer proof.[^48][^8]

Arcadia appears stronger for multi-site enterprises and channel partners that want utility-data plumbing via APIs or managed data services, and potentially relevant to ESCO/OEM channel models because of its data-layer role; confidence: medium.[^44][^15][^47]

Product surface

Arcadia publicly markets utility bill management, interval data, the Arc technology platform, and AI-powered enterprise solutions.[^46][^15][^47][^44]

This makes Arcadia one of the clearest API/data-normalization players in the set, especially after Urjanet, but the public record reviewed does not show native regulatory/PUC monitoring, campus chargeback, or a mature conversational bill-Q&A product with public demos and documented references.[^49][^1][^47]

The company uses AI language around enterprise solutions, but the evidence available is closer to recommendations and optimization around energy costs than to a deeply documented LLM chat interface over bill data; confidence: medium.[^14][^44]

Pricing and packaging signals

Arcadia does not publish standard public pricing for Arc or enterprise utility-data products in the material reviewed, so exact pricing is Unverified.[^50][^15]

The packaging signal is enterprise/API-style custom commercial terms rather than seat-based SMB pricing, and reseller or embedded-platform patterns are plausible because of the data-platform orientation; confidence: medium.[^47][^44]

Partners and dependencies

Arcadia's major dependency and advantage is the Urjanet acquisition, which significantly strengthened its global and commercial utility-data coverage.[^1][^7]

This also means buyers should diligence data custody, migration, and continuity assumptions in any Urjanet-to-Arc transition, especially if slide decks still treat Urjanet as fully separate; many decks will now be outdated on that point.[^51][^7][^1]

Recent corporate events

Arcadia acquired Urjanet in 2022, a fact still crucial to current positioning because it underpins the enterprise utility-data thesis.[^52][^7][^1]

In 2024 Arcadia announced a $50 million growth funding round focused on community solar expansion and AI innovation.[^53][^54][^55]

In 2025–2026 it more explicitly launched/marketed enterprise solutions and Arc as the technology platform, signaling a continued enterprise pivot beyond its earlier consumer-energy association.[^14][^46][^44]

Reputation and risk signals

The main positive signal is strategic relevance as a utility-data normalization layer, especially post-Urjanet.[^51][^1]

The main risk signals are ambiguity in market identity, overlap between consumer/community-solar and enterprise narratives, and evidence of layoffs reported in public professional posts in early 2024 that should be verified in diligence but not overstated as a formal disclosed restructuring event; confidence: low-to-medium on layoff significance because the sourced evidence is secondary.[^56][^54][^53]

Differentiation vs target bundle

- **Does:** Utility bill/interval data normalization, API-first data access, enterprise energy-cost visibility, data plumbing that can power downstream alerts.[^15][^44][^47]
- **Partially does:** AI-powered enterprise recommendations/optimization narratives.[^14][^44]
- **Does not clearly document:** Higher-ed chargeback, regulatory/PUC docket monitoring, a public natural-language bill-Q&A layer, or a complete campus allocation workflow.[^46][^44][^15]

Adjacent players

Likely shortlist adjacencies include UtilityAPI, Bayou Energy, and Schneider Electric/ENGIE Impact depending on whether the RFP is API-centric or broader managed energy intelligence.[^12][^43][^57][^58]

WatchWire (by Tango)

Positioning

WatchWire positions itself as sustainability and energy data management software, now operating under Tango ownership and integrated into a broader real-estate and facilities software narrative.[^11][^59][^2]

Its core buyer is a real estate, facilities, or sustainability team that needs utility data collection, ESG reporting, and portfolio visibility; it replaces manual sustainability data wrangling, fragmented building data collection, and consultant-heavy reporting processes.[^60][^61][^2]

Primary customer segments

WatchWire is strongest in CRE and portfolio sustainability reporting, where the product and Tango narrative align most clearly.[^59][^2][^60]

It has some educational-facilities messaging and public material for educational facilities, but the higher-ed evidence appears materially weaker than EnergyCAP's public campus proof.[^62][^63]

Healthcare and multi-site retail/QSR appear plausible but not especially well documented in primary public material reviewed here, while the OEM/ESCO channel also appears weak relative to direct real-estate/facilities selling.[^64][^2][^59]

Product surface

WatchWire publicly emphasizes utility data management, portfolio dashboards, sustainability reporting integrations, building performance standards support, and ESG workflows.[^65][^61][^2]

Public evidence is stronger for reporting, integrations, and portfolio sustainability than for deep tariff intelligence, payment, or bill-level conversational AI.[^61][^65][^60]

API depth exists via integrations, but the public brand is UI/reporting led; no strong public evidence was found for an LLM/chat layer, regulatory docket monitoring, or campus chargeback workflows.[^66][^61]

Pricing and packaging signals

No public standard pricing was found in primary company materials reviewed; Capterra indicates pricing exists through quote-based engagement, but exact structure is Unverified and should not be overstated.[^67][^68]

The commercial posture appears enterprise quote-based, often as part of larger portfolio sustainability software buying rather than simple utility-bill software licensing; confidence: medium.[^68][^2][^59]

Partners and dependencies

WatchWire's strongest visible dependency is now Tango ownership and positioning within Tango's broader real-estate platform.[^69][^11][^59]

Its sustainability integrations suggest it interoperates with external reporting ecosystems rather than controlling the whole utility-data stack itself, although exact custody architecture was not clearly documented in public material reviewed here; confidence: medium.[^60][^61]

Recent corporate events

Tango acquired WatchWire in August 2023, and subsequent 2024 messaging emphasized integrated energy and sustainability leadership within Tango.[^11][^59][^66]

A 2025 TIMIA financing announcement tied to product development and customer success is a notable capital signal, though not equivalent to a venture round or strategic M&A.[^64]

Reputation and risk signals

Review-site signals exist on Capterra but are limited in sample size, so they are useful directional inputs rather than definitive diligence evidence.[^67][^68]

The biggest diligence risk is that WatchWire can be overread as a deep utility-bill intelligence platform when its public differentiation is more clearly in CRE sustainability, ESG reporting, and compliance-oriented portfolio management.[^2][^65][^60]

Differentiation vs target bundle

- **Does:** Portfolio energy visibility, sustainability/ESG reporting, CRE-oriented data management, support for compliance/reporting workflows.[^65][^2][^60]
- **Partially does:** Utility data collection and integration across portfolios.[^61][^66]
- **Does not clearly document:** Strong tariff engine, bill-Q&A chat, regulatory/PUC monitoring, payment, or higher-ed chargeback.[^63][^2][^65]

Adjacent players

Likely shortlist adjacencies include Measurabl, Schneider Electric Resource Advisor, and EnergyCAP when the RFP spans ESG/resource reporting and portfolio utility visibility.[^43][^70][^25]

ENGIE Impact

Positioning

ENGIE Impact sells managed energy, sustainability, utility expense management, data management, and procurement services for large enterprises, especially those wanting outsourced execution rather than software-only self-service.[^3][^71][^72][^12]

It often replaces a combination of internal energy teams, bill-processing staff, consultants, and disparate software tools by wrapping software, data operations, and advisory services into a managed-service model.[^73][^71][^12]

Primary customer segments

ENGIE Impact is strongest in large multi-site enterprise, retail, restaurant, and complex global corporate portfolios.[^74][^75][^3]

It also has healthcare-adjacent credibility through ENGIE North America healthcare references, though the public case evidence is more mixed than its retail/enterprise breadth.[^76][^77]

Higher-ed is not a standout public strength in the material reviewed, and OEM/ESCO channel relevance is indirect because ENGIE is itself a major energy-services organization rather than a pure software partner program.[^78][^79][^3]

Product surface

ENGIE Impact publicly markets utility expense management, expense and data management, sustainability data, benchmarking, and energy supply/resource management.[^71][^80][^81][^12][^73]

This means strong coverage for bill management, data operations, portfolio dashboards, and ESG/sustainability reporting, but less evidence of productized tariff-engine depth comparable to Energy Toolbase or campus allocation depth comparable to EnergyCAP.[^81][^12][^71]

There was no strong public evidence in the reviewed sources for an LLM/chat interface, public regulatory/PUC docket monitoring product, or bill-intelligence experience centered on conversational Q&A.[^72][^81]

Pricing and packaging signals

Public list pricing was not found; ENGIE Impact appears to sell through enterprise contracts and services-based engagements.[^3][^12]

The commercial narrative is consistent with FTE replacement and outsourced operations, especially for utility expense management and resource-data administration; confidence: high on managed-service orientation, low on exact pricing ranges.[^12][^73][^71]

Partners and dependencies

ENGIE Impact appears less dependent on third-party aggregators in its public narrative than API-centric vendors because it combines services, process, and platform.[^72][^12]

Its key dependency is organizational capacity and delivery model rather than a narrow single-product module, which can be a strength for large outsourcing buyers and a weakness for customers wanting lightweight API-first embedding.[^73][^81][^3]

Recent corporate events

Within the last 18–24 months, the clearest public events are ongoing product/platform messaging, the 2024 net-zero thought leadership push, and continued capability marketing around data and energy management rather than a major standalone financing or M&A event at the ENGIE Impact unit level.[^82][^3][^72]

Because ENGIE Impact sits within ENGIE, slides that frame it as an independent startup-like software company are likely misleading or outdated.[^83][^84]

Reputation and risk signals

The strongest positive signal is enterprise scale and breadth of service coverage, particularly for buyers who prefer outsourced utility bill and resource management.[^3][^12]

The main risk is that software evaluators can mistake ENGIE Impact for a modern product-led bill-intelligence platform when much of the value proposition is managed-service execution, making UX, API flexibility, and speed-to-configure harder to benchmark from public material alone.[^12][^73][^72]

Differentiation vs target bundle

- **Does:** Outsourced bill management, utility expense workflows, broad portfolio visibility, sustainability reporting, procurement/resource management.[^71][^73][^12]
- **Partially does:** Analytics and operational alerting through managed services.[^81][^3]
- **Does not clearly document:** Conversational bill Q&A, regulatory/PUC docket monitoring, campus chargeback, or a public tariff-engine-first product motion.[^72][^81]

Adjacent players

Likely shortlist adjacencies include Schneider Electric Resource Advisor, EnergyCAP, and WatchWire depending on whether the buyer wants services-heavy outsourcing or software-led reporting and bill control.[^25][^43][^2]

Energy Toolbase

Positioning

Energy Toolbase is a commercial and industrial solar-plus-storage software company whose differentiator is project modeling, rate/tariff analysis, proposal generation, and controls/monitoring around distributed energy projects.[^85][^86][^87][^5]

It is bought by developers, EPCs, installers, and energy professionals rather than by finance teams seeking enterprise utility bill accounting; the job it replaces is spreadsheet-based project modeling and consultant-heavy tariff/storage dispatch analysis.[^88][^5][^89]

Primary customer segments

Energy Toolbase is strongest in the OEM/ESCO/channel ecosystem and C&I project developer market.[^5][^89]

It is weak as a direct fit for higher education, healthcare operations, multi-site restaurant/retail bill management, or CRE sustainability reporting except where those entities are evaluating onsite DER economics.[^86][^87][^5]

Product surface

The public product set includes ETB Developer, ETB Monitor, ETB Controller, and consulting around utility rates, net metering, and storage economics.[^87][^88][^86][^5]

This makes Energy Toolbase unusually strong on tariff/rate analysis and dispatch-oriented economic modeling, but weak on utility bill ingestion, payment, enterprise portfolio dashboards, and generalized bill-intelligence workflows.[^89][^87][^5]

No strong public evidence was found for LLM/chat, bill-Q&A, or regulatory/PUC monitoring as a product category.[^90][^91][^92]

Pricing and packaging signals

Energy Toolbase is one of the few profiled vendors with public pricing signals: it announced a price adjustment for ETB Developer effective June 1, 2024, and maintains a pricing page, though complete enterprise commercial nuance remains limited in public material.[^93][^94]

This supports high confidence that at least parts of the platform have explicit packaged pricing, likely more transparent than enterprise utility-management incumbents.[^94][^93]

Partners and dependencies

Energy Toolbase is deeply embedded in the project-development and hardware/control ecosystem, with ETB Controller and monitoring elements implying dependence on project and asset workflows rather than utility-data aggregation at portfolio-accounting scale.[^95][^86][^87]

Its data-custody model appears centered on project simulation, tariff databases, and asset operations rather than serving as the normalized bill system of record for enterprise finance teams.[^5][^89]

Recent corporate events

Recent public signals include 2024 pricing changes for ETB Developer, ongoing proposal feature launches in 2025, and continued utility-rates content that reinforces the tariff-analysis brand.[^96][^90][^93]

These are product-commercial events rather than financings or M&A in the reviewed source set.[^93][^96]

Reputation and risk signals

The main positive signal is specificity and credibility in tariff/rate analysis for DER economics.[^88][^89]

The main risk is category mismatch: slide decks that place Energy Toolbase directly beside enterprise bill-intelligence platforms without qualification can mislead decision-makers because the product is built around distributed energy project economics, not generalized utility-bill intelligence for campus or portfolio finance teams.[^86][^87][^5]

Differentiation vs target bundle

- **Does:** Tariff/rate analysis, economic modeling, proposal generation, DER controls/monitoring.[^87][^86][^5]
- **Partially does:** Alerts/optimization in the context of storage controls and project operation.[^86][^87]
- **Does not clearly document:** Bill ingestion/OCR at enterprise scale, campus chargeback, conversational bill Q&A, regulatory/PUC monitoring, or broad multi-site bill-accounting workflows.[^89][^5]

Adjacent players

Likely shortlist adjacencies include Aurora Solar, OpenSolar, and other DER modeling stacks rather than EnergyCAP/WatchWire in many deals; for tariff-engine conversations specifically, buyers may also compare against internal consultant models or specialized storage-optimization platforms. Public evidence here is directional, so exact shortlist names beyond Energy Toolbase’s immediate public materials remain partly Unverified.[^97][^5]

BrainBox AI in relation to Trane Technologies

Positioning

BrainBox AI sells AI-driven HVAC and building optimization, and Trane Technologies completed its acquisition of BrainBox AI in January 2025.[^4][^98][^99]

The job it replaces is not utility bill accounting; it replaces manual BAS tuning, reactive HVAC optimization, and some building-engineer decision support around HVAC performance and energy efficiency.[^100][^101][^102]

Primary customer segments

BrainBox AI/Trane is strongest in commercial buildings, building operators, and smart-building optimization contexts, with natural relevance to CRE and large facilities portfolios.[^101][^102][^100]

It is weaker as a direct fit for higher-ed utility accounting, healthcare bill management, restaurant/retail utility intelligence, or OEM/ESCO channel bill software; however, it is adjacent where building optimization is part of the same energy-management budget.[^100][^101][^4]

Product surface

Public evidence supports AI control, autonomous building optimization, and ARIA, a generative-AI virtual building assistant/virtual building engineer.[^102][^103][^101][^100]

This gives BrainBox AI the strongest public generative-AI/chat claim among the six, but the chat pertains to building operations, not normalized utility-bill Q&A or tariff-truth workflows.[^104][^105][^102]

There was no public evidence in reviewed sources for utility bill ingestion, payment, regulatory monitoring, or campus chargeback as core BrainBox AI product capabilities.[^101][^102][^100]

Pricing and packaging signals

No public pricing was found in the reviewed material; commercial terms are Unverified.[^106][^101]

The likely pricing narrative is ROI tied to HVAC savings and operational efficiency rather than FTE replacement in utility accounting; confidence: medium.[^102][^100]

Partners and dependencies

Post-acquisition, the key dependency is Trane Technologies and Trane’s smart-building platform/channel.[^98][^4]

That strengthens enterprise distribution but also means older decks referring to BrainBox AI as a standalone independent company are now outdated.[^99][^4]

Recent corporate events

The major event is the completion of Trane Technologies' acquisition of BrainBox AI on January 2, 2025.[^4][^98][^99]

Before that, BrainBox AI launched ARIA in 2024 and received high-profile recognition for it later that year.[^103][^104][^102]

Reputation and risk signals

The strongest positive signal is clear product-market fit in HVAC optimization and one of the better-documented AI narratives in the building-operations category.[^104][^102]

The main diligence risk is category drift: vendors and analysts may over-associate "AI for buildings" with utility-bill intelligence, when BrainBox AI is fundamentally a controls and building-ops product, not a bill-intelligence stack.[^100][^101][^102]

Differentiation vs target bundle

- **Does:** Building optimization, AI control, generative assistant for building operations.[^101][^102][^100]
- **Partially does:** Proactive operational alerts within HVAC/building automation contexts.[^100][^101]
- **Does not clearly document:** Utility bill intelligence, tariff truth, regulatory/PUC monitoring, chargeback, or normalized bill-chat workflows.[^102][^100]

Adjacent players

Likely shortlist adjacencies include Trane's own smart-building stack, Johnson Controls/OpenBlue-type optimization offers, and building-analytics vendors rather than EnergyCAP or Arcadia in pure bill-intelligence RFPs; exact shortlist evidence is partly Unverified from the sources reviewed here.[^4][^101][^100]

Other platforms large corporations use

Relevant adjacent platforms

Large corporations also commonly use Schneider Electric EcoStruxure Resource Advisor for enterprise resource and sustainability management, especially where global reporting and advisory integration matter.[^107][^108][^43]

In CRE-heavy contexts, Measurabl and Gridium are important adjacent names: Measurabl is especially visible in real-estate sustainability workflows, while Gridium positions around energy-management-as-a-service for real estate.[^109][^70][^110]

These platforms matter because many enterprise RFPs are not pure “utility bill intelligence” buys; they are broader resource-management, ESG, or building-operations decisions where bill intelligence is one layer of the stack.[^70][^110][^43]

Cross-cutting questions

Who actually competes with conversational bill Q&A plus tariff truth plus proactive dollar-ranked alerts?

The public evidence suggests no perfect like-for-like incumbent. The closest partial competitors are EnergyCAP for utility-accounting depth plus recent Watts AI claims, Arcadia for normalized bill and interval data with enterprise AI positioning, and BrainBox AI for documented generative AI but in building operations rather than bills.[^6][^102][^14]

Specialized regulatory/PUC monitoring appears to come from adjacent providers such as Halcyon and Stateside rather than from the six profiled vendors, which implies that the regulatory-monitoring element is usually separate from utility-bill intelligence today.[^111][^112][^13]

That means a conversational layer over normalized bills plus tariff truth plus proactive alerts is more likely to compete against combinations of platforms than against a single documented incumbent product.[^13][^28][^44]

Higher education: which of the six have documented campus traction?

EnergyCAP clearly has documented higher-ed traction as of early 2026 through its higher-ed page and public university case studies including Kansas and Arizona.[^21][^10][^9]

WatchWire has some educational-facilities messaging, but the public evidence found is lighter and more generic than EnergyCAP’s named campus proof.[^62][^63]

Public higher-ed proof was not clearly documented in the reviewed sources for Arcadia (arcadia.com), ENGIE Impact, Energy Toolbase, or BrainBox AI in the specific utility-bill-intelligence context; these should be treated as weak/Unverified for documented campus wins absent additional diligence material.[^5][^44][^12][^4]

Gaps and conflicts

A recurring conflict is “AI” language versus documented workflow depth. EnergyCAP’s Watts AI is publicly framed around faster reporting and assistance, not clearly around a mature autonomous copilot for bill Q&A plus regulatory monitoring.[^30][^6]

Arcadia’s AI-powered enterprise messaging appears real but still best evidenced as data-platform and energy-cost optimization positioning, not a fully documented conversational intelligence product.[^44][^14]

BrainBox AI's autonomous/AI claims are credible within HVAC optimization, but those claims should not be conflated with utility-bill intelligence.[^104][^102]

ENGIE Impact can also be marketed as "full stack," yet much of the public value proposition is services-led rather than a transparent product-led software stack, which matters in head-to-head product comparisons.[^73][^12][^72]

Factual corrections likely needed in typical decks

- Arcadia should be identified as arcadia.com, not the healthcare analytics company arcadia.io.[^8][^1]
- Urjanet should not be shown as a separate current competitor without noting Arcadia's acquisition.[^7][^1]
- BrainBox AI should not be shown as independent without noting Trane Technologies completed the acquisition in January 2025.[^98][^4]
- WatchWire should be labeled as WatchWire by Tango / under Tango ownership, not standalone.[^2][^11]
- ENGIE Impact should be framed as part of ENGIE, not as a startup-style independent software vendor.[^84][^83]
- EnergyCAP's AI positioning should be softened unless the deck can substantiate a full conversational bill-intelligence layer beyond Watts AI reporting assistance.[^6][^30]

Feature comparison

Company	Bill intel	Tariff	API	NL/chat	Proactive alerts	Regulatory/ PUC	Chargeback
EnergyCAP	Strong utility accounting and bill management[^18][^17]	Documented tariff analytics[^28]	Public developer API[^29]	Limited/early Watts AI evidence[^6][^30]	Analytics/ anomaly support documented, but autonomous dollar-ranked alerting not clearly public[^27]	No clear public evidence[^6]	Strongest in set for campus-style allocation/ accounting orientation[^17][^20]
Arcadia	Strong normalized bill and interval data[^15][^47]	Partial, more cost/ procurement than explicit tariff-engine marketing[^44][^15]	Strong API/ data-platform posture[^47][^49]	AI messaging present, chat layer not clearly documented[^14][^44]	Possible via enterprise solutions, but not well documented publicly[^44]	No clear public evidence[^46]	No clear public evidence[^15]
WatchWire	Moderate utility data collection/ reporting[^2][^61]	Weak public evidence for deep tariff engine[^65]	Integration-led, API depth not central public message[^61]	No clear public evidence[^66]	Reporting/ compliance support, not strong proactive savings-alert proof[^65]	No clear public evidence[^2]	No clear public evidence[^63]
ENGIE Impact	Strong managed utility expense management[^12][^73]	Moderate through services, weak public productized tariff-engine proof[^3][^114]	Unclear/ secondary in public narrative[^72]	No clear public evidence[^81]	Likely via managed services, not documented as product-native AI alerting[^3][^12]	No clear public evidence[^71]	No clear public evidence for campus chargeback[^12]
Energy Toolbase	Weak for enterprise bill intelligence[^5]	Strongest public tariff/ rate analysis in set[^89][^88]	Not core enterprise bill API story[^5]	No clear public evidence[^90]	Present in DER controls/ monitoring contexts[^86][^87]	No clear public evidence[^91]	No clear public evidence[^5]

Company	Bill intel	Tariff	API	NL/chat	Proactive alerts	Regulatory/ PUC	Chargeback
BrainBox AI / Trane	Weak for bill intelligence[^100][^102]	No clear bill tariff engine evidence[^101]	Not relevant to bill-data API in reviewed sources[^100]	Strongest documented generative assistant, but for building ops not bills[^102][^103]	Strong in HVAC operational optimization alerts[^100][^101]	No clear public evidence[^102]	No clear public evidence[^4]

Update recommendations

- Clarify that Arcadia here means arcadia.com, not Arcadia.io.[^8][^1]
- Remove Urjanet as a standalone current peer unless the deck explicitly says “Arcadia (via Urjanet acquisition).”[^1][^7]
- Update BrainBox AI ownership to Trane Technologies effective January 2025.[^98][^4]
- Update WatchWire ownership to Tango and avoid presenting it as independent.[^59][^11]
- Soften claims that any incumbent already delivers the full bundle of conversational bill Q&A, tariff truth, proactive dollar-ranked alerts, regulatory monitoring, and chargeback in one product; public evidence did not support that for any profiled vendor.[^13][^6][^44]
- Emphasize EnergyCAP as the strongest documented higher-ed incumbent rather than a generic enterprise peer.[^10][^21][^9]
- Reframe ENGIE Impact as services-led outsourcing plus platform, not a product-led AI-native bill-intelligence vendor.[^12][^73][^72]
- Reframe Energy Toolbase as tariff/DER modeling infrastructure, not a general portfolio utility-bill intelligence platform.[^89][^5]
- Separate building-optimization AI from bill-intelligence AI when discussing BrainBox AI/Trane.[^102][^104]
- Add an “adjacent regulatory intelligence” category with Halcyon or Stateside because regulatory/PUC monitoring appears to live outside most current utility-bill platforms.[^112][^115][^13]

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109. Measurabl 2024 Updates | Here's How 2024 Measured Up - Stay ahead with Measurabl 2024 updates. Learn about the latest features and improvements shaping the...

110. Gridium | Real Estate's Energy-Management-as-a-Service Platform - The Energy-Management-as-a-Service Platform for Real Estate. Turn your energy data into your next co...

111. Introducing Docket Profiles (pt 2/2) - Halcyon - Introducing Docket Profiles (pt 2/2) - Halcyon Docket profiles provide a comprehensive overview of p...

112. State Utilities Monitoring | Public Service Activity Tracking - Our state utilities monitoring and tracking services ensure you will stay on top of rulemaking activ...

113. Customer stories | Arcadia - Using Arcadia's utility data platform, Subcontractor Hub helps users access accurate and up-to-date ...

114. Energy Supply Management - ENGIE Impact - Our energy supply management experts help you navigate complex markets and regulations and optimize ...

115. Halcyon Raises $21 Million Series A To Bring AI-Powered ... - Halcyon Platform — AI-powered software that lets users search and query across thousands of regulato...

DOCUMENT 8: Compass — EnergyCAP and the Utility Bill Intelligence Competitive Set

Source file: Compass — EnergyCAP and Utility Bill Intelligence Competitive Set.md

EnergyCAP and the utility bill intelligence competitive set

EnergyCAP is the unambiguous incumbent to beat in higher-ed and public-sector utility bill management, but its product architecture, AI surface, and go-to-market still read like a 2019 portfolio reporting tool wrapped in a 2025 marketing refresh. That creates a clear opening for an AI-native, conversation-first challenger like Clear Current Technologies — but only in segments where EnergyCAP's campus-chargeback moat does not apply. EnergyCAP was founded in 1980 (as FASER/OmniComp), survived the Enron era, was recapitalized by Resurgens Technology Partners in March 2021, and bolted on Irish analytics vendor Wattics in August 2022 to form today's four-product suite (UtilityManagement, SmartAnalytics, CarbonHub, Utility Company Platform). Against a market that is simultaneously being re-platformed by Schneider Electric's Resource Advisor+ (launched January 20, 2026 with agentic AI), pressured by AI-first CRE entrants (Cortex, Gridium), and disrupted by the utility-bill affordability narrative around AI data centers, the strategic question is not "can EnergyCAP be displaced everywhere" but "where is EnergyCAP structurally vulnerable in the next 18 months."

EnergyCAP, in one paragraph

EnergyCAP sells a **utility bill management "system of record"** — ingestion, audit, accrual, chargeback, GHG accounting, and ENERGY STAR Portfolio Manager submission — on an annual subscription priced by **number of meters (points of service)** plus optional services (Bill CAPture OCR ingestion, Bill Pay, managed services). It primarily replaces the incumbent in-house **spreadsheet-plus-consultant stack** used by energy managers and facilities finance staff at multi-building portfolios. The company claims customers identify errors in roughly **1 in 5 bills**, catch "nearly 5%" of utility cost in anomalies, and typically save **10%+ annually**. Headquarters is Boalsburg, PA, with additional offices in Greenwood Village, CO and Dublin, Ireland; PitchBook lists **~112 employees**. It is a **five-time ENERGY STAR Partner of the Year**, **SOC 2 Type II** (most recent audit April 2024, per EnergyCAP's security page), and **FedRAMP Low-Impact SaaS** authorized — a meaningful procurement advantage over most VC-stage challengers.

Segment strength is uneven and highly predictable

EnergyCAP's segment positioning is not symmetric. The company is deeply entrenched in higher education and the broader public sector; it is weaker in places where real-time operations and tenant-facing workflows dominate.

The **higher-education stronghold is real and documented**. Published success stories include **Carnegie Mellon University** (700+ monthly vendor bills, 120+ internal submeters feeding general ledger), **University of Kansas** ($2.3M/year in chargeback revenue, 1,100+ meters across multiple campuses), **UC San Diego** (caught a reported $1 million billing error, cut labor 95%), **University of New Mexico** (Lobo Energy), **Oregon State**, and others. CEO-founder Steve Heinz has been pitching "campus chargebacks" as EnergyCAP's national leadership position

at AEE conferences for years, and the product's bill-split/workflow tooling genuinely reflects that domain depth. **K-12, state/federal government, and municipalities** are similarly strong (City of Tucson, Connecticut DEEP, Coral Gables). Healthcare and manufacturing are supported but feel like secondary verticals in the marketing. **Multi-site retail/QSR and commercial real estate are the weakest** segments — no flagship logos surface, and the operational-optimization workflows those segments want (real-time HVAC scheduling, tenant sustainability dashboards, ESG disclosure for REITs) sit in the Wattics-era SmartAnalytics module, which feels bolted on rather than native.

The product surface is broad but dated in the places that matter

EnergyCAP's product catalog is wider than any pure-play competitor, yet the **conversational-AI layer is nascent and the architecture is still fundamentally bills-first, not operations-first**. The core UtilityManagement platform handles bill capture (OCR-based Bill CAPture, EDI 810 ingestion, vendor-direct feeds), tariff/rate engines, anomaly audits, accruals, budgeting, Portfolio Manager sync, and chargeback/bill-split workflows — the last of which is genuinely best-in-class for multi-tenant campus scenarios with submeters, virtual meters, and Visual Basic scripting. SmartAnalytics (the 2022 Wattics acquisition) adds interval-data monitoring, M&V, and IPMVP project tooling. CarbonHub calculates Scope 1/2/3 using market or location factors. The Utility Company Platform is a separate product sold to utilities themselves for benchmarking compliance. Bill Pay is offered as a managed service with claimed "near-100% elimination of late fees."

The newest surface is **"Watts AI," EnergyCAP's AI engine** introduced during 2025, positioned as the conversational Q&A and recommendation layer on top of utility data. Marketing frames it as "AI-powered analytics flags over 1 million billing errors annually, benchmarks 300,000+ buildings." Compared with Schneider's **Resource Advisor+ Sera** (launched January 2026 with an explicit multi-agent architecture and "SE Advisory Intelligence" knowledge layer) or Schneider's earlier **Resource Advisor Copilot** (GA late 2023/early 2024), Watts AI reads as catch-up positioning rather than a differentiated bet. No public documentation confirms PUC docket monitoring, dollar-ranked proactive alerts, or LLM-based tariff explanation — the exact capabilities that define Clear Current's wedge.

Corporate history and ownership are stable but unspectacular

Ownership is private equity, not venture capital, and has been for five years. Resurgens Technology Partners (with Webster Equity Partners as a co-investor per PitchBook) took control in March 2021, renaming the company EnergyCAP, LLC. The single meaningful M&A event under PE ownership was the **August 2022 acquisition of Wattics**, a Dublin-based energy analytics vendor that became SmartAnalytics and established the Ireland office. No new funding round, platform re-platforming announcement, or CEO change has been disclosed publicly in the 2024–early 2026 window. Note: **do not confuse EnergyCAP with Energy Capital Partners (ECP)**, the Summit, NJ private equity firm that Bridgepoint acquired for $1.05B in August 2024; decks

occasionally conflate the two because of the similar name, and it is a common factual error. The current platform generation (EnergyCAP 7 → EnergyCAP UtilityManagement cloud) migration has been essentially complete for the last two years, and all new sales are on the cloud SKU.

The competitive landscape around them

The market that EnergyCAP actually competes in splits into four archetypes, and the AI-native conversational-insight archetype that Clear Current targets is still under-served. The comparison below captures the players most likely to show up in the same RFPs or analyst conversations.

Vendor	Core archetype	Ownership / status	Strongest segment	AI posture as of 2026
EnergyCAP	Bill-management system of record	PE (Resurgens, 2021)	Higher ed, government, K-12	"Watts AI" Q&A + ML audits; retrofit onto bills-first core
Schneider Resource Advisor+	Enterprise ESG + energy	Public (Schneider SE Advisory Services)	Global F500, CSRD filers	"Sera" multi-agent platform, GA Jan 2026
Accruent / Observe (ex-SiteLine)	Facility + asset mgmt with energy module	Fortive (acquired 2018, $2B)	Healthcare, multi-site retail	Asset-centric AI; not utility-bill native
Urjanet / Arcadia Arc	Bill-data ingestion API	Arcadia (acquired 2022)	OEM data pipe to other SaaS	Data layer, not a user-facing app
Gridium	CRE energy analytics + rate engine	Independent, Menlo Park	California/Texas CRE operators	ML tariff optimization; no conversational layer
Cortex Sustainability Intelligence	CRE decarbonization + BMS optimization	VC-backed (~$11.6M, Series A May 2021, 4490 Ventures)	NYC/Class-A office (ESRT, RXR, Silverstein, Savanna)	"Cortex Push" AI automation to BMS
WatchWire / ENGIE Impact / NG Insight	ESG/sustainability reporting suites	Various (ENGIE corporate; NG Insight under NORESCO/Carlyle, Unverified current status)	Fortune 1000, REITs	Reporting-first; limited conversational AI
Measurabl / Goby (Conservice ESG)	ESG disclosure for CRE	Measurabl independent; Goby inside Conservice	REITs, GRESB reporters	ESG workflow automation
Billee Technologies	AI utility billing for multifamily	Seed ($9.15M, July 2025, RET Ventures)	Multifamily apartments	AI-native, bills + work orders

The **most structurally threatening competitor for higher ed is still EnergyCAP itself** — no AI-native challenger has published campus logos at CMU/UCSD scale. The most structurally threatening competitor **for enterprise and ESG buyers is Schneider Resource Advisor+** because it pairs agentic AI with decades of procurement/advisory relationships. The most threatening competitor **for CRE portfolios is Gridium for analytics plus Cortex for BMS-connected optimization**, neither of which competes for campus chargeback RFPs. Gridium's rate engine — building a "synthetic utility bill under each possible rate" — is the closest public competitor to a tariff-truth conversational feature. **"Mantle" as an AI-first energy analytics vendor does not appear to be a mature commercial player** in the utility-bill space; web searches return a cap-table SaaS by the same name and should be treated as Unverified in any brief claiming it as a peer.

Where the slide decks tend to be wrong

Several factual errors recur in this market's marketing material and should be corrected before any Clear Current deck ships. **EnergyCAP is not owned by Serent Capital, Energy Capital Partners, or Vista** — it has been a Resurgens Technology Partners portfolio company since March 2021, co-invested by Webster Equity. **Urjanet is a brand inside Arcadia**, not an independent vendor; anyone diagramming it as a separate competitor is working from 2021 slides. **Accruent is a Fortive subsidiary**, sitting alongside Gordian and ServiceChannel in Fortive's Facility & Asset Lifecycle Group — not a standalone PE-backed company, and not a spin-out. **Schneider's energy and sustainability software is no longer branded "Sustainability Business"**; it sits inside **SE Advisory Services**, launched as a dedicated consulting practice led by Steve Wilhite, with Resource Advisor+ as the flagship AI-first product launched January 20, 2026. **Cortex rebranded from Cortex Building Intelligence to Cortex Sustainability Intelligence** and its last disclosed raise was a $6M Series A in May 2021 (4490 Ventures) for ~$11.6M total — any deck quoting "recent Series B" is wrong as of April 2026.

Analyst consensus (IDC MarketScape 2025, Verdantix Green Quadrant 2025) names Schneider as a Leader across ESG/sustainability software, not EnergyCAP. EnergyCAP's awards come primarily from **mid-market review sites** (Info-Tech/SoftwareReviews Data Quadrant, G2 Market Leader) and industry-specific recognitions (ENERGY STAR Partner of the Year) — real signals for buyers, but a weaker tier than Gartner/Forrester/Verdantix enterprise leader placements that Schneider claims.

The wedge Clear Current should press

EnergyCAP's architecture reveals three durable weaknesses. **First, the product is still bills-first, not conversation-first** — Watts AI is new, retrofit, and rides on top of a tree-view navigation metaphor ("meters, buildings, organizational divisions, accounts, cost centers, vendors, and rate schedules") designed for energy managers, not executives. A dollar-ranked proactive alert feed and natural-language Q&A over normalized bills is a cleaner first-run experience. **Second, PUC docket monitoring and tariff truth are genuinely absent** — EnergyCAP has rate schedules as a configuration object, but it does not ingest state PUC filings or translate pending rate cases into forward-looking dollar impact per building, and neither does Schneider. That is an

uncontested feature today. **Third, EnergyCAP's chargeback moat is narrow** — it is excellent for campus chargebacks with submeters and a central plant, but it is over-engineered for multi-site retail/QSR and inadequate for REIT tenant-facing ESG workflows that Measurabl, Goby, and Cortex dominate. Any Clear Current RFP strategy that runs head-on at a university's chargeback replacement is a slog; running at multi-site retail, healthcare systems, and CRE portfolios where EnergyCAP is installed but under-loved is the faster sale.

What to watch in the next four quarters

Three events could reorder this market. **Resource Advisor+ adoption curves through 2026** will determine whether agentic AI becomes table stakes or stays enterprise-only — if Schneider signs mid-market logos, EnergyCAP's "Watts AI" positioning erodes quickly. **A Resurgens exit or EnergyCAP recap** is plausible in 2026–2027 after five years of hold; any M&A disruption creates a six-to-twelve-month window where customers reconsider. **Utility-bill affordability politics** — the PJM capacity price spike from $60/kWh (2024) to $300+/kWh (2025) and the $31B of 2025 rate-increase filings — is creating CFO-level demand for exactly the dollar-ranked, tariff-aware, docket-monitoring capabilities that incumbents do not yet ship. That is the macro tailwind Clear Current should use to reframe the category from "utility bill software" to "utility cost intelligence," and it is the reframe EnergyCAP cannot easily follow without rebuilding its data model.

DOCUMENT 9: Compass — Arcadia Deep Competitive Report

Source file: Compass — Arcadia Deep Competitive Report.md

Arcadia Competitive Intelligence Report

Prepared for: Clear Current Technologies Date: April 2026 Classification: Internal Strategic Use

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Executive Summary

Arcadia is the most well-funded AI-native competitor in the commercial energy data and enterprise energy management space, having raised approximately $649–658 million in total capital since 2014. With a January 2025 pivot into enterprise solutions, an aggressive acquisition strategy, and the industry's most comprehensive utility data infrastructure, Arcadia represents the primary AI-native threat to Clear Current Technologies. However, Arcadia's enterprise product is less than 18 months old, is built on a data and API platform architected primarily for energy tech developers (not energy managers), and exhibits clear structural gaps in the vertical markets most relevant to Clear Current — particularly higher education, campus district energy systems, and domestic utility billing complexity. This report documents Arcadia's funding, products, customers, strategy, hiring, partnerships, and exploitable weaknesses in granular detail.[^1][^2]

Intelligence confidence rating by section:

- Funding history: **HIGH** (primary press releases, CB Insights, Growjo)
- Product architecture: **HIGH** (primary source: Arcadia's own llms.txt, press releases, blog posts)
- Customers: **MEDIUM-HIGH** (verified customer stories; some customer list items from llms.txt marketing claims)
- Revenue/headcount: **LOW-MEDIUM** (no public financials; third-party estimates conflict widely)
- Pricing: **LOW** (no published pricing; no verified procurement data found)
- Strategic weaknesses: **HIGH** (triangulated from job postings, product gaps, no higher-ed customers identified)

Section 1: Company Overview and Funding History

Founding Story

Arcadia was founded in 2014 in Washington, D.C. by Kiran Bhatraju (CEO), Ryan Nesbitt (President), and Kate Henningsen (COO). Bhatraju, a University of Pennsylvania graduate and former Capitol Hill staffer, founded the company after observing the complexity of U.S. energy markets and the lack of consumer access to clean energy options. The company's original mission was to connect residential consumers to renewable energy — specifically community solar — without upfront costs.[^3]

Co-founder Ryan Nesbitt departed in 2018 to pursue web3 investing. Patrick McCamley joined as CCO in 2020 and replaced Nesbitt as President in 2022. Kate Henningsen remains as COO. Prior to founding Arcadia, Bhatraju founded American Efficient (2010), a company that sold energy management software to utilities — directly relevant context for understanding how Arcadia thinks about the enterprise energy market.[^3]

Complete Funding History

Round	Date	Amount	Lead Investor(s)	Notable Participants
Seed/ Early	2014–2016	$1.9M	Undisclosed	BoxGroup, Wonder Ventures [^4]
Series A	2017	~$6M	EIP	BoxGroup, Wonder Ventures [^4]
Debt	Jan 2018	$5.5M	Undisclosed	— [^4]
Series B	Aug 2018	$25M	G2 Venture Partners	ValueAct, EIP, BoxGroup, McKnight Foundation, Cendana Capital [^5][^4]
Series C	Dec 2019	$30M	G2 Venture Partners	Macquarie Group, EIP, Mitsui USA, ValueAct Spring, Seek Ventures [^6][^4]
Series C-1	Dec 2020	$21M	Undisclosed	Existing investors (undisclosed at time) [^7]
Series D	Sep 2021	$100M	Tiger Global + Drawdown Fund	Wellington, Reimagined Ventures, Camber Creek, MCJ Collective, EIP, G2VP, Inclusive Capital, BoxGroup [^7]
Series E	May 2022	$200M	J.P. Morgan Asset Management	Broadscale Group, Salesforce Ventures, Tiger Global, Wellington, Camber Creek, Drawdown Fund, MCJ Collective, Keyframe Capital, Triangle Peak [^8]
Venture	Dec 2022	$125M	Magnetar Capital	Existing investors [^5]
Growth	Apr 2024	$50M + $30M JPMorgan credit facility	Macquarie Asset Management (new equity)	EIP, JPMorgan AM, BoxGroup, G2VP, Camber Creek, Triangle Peak, Broadscale [^9][^10]

Total raised: ~$649.9M (CB Insights) to ~$658M (Growjo)[^2][^1]

Last known valuation: $1.5 billion (May 2022 Series E). No valuation disclosed in the April 2024 round.[^1]

Key investors still active as of 2024: G2 Venture Partners, J.P. Morgan Asset Management, Macquarie Asset Management, Energy Impact Partners, Camber Creek, Tiger Global, Broadscale Group, Triangle Peak Partners.

Headcount and Revenue Estimates

Arcadia does not publish revenue or headcount. Available third-party estimates conflict significantly, likely because Arcadia has restructured around its Perch Energy community solar spinout, and analysts are estimating the combined entity differently.

Source	Headcount	Revenue Estimate	Date
Contrary Research	~620 employees	Not disclosed	Sep 2023 [^3]
ZoomInfo	763 employees	$160.3M	2025 [^11]
Growjo	856 employees	$270M est.	2025 [^1]
Indeed (company-reported)	201–500 range	$5–25M	2025 [^12]
CB Insights	Not disclosed	Not disclosed	2025 [^2]

Assessment: The Growjo $270M figure at 856 employees yields ~$315K revenue per employee, consistent with a data/SaaS business. The Indeed self-reported figure of $5–25M is clearly an underestimate, likely reflecting only one business unit. The ZoomInfo $160.3M figure is likely closer to reality for the enterprise/platform business pre-Perch merger. No confirmed public figure exists.

No layoffs or restructuring news was identified in the 18-month period from October 2024 to April 2026. The Perch Energy community solar spin-out (March 2025) effectively moved a large employee segment into a separate legal entity, which may have reduced Arcadia's parent headcount without a formal layoff announcement. [^13]

Section 2: Product Architecture — What They Actually Built

Arcadia's product stack has two distinct layers: (1) the Arc developer/API platform, and (2) the January 2025 Enterprise Solutions suite built on top of that platform. Understanding this architecture is critical for Clear Current's positioning.

The Arc Developer Platform (B2B API Infrastructure)

The Arc platform, launched in October 2021, was built as a "Plaid for energy" — an API-first infrastructure layer that lets any company access standardized utility data regardless of which utility serves a given account. Its customers are energy tech companies and developers, not energy managers. The platform's five sub-products are: [^14][^3]

Plug: The core data-access API. Connects to 9,500+ utilities across 52 countries and delivers utility billing data (bill images, line-item charges, usage) and interval data (15/30/60-minute consumption) in standardized format. As of January 2026, covers nearly 400 utilities for interval data (expanded from 60 at the start of Q4 2024). This rapid interval coverage expansion is a significant platform investment signal.[^15][^16]

Signal: The tariff intelligence engine. Maintains 25,000+ North American tariff structures (30,000+ globally), processes 75,000 rate updates per month, and performs approximately 1 billion tariff calculations annually. This is the most mature component of the Arc stack — Genability (a tariff database company acquired by Arcadia) was the foundation for Signal. Ford uses Signal specifically for EV home-charging rate optimization — not for enterprise energy management.[^17][^18][^19][^20]

Switch: Solar and storage savings modeling tool. Achieves 99.5% accuracy per independent NREL review. Not relevant to Clear Current's competitive analysis.[^20]

Bundle: Energy billing workflow product for community solar and retail energy providers. Not relevant to enterprise energy management.[^3]

Spark: Advanced analytics and EV/smart device optimization. Developer-facing.[^3]

How data gets into the platform: Arcadia primarily uses utility portal credential scraping (users provide utility login credentials via Arcadia's "Connect" embeddable widget), supplemented by direct EDI feeds established through Urjanet's network, and Green Button Connect where available. PDF bill ingestion is part of the bill parsing workflow. Interval data is pulled directly via AMI meter portals for the ~400 utilities where this is now available.[^16][^15][^3]

The January 2025 Enterprise Solutions Suite

On January 29, 2025, Arcadia launched three enterprise modules built on the Arc platform foundation:[^21][^22]

Module 1: Utility Bill Management

This is the most developed enterprise module and most directly competitive with Clear Current. It includes:

- **Automated data capture:** Pull billing data from 9,500+ utilities via Plug, validated against the tariff database
- **Automated Tariff Assignment (ATA):** Processes 500,000+ meter-tariff assignments automatically; achieves 80% full-auto match rate, 20% requires expert analyst review[^17]
- **Cost Validation Service (CVS):** Recreates bills using interval data + tariff database + calculation engine; achieves 98.4% median bill recreation accuracy at portfolio scale[^17]
- **Bill auditing:** Identifies billing errors by comparing calculated-correct charges to actual billed charges[^17]
- **Tariff optimization:** Identifies alternative rate structures that would lower cost[^21]
- **Automated bill payment:** End-to-end payment processing[^21]

- **Budgeting & forecasting:** Uses validated historical data for forward projections[^17]

Key claim (marketing, verify independently): "One hyperscaler identified over $10 million in billing errors within a few months of using Arcadia to aggregate and standardize utility bills" — stated by Kiran Bhatraju on LinkedIn, January 2026. Customer not named.[^23]

Module 2: Energy Procurement Advisory

This is a services-plus-software hybrid built on the February 2025 acquisition of RPD Energy:[^24][^25]

- Co-optimization across all supply options: flexible retail supply agreements (0–100% clean), community solar, direct purchases from renewable generators, behind-the-meter solar and storage, demand-side management, demand response, utility tariff optimization, RECs and offsets[^26]
- Rohit Marwaha (formerly Constellation Energy) brought in to lead[^24]
- RPD Energy's customer base (Iron Mountain, Adobe, Intuit, UPS) inherited[^24]
- Scenario modeling for strategic procurement decisions[^27]

Critical distinction for Clear Current: This module is primarily a procurement advisory and execution service, not a contract analytics or billing error detection product. It helps customers *buy* energy; it does not help them audit *what they've been billed*.

Module 3: Sustainability Reporting

- Scope 1 and 2 emissions tracking from standardized utility data[^20]
- Supports GRI, SASB, TCFD, CSRD, and SEC climate disclosure frameworks[^20]
- Audit-ready data with source traceability[^20]
- Integrates with Cority (EHS/ESG platform) and Sweep (EU sustainability)[^28][^29]

CORRECTION on "Supply Management" module: The Part 4 research referred to a "Supply Management" module. This does not appear to be a separate product. The three official modules as named in Arcadia's January 2025 press release and current website are: Utility Bill Management, Energy Procurement Advisory, and Sustainability Reporting. The earlier reference likely conflated "Energy Procurement Advisory" with "Supply Management."[^30][^21]

Customer UX: Dashboard or API?

The enterprise UX is primarily a **dashboard** — Arcadia describes its offering as delivering "actionable insights" through a centralized platform view. However, the platform's DNA is API-first; Cox Enterprises, for example, uses Arcadia as a **data feed into their own in-house Energy Dashboard**, explicitly choosing Arcadia over competitors that "wanted to put us on their dashboard". This data-connectivity use case (API-out rather than dashboard-in) is a core pattern in Arcadia's current enterprise base.[^31]

There is **no documented natural language AI chat interface** for energy managers. Arcadia's AI capabilities are embedded in automated processes (ATA, CVS, gap-filling) rather than user-facing chat.[^20]

Section 3: The Urjanet Acquisition — What They Bought

Arcadia acquired Urjanet in May 2022. The acquisition was the single most strategically important move in Arcadia's history because it transformed Arcadia's data coverage from residential-focused to fully commercial.[^32][^33]

What Urjanet was before the acquisition: Atlanta-based Urjanet was the largest global utility data aggregator in the world, providing automated access to utility billing data from 6,500+ utility, telecom, and cable providers worldwide. Urjanet's flagship product at the time of acquisition was *Utility Cloud*, which provided on-demand access to credentialed utility account information for ESG reporting and energy management. Urjanet's commercial focus was notable: approximately **30% of the Fortune 500's utility bills** were being captured through Urjanet's technology prior to acquisition.[^34]

What Arcadia gained from Urjanet:

- Expansion from Arc's original 125 residential providers covering ~80% of U.S. residential accounts to 9,500+ utilities covering 95%+ of both residential **and commercial** accounts in the U.S.[^35][^32]
- Global coverage expanded to 52 countries[^33]
- Electric, gas, water, and waste data (not just electric)[^33]
- The Fortune 500 commercial enterprise client relationships Urjanet had built[^34]
- Connection to 1.35 million utility accounts and $20B+ in annual utility bills[^32]

Coverage expansion since acquisition: Since the Urjanet acquisition in May 2022, Arcadia has expanded interval (AMI) data coverage from approximately 60 utilities to nearly 400 utilities as of January 2026. Commercial account utility bill coverage remains claimed at 95%+.[^16][^28]

Commercial account interval data gap: The 95% commercial coverage claim applies to **billing data**. Interval data coverage (which enables demand charge optimization and real-time rate analysis) is far more limited — Arcadia's own documentation claims coverage of "over 75% of AMI meters for interval data", but this is portfolio-wide and does not mean 75% of commercial accounts specifically have interval data available. The expansion to ~400 utility providers for interval data is significant progress from the prior year's 60, but many mid-size and smaller commercial utilities where Clear Current's target customers have facilities remain uncovered.[^16][^17]

Section 4: Customers and Use Cases

Documented Enterprise Customers

Customer	Use Case	Data Source	Notes
Cox Enterprises	Utility data aggregation for 40,000 accounts (~$120M annual spend); energy dashboard data feed; billing optimization	Primary (case study) [^31]	Chose Arcadia specifically to feed their *own* internal dashboard — not to use Arcadia's dashboard. Found $43K+ in savings from zero-kWh accounts and rate optimization.
Fox Corporation	Utility data collection for emissions reporting; ESG/sustainability	Marketing (llms.txt)[^20]	Use case is ESG reporting, not billing audit
Arconic	Fortune 500 manufacturer; utility data + energy insights	Customer story page[^31]	Details sparse
Iron Mountain	Renewable energy procurement (via RPD Energy)	Press release[^24]	Procurement advisory, not billing management
Adobe	Renewable energy procurement (via RPD Energy)	Press release[^24]	Procurement advisory
Intuit	Clean energy procurement (via RPD Energy)	Press release[^24] [^31]	Community solar + retail contract
UPS	Clean energy procurement (via RPD Energy)	Press release[^24]	Procurement advisory
Conagra	Listed as customer	llms.txt[^20]	No case study available
Penske	Listed as customer	llms.txt[^20]	No case study available
Unnamed hyperscaler	Billing error identification ($10M+)	CEO LinkedIn post[^23]	**Not independently verified** — marketing claim only

Ford: What the Relationship Actually Is

The Ford relationship is a **developer API (Signal) use case for EV charging optimization**, not an enterprise energy management relationship. Ford uses Signal's tariff database to help EV owners identify the cheapest home-charging tariffs. This is entirely different from enterprise utility bill management or procurement advisory. Arcadia markets this prominently because Ford is a recognizable brand, but it does not indicate Ford uses Arcadia for corporate facility energy management.[^18]

Developer/API-Only Customers (Not Enterprise Competitors)

These are customers of the Arc developer platform — they should not be mistaken for enterprise energy management customers: Oracle, Aurora Solar, EVgo, Enphase, Stem, Stable Auto, Backpack, Station A, WatchWire by Tango (which uses Arcadia as a data *source* for its own energy management product).[^31][^3]

The WatchWire by Tango relationship is noteworthy: WatchWire is a direct competitor to EnergyCAP in commercial real estate energy management, and it uses Arcadia's Plug API as a utility data feed — suggesting Arcadia is *upstream supplier* to some of Clear Current's competitors as well as being a direct competitor itself.

Higher Education Customers

No documented higher education customers exist in any Arcadia customer story, case study, press release, or marketing material. Arcadia's enterprise customer base appears to be predominantly corporate (manufacturing, media, automotive services, tech) rather than institutional. This is a significant gap.

User Reviews

No G2 or Capterra reviews exist for Arcadia's enterprise energy management product (the product launched January 2025 and has no G2/Capterra profile). Trustpilot shows 1,159 reviews at 2.6 stars and BBB shows 25+ complaints, but **all reviews are about Arcadia's consumer community solar product** (billing transparency issues, unauthorized charges, poor customer support) — not the enterprise platform. These reviews should not be conflated with the enterprise product experience but do indicate operational debt in the consumer business that was adjacent to the enterprise business until the community solar spinout.[^36][^37][^38]

Section 5: Pricing and Commercial Model

Published Pricing

Arcadia **does not publish pricing** for any enterprise solution. No contract values, pricing tiers, or benchmark figures were identified in any press release, procurement record, analyst report, or third-party source.

Arc Developer Platform Pricing (Inferred)

The Arc developer platform (Plug, Signal, Switch) uses API consumption-based pricing. Signal's pricing has historically been compared to Genability's pricing by developers, suggesting a per-call or volume-tiered model. No specific figures are publicly available.[^19]

Enterprise Solutions Pricing (Inferred)

Based on the use case profile — multi-site enterprises with thousands of accounts — the enterprise pricing model is likely:

- **Per-account or per-meter** pricing for Utility Bill Management (analogous to EnergyCAP's $40–52/meter/year model)
- **Percentage of procurement spend or advisory fee** for Energy Procurement Advisory (standard in energy broker/advisory models)
- **Flat or tiered SaaS fee** for Sustainability Reporting, potentially bundled

Cox Enterprises' case study ($120M annual energy spend, 40,000 accounts) represents a very large enterprise contract. Arcadia's positioning ("built on a decade of providing utility bill data automation for leading Fortune 500 brands") suggests the enterprise tier targets $50M+ energy spend portfolios. This implies pricing that would be out of reach for Clear Current's mid-market targets (e.g., community colleges, regional health systems, mid-size commercial real estate).[^31]

Competitive Pricing Position vs. EnergyCAP

At EnergyCAP's ~$40–52/meter/year benchmark, a 5,000-meter university would pay $200K–260K/year. Arcadia's enterprise pricing is likely *higher* given the data infrastructure cost and premium AI positioning, making it similarly inaccessible to budget-constrained public institutions.

Section 6: Geographic Focus and Market Coverage

U.S. Coverage

Arcadia claims 95%+ coverage of U.S. utility billing accounts for both residential and commercial. However, gaps exist:[^28]

- **Interval data:** Only ~400 of thousands of U.S. utilities support interval data pull as of January 2026, though this covers the largest utilities serving most commercial load[^16]
- **Manual utility portals:** Some utilities (particularly smaller municipal utilities, rural co-ops, and state agencies) may not have machine-readable portal access
- **Data freshness:** Utility portal scraping can lag or fail when utilities change their login portals — a documented operational challenge in the utility data aggregation space

ERCOT/Texas Deregulated Market

With the RPD Energy acquisition (Houston-based), Arcadia now has procurement advisory capabilities in ERCOT. However, ERCOT has no utility tariff structure in the traditional sense (prices are market-based, not regulated rate schedules), which means Arcadia's tariff database advantage (the core of Signal and ATA/CVS) is less applicable there for supply pricing. Arcadia's tariff intelligence would still apply to transmission and distribution charges from wires-only utilities (Oncor, CenterPoint, etc.) within ERCOT's footprint, but the supply pricing complexity of 100+ retail electricity providers is a different analytical problem than tariff classification. [^39][^24]

International

Arcadia claims coverage in 52 countries for utility billing data. The Sweep partnership (April 2024) was explicitly described as Arcadia's "first EU-based software vendor partner" and supports CSRD compliance. Enterprise solution availability outside the U.S. is not documented — the enterprise pivot appears U.S.-focused. [^40][^28][^20]

Section 7: Strategic Weaknesses and Competitive Gaps

This section is the most actionable for Clear Current's go-to-market strategy. The following gaps are assessed with HIGH confidence based on product documentation, customer evidence, and job postings.

Gap 1: No Higher Education Customers or Vertical Expertise

Zero documented higher education customers exist. Arcadia's customer stories, case studies, press releases, and marketing materials show corporate enterprises (Cox, Fox, Arconic) and Fortune 500 procurement relationships. There is no evidence of university, community college, or K-12 school district customers anywhere in Arcadia's documented customer base.[^41][^20]

Higher education has distinctive requirements — district energy systems, cost allocation across schools and departments, public procurement compliance, fiscal year reporting that doesn't match calendar year, and ENERGY STAR Campus reporting — that Arcadia's corporate-oriented platform does not address. **This is Clear Current's clearest vertical differentiation opportunity.**

Gap 2: No Campus Chargeback or Cost Allocation

Arcadia's enterprise UX is designed for a single corporate entity managing its utility spend. There is no evidence of chargeback functionality (allocating energy costs to sub-tenants, academic departments, or campus buildings with district energy corrections). This is table stakes for universities and multi-tenant commercial real estate but absent from Arcadia's documented feature set.[^30]

Gap 3: No Regulatory Intelligence

Arcadia monitors tariff *changes* (75,000 updates/month) but there is no documented feature for proactive regulatory intelligence — monitoring PUC filings, tracking proposed rate cases, alerting customers to pending rate hike proceedings, or providing context on why rates are changing. This is a pain point for sophisticated commercial energy buyers, especially utilities with active rate cases in California, New York, and Illinois.[^17]

Gap 4: No Natural Language AI Chat Interface

Arcadia's AI capabilities are embedded in automated back-end processes (ATA matching, CVS calculation, gap-filling, data standardization). The user experience is a dashboard and data API. There is **no documented conversational AI interface** — no ability for an energy manager to ask "why did my bill spike 18% in October" or "what would my savings be if I shifted to TOU rate schedule E-19S?" in natural language. This is a genuine differentiation opportunity for a platform built for energy managers rather than data engineers.[^42][^17]

Gap 5: Legacy Platform Migration in Progress

Job postings as of late 2025 and early 2026 explicitly reveal that Arcadia is in the middle of a complex legacy platform migration — transitioning enterprise customers from older Urjanet/Arc workflows to a new flagship platform. One active Senior Product Manager job description states the primary responsibility is "migrating complex customer workflows and data from a legacy platform to our flagship platform" and "working closely with enterprise customers to ensure they have consistent and clear communication about migration processes". This is a significant operational distraction and a sales vulnerability window.[^43][^44]

Clear Current implication: Arcadia's enterprise team is simultaneously selling new accounts, migrating existing accounts from legacy systems, and building new AI features. This creates focus fragmentation that an agile startup can exploit.

Gap 6: Enterprise Pivot is Very New

The Enterprise Solutions suite launched January 29, 2025. As of April 2026, it is 14 months old. The procurement advisory product was strengthened by the RPD Energy acquisition in February 2025 but the bill management and sustainability reporting modules are still maturing. Customer references for the enterprise platform are sparse — only Cox Enterprises and Arconic have detailed case studies for utility data management.[^21][^31]

Gap 7: No Dedicated Energy Manager-Facing UX

Arcadia's platform was built for developers and data engineers. Cox Enterprises explicitly chose Arcadia as a **data feed** because competing products "wanted to put us on their dashboard" — i.e., they found Arcadia's dashboard insufficient and preferred to build their own. This is a structural limitation: the platform is architected for API consumers, not energy managers who want actionable insights without engineering overhead.[^31]

Gap 8: Consumer Business Reputation Drag

Arcadia's consumer Trustpilot rating is 2.6/5 stars across 1,159 reviews and has 25+ BBB complaints. These reviews describe billing confusion, unauthorized charges, poor customer support, and lack of billing transparency in Arcadia's community solar product. While these are consumer complaints rather than enterprise reviews, they exist in the same Google search results as Arcadia's enterprise marketing. For procurement committees evaluating vendors, this public reputation risk matters.[^38][^36]

Gap 9: No Deregulated Supply Contract Audit Capability

Arcadia's Energy Procurement Advisory helps clients *structure* energy supply contracts. But there is no documented capability to **audit existing supply contracts** for billing errors — a separate and high-value problem in deregulated markets where retail electricity providers bill on complex index-based, block-and-index, or swing-tolerance structures. This is a clear opportunity for Clear Current in deregulated markets (ERCOT, PJM, NEISO, NYISO).

Section 8: Hiring Patterns and Product Roadmap Signals

Current Open Roles (Greenhouse.io, April 2026)

Arcadia is currently hiring across three clear functional areas:[^45]

Engineering/AI (heaviest hiring area):

- Director, Data Engineering & Analytics ($183,000–329,000 at Indeed)[^12]
- Senior AI Engineer
- Senior Engineering Manager - AI
- Staff Software Engineer ($165,000–295,000)[^12]
- Software Engineer II - Salesforce (signals a Salesforce CRM integration)
- Data Engineer III (Perch Energy subsidiary)

Enterprise Sales:

- Account Executive, Enterprise Solutions

Product:

- Senior Product Manager - AI
- Senior Product Manager - Tariff Data
- Senior Product Manager - Tooling and Telemetry
- Senior Product Operations Manager

What the Hiring Pattern Reveals

AI infrastructure is the top priority: The concentration of AI engineering roles (Senior AI Engineer, Senior Engineering Manager - AI, Senior Product Manager - AI) confirms Arcadia is investing heavily in AI-embedding within its existing data products. The stated direction from CEO Kiran Bhatraju — "Building energy superintelligence in 2026" and describing the "Arcadia Data Model" as an enterprise system of record — is being staffed.[^45][^42]

Tariff data is a sustained investment area: A Senior PM for Tariff Data signals ongoing expansion and improvement of the Signal tariff database — Arcadia's primary technical moat.[^46][^47]

Legacy platform migration is a current constraint: Multiple product and engineering roles explicitly mention migrating customers from legacy platforms — this is not future roadmap but current operational burden. One PM description says the role will eventually "transition to contributing to new features...that streamline complex commercial bill pay and energy procurement processes" — but only after migration is complete.[^44][^43]

Enterprise sales hiring is thin: Only one Account Executive, Enterprise Solutions position is open. This suggests either (a) Arcadia's enterprise sales motion is more land-and-expand via data partnerships than field sales, or (b) they are selling primarily through existing customer relationships from Urjanet and RPD Energy rather than building a large outbound sales force.

Domain specialists vs. technologists: The Account Executive role explicitly requires "a background in retail energy markets, retail energy brokerage or corporate energy procurement" — suggesting Arcadia recognizes it needs energy domain expertise in sales. However, the engineering and product roles show no requirements for energy domain knowledge. This creates a potential credibility gap in customer-facing technical discussions.[^48]

Salesforce integration: The Software Engineer II - Salesforce role confirms Arcadia is building deeper Salesforce CRM integration, likely for enterprise customer management and sales pipeline tracking — a sign of scaling enterprise go-to-market infrastructure.[^45]

No energy domain specialists in engineering: There are no postings for Certified Energy Managers (CEM), utility regulatory analysts, or former utility executives in engineering or product roles. The energy domain expertise sits in the 50+ tariff analysts (operations) and the RPD Energy acquisition team (procurement), not in product development.[^17]

Section 9: Partnerships and Integrations

Documented Partnerships (2024–2026)

Partner	Type	Date	Strategic Significance
Cority	EHS/ESG software integration	Feb 2025[^28][^49]	Positions Arcadia as utility data layer inside enterprise EHS platforms; expands to Cority's customer base
Sweep	EU sustainability platform	Apr 2024[^29][^40]	First EU partnership; targets CSRD compliance; expands to European enterprises
Perch Energy	Community solar JV (spinout)	Mar 2025, closing Q2 2026[^13][^50]	Separates community solar from enterprise platform; largest community solar platform in U.S.; 3GW across 1,000 projects
RPD Energy	Acquisition	Feb 2025[^24]	Insources renewable procurement advisory capability; brings Rohit Marwaha (Constellation veteran) and blue-chip customer relationships
WatchWire by Tango	Data supply (Arc Plug customer)	Ongoing[^31]	Arcadia supplies utility data to a competing energy management platform — upstream supplier relationship
Verse	Data supply (Arc Plug customer)	Ongoing[^31]	Same pattern — Arcadia supplies data to a competing energy analytics startup
SoftwareOne Marketplace	Distribution	Listed 2024[^22]	Listed on SoftwareOne Marketplace for enterprise discovery
300+ unnamed partners	Partner network	Ongoing[^20]	Mostly developers building on Arc APIs; some energy consultants and retail energy providers

Notable Integration Gaps

- **No documented ERP integrations:** No SAP, Oracle ERP, Microsoft Dynamics, or Infor integrations are mentioned in any product documentation or press release. This is a gap for enterprise buyers whose finance teams work in ERP systems.
- **No utility-side partnerships:** Arcadia does not appear to sell to utilities (the Plug product relies on credential scraping and Urjanet's pre-built connections rather than utility partnerships). Bhatraju's pre-Arcadia company sold to utilities; his explicit pivot was away from utilities.[^3]
- **No CMMS integrations:** No documented integration with facility management / CMMS platforms (Maximo, Archibus, FM:Systems), which is a key integration point for facility-manager-driven buyers.
- **No higher education consortia partnerships:** No documented partnerships with NACUBO, APPA, E&I Cooperative Services, or Educause, which are the primary distribution channels for higher education software.

Section 10: Leadership and Founding Team

Key Executives

Name	Role	Background	Notable
Kiran Bhatraju	Founder & CEO	UPenn → Capitol Hill → American Efficient (utility software) → Arcadia 2014[^3]	Public voice of Arcadia; prolific LinkedIn poster and newsletter contributor on energy strategy
Kate Henningsen	Co-Founder & COO	Corporate litigation lawyer → Arcadia 2015[^3]	Operations and legal; low public profile
Paul Mulé	CFO	—	Current CFO per Growjo[^1]; replaced John Rucker (joined 2021 from Slice)
Barbara Clay	CLO	—	Chief Legal Officer per Arcadia llms.txt[^20]
Patrick McCamley	President	Skyline Partners International (partner) → Arcadia CCO 2020 → President 2022[^3]	Commercial leadership
Curtis Snyder	SVP & GM, Arc Enterprise Platform	—	Leads enterprise platform; named at Growjo[^1]
Joel Gamoran	VP & GM, Energy Services	—	Energy services division leadership[^1]
Rohit Marwaha	Procurement Leadership	Constellation Energy veteran → RPD Energy → Arcadia Feb 2025[^24]	Most senior energy-domain hire; heads procurement advisory

CEO Public Statements and Strategic Direction (2024–2026)

Bhatraju's public communications show a clear strategic narrative: **"energy scarcity"** as the organizing frame for the enterprise pivot. Key quotes and publications:

- January 2025 enterprise launch: "We're about to enter a world of energy scarcity... Now is the time for businesses to rethink, manage, and streamline their energy strategies"[^21]
- March 2025 Latitude Media article: "The old model is broken — reliant on manual processes and narrow solutions that lack data-driven insights"[^51]
- May 2025 MCJ Newsletter: Detailed piece on corporate energy strategy in the face of AI-driven demand growth and grid constraints[^52][^53]

- January 2026 LinkedIn: "Building energy superintelligence in 2026" — describes creating the "Arcadia Data Model" as an "enterprise system of record"[^42]
- March 2026: Named TIME/Statista Top GreenTech Companies for third consecutive year[^54]

Assessment: Bhatraju is positioning Arcadia as the enterprise system of record for energy data — a platform play, not a point solution. This is an ambitious positioning that directly competes with EnergyCAP's installed base and with the emerging category of AI-native energy management. However, the "system of record" positioning requires deep enterprise workflow integration that Arcadia's current product does not yet fully deliver.

Section 11: Positioning and Messaging

Pre-Pivot Positioning (Before Jan 2025)

"Global utility data and community solar platform" — emphasizing community solar access and developer APIs[^10]

Post-Pivot Positioning (Jan 2025–Present)

"Global utility data and energy solutions platform" — emphasizing enterprise energy management and the "energy scarcity" macro narrative[^55][^21][^20]

Current Messaging Pillars (from website and recent materials)

1. **Comprehensive data:** "9,500+ utilities, 52 countries, 95%+ US utility account coverage, 30K+ tariff structures"[^20]
2. **AI-powered accuracy:** "98.4% median bill recreation accuracy... 99% at portfolio scale"[^17]
3. **End-to-end solutions:** "From raw data access via APIs to full-service enterprise energy management"[^20]
4. **Energy scarcity framing:** "Businesses need to navigate increased energy complexity" — positioning energy management as a strategic, not operational, function[^56][^21]
5. **Decade of data:** "Built on a foundational data platform developed and scaled for over a decade"[^21]

Competitive Differentiation from EnergyCAP

Arcadia does not mention EnergyCAP by name in any public materials found. The implicit differentiation is:

- **Coverage scale** (95%+ US accounts vs. EnergyCAP's bill entry model)
- **AI automation** vs. EnergyCAP's more manual data entry workflow
- **Procurement advisory** — EnergyCAP does not offer procurement services
- **ESG/Scope 2 integration** — positioned for modern sustainability reporting requirements

How Clear Current Should Respond to Arcadia's Positioning

Arcadia's messaging creates a specific vulnerability: it emphasizes data scale and platform breadth, implying that any energy management problem can be solved with the right data infrastructure. This leaves space for Clear Current to own **domain-specific workflow depth** — the actual energy manager experience of managing district energy allocations, navigating PUC proceedings, understanding ratchet clause mechanics, or getting a plain-language explanation of why a demand charge spike occurred. Arcadia's platform knows what the data says; Clear Current can know what the data *means* to an energy manager.

Section 12: Recent News and Strategic Moves (April 2025 – April 2026)

Date	Move	Significance
Jan 2025	Enterprise Solutions launch (Utility Bill Management, Energy Procurement Advisory, Sustainability Reporting)[^21]	The enterprise pivot; defines Arcadia as direct competitor
Feb 2025	RPD Energy acquisition; Rohit Marwaha joins[^24]	Bolsters procurement advisory; insources Constellation-level energy expertise
Feb 2025	Cority partnership (EHS/sustainability integration)[^28]	ESG channel expansion
Mar 2025	Perch Energy community solar JV announced (closing Q2 2026)[^13]	Spin-out community solar to focus enterprise resources
Mar 2025	Named TIME Top GreenTech for second year[^57]	Brand recognition; no financial significance
Q4 2025–Q1 2026	Interval data expanded from 60 to ~400 utilities[^16]	Major platform infrastructure investment; enables demand charge optimization at scale
Jan 2026	First Annual Commercial Electricity Rate Report published (81 utilities, 5 building types, 2020–2025)[^58][^59]	Content marketing targeting enterprise energy buyers; 97.5% of facilities saw rate increases; median CAGR 5.9%
Jan 2026	"Energy Data Model" blog post; CEO announces "energy superintelligence" direction[^42]	Signals platform product direction for 2026
Mar 2026	Named TIME Top GreenTech for third year[^54]	

Analyst Coverage Assessment

- **Verdantix:** Arcadia does **not** appear in the 2026 Verdantix Green Quadrant: Enterprise Carbon Management Software, which recognized Cority, IBM, Schneider Electric, Sphera, Sweep, UL Solutions, Watershed, and Wolters Kluwer as leaders. This is notable given that Arcadia partners with Cority and Sweep, both of which

are recognized. Arcadia's positioning is as a data infrastructure layer *beneath* these platforms, not as a carbon management platform itself.[^60]

- **Gartner:** No Gartner Magic Quadrant mention found for any relevant category.
- **Forrester:** No Forrester Wave mention found.
- **TIME/Statista:** Three consecutive years on Top GreenTech list — editorial recognition, not analyst assessment.[^54]

Strategic Implications for Clear Current Technologies

Where Clear Current Can Win Against Arcadia

1. Higher education as an uncontested vertical: Arcadia has zero documented higher education customers. The specific requirements of universities — district energy cost allocation, departmental chargeback, ENERGY STAR Campus, fiscal year reporting, public procurement compliance — are absent from Arcadia's product. A focused higher education play gives Clear Current a beachhead where Arcadia cannot compete without significant product development.

2. Energy manager UX vs. data engineer UX: Arcadia built for developers. Clear Current can build for energy managers — the person who actually receives a 3,000-line utility bill, doesn't know why the demand charge is 40% higher than last month, and needs a system to tell them in plain language. A conversational AI interface for non-technical energy managers is a structural advantage over Arcadia's API-first DNA.

3. District energy and campus billing complexity: Arcadia has no documented capability for district energy cost allocation — the complex problem of fairly distributing chilling plant, steam plant, and central utility costs to individual buildings or departments. This is table stakes for university and hospital energy management.

4. Regulatory intelligence: No competitor — including Arcadia — offers proactive PUC monitoring and rate case intelligence. This is an unoccupied product category that creates significant value for large commercial energy buyers facing rate increases.

5. Mid-market and institutional pricing: Arcadia's enterprise tier appears priced for Fortune 500 complexity. A cleaner, more affordable product built for mid-to-large organizations ($5M–$50M annual energy spend) with purpose-built vertical workflow leaves the door open for Clear Current to price competitively in markets Arcadia ignores.

6. The migration window: Arcadia is actively migrating enterprise customers from a legacy platform while simultaneously trying to launch new enterprise features and acquire new enterprise customers. This is a 12–24 month window of competitive vulnerability. Arcadia's attention is divided; Clear Current's can be focused.[^43][^44]

7. Deregulated supply contract auditing: Arcadia's Utility Bill Management audits utility bills from regulated utilities. It does not appear to audit third-party retail electricity provider contracts in deregulated markets for billing accuracy against contract terms (wrong index pricing, incorrect swing tolerances, volumetric billing errors). This is a specific capability gap in ERCOT, PJM, and other deregulated markets.

Where Arcadia Has Durable Advantages

- **Data coverage breadth:** 9,500+ utilities and 95%+ commercial account coverage is a genuine moat. Replicating Urjanet's network would take years and hundreds of millions of dollars.[^34]
- **Tariff database depth:** 30,000+ tariff structures, 75K monthly updates, 50+ tariff analysts. This is the core Signal product and represents a decade of investment.[^17]
- **Capital position:** ~$650M raised, $1.5B last known valuation, JPMorgan credit facility. Arcadia can sustain losses to expand into new verticals.[^5][^1]
- **Fortune 500 brand relationships:** RPD Energy's existing customers (Iron Mountain, Adobe, Intuit, UPS) and Urjanet's Fortune 500 installed base give Arcadia enterprise credibility Clear Current must earn.

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DOCUMENT 10: Compass — Six-Thread Secondary Brief

Source file: Compass — Six Thread Secondary Brief.md

Six-thread research brief for Clear Current Technologies

Prepared: April 16, 2026 | **Deliverables due:** April 20, 2026 **Purpose:** Fill intelligence gaps for Engagement Trigger Map, Seasonal Energy Management Calendar, User Journey Maps, Opportunity Heatmap, and Product Module Recommendations.

Note on execution: Three of six subagent threads (1, 2, 4, 5) completed successfully; Threads 3 and 6 experienced infrastructure errors during subagent delegation. Thread 3 (switching/churn) was compiled from direct search results and triangulated from Thread 1's competitive data; Thread 6 (LL97/BPS calendar) was completed via direct web research. All findings below include source citations and source-quality flags.

Thread 1 — Corporate multi-site energy manager persona

Key findings

Role ownership is fragmented; a dedicated "Energy Manager" title is rare below 500 sites. In 100–500 location organizations, utility bill ownership splits across **AP/Finance (pays the bill), VP of Operations or Director of Facilities (handles exceptions), and an outsourced bill-audit vendor** (ENGIE Impact, UtiliSave, Cass, SpyGlass, Spring Global). Named Energy Manager roles exist at <30% of mid-market multi-site operators based on LinkedIn title prevalence. A qualified in-house energy manager commands **$120–160K base, ~$200K fully loaded** (GWT2Energy, serving Panda Express, Red Robin, Taco Bell) — which is why outsourcing is the default below ~1,000 units.

Convenience retail archetype: Wawa created a formal energy group only after **12–15% YoY utility cost increases for five consecutive years**; the lead was titled "Energy & Petroleum Operations Manager" (Scott Boorse), not Energy Manager. Hotels at enterprise scale (Marriott, Hilton) run a **split model**: corporate VP of Global ESG sets targets; each property's Director of Property Operations/Engineering is accountable on kBTU/m² and CO_2e. Hilton has LightStay (ISO 50001-certified across 6,700+ hotels).

Workflow: A 200-site operator processes **400–1,000+ monthly utility invoices** across electric, gas, water, sewer, trash, telecom. Manual invoice entry consumes **~60% of AP staff time**; centralized automation cuts approval cycles from **20.8 days to 2–3 days** (Facilio). ENGIE Impact reports **~17% of utility invoices contain an exception or error**; UtiliSave's audit-hit rate is **94%** (16,000+ properties audited; $650M refunds recovered since 1991). EnergyCAP markets that **1 in 5 bills contain errors** and that 98% of users take monthly action on flagged bills, yielding ~7.5% YoY cost savings (vendor-attested).

Tool landscape by tier: EnergyCAP dominates government/education/healthcare/mid-market corporate, priced ~$2,600/yr (100 meters) to $9,000–$50K+/yr at portfolio scale; implementation $5K–$100K+. WatchWire leads REITs, retail/franchise, PE, data centers (~$100/mo entry). ENGIE Impact is the enterprise outsourced standard. Cass Information Systems dominates AP-centric retail/QSR. **Spreadsheets remain the default for 100–300 location operators that haven't adopted a platform — Clear Current's sweet spot.**

Procurement thresholds: Director $5–10K; VP/Sr. Director $25–50K; CFO $50–100K+; Board/multi-year >$250K. **Most energy SaaS deals for 100–500 sites land in the $15–75K ARR range** (EnergyCAP Enterprise starts $9K/250 meters; a 300-site operator with 600–900 meters lands $25–50K). This puts decisions at VP/Director level with CFO sign-off — not board.

Top buying triggers (ranked by case-study frequency): (1) **double-digit utility rate shock** — Wawa's 12–15% YoY trigger is the canonical example; (2) **M&A/portfolio chaos** where acquired sites' utility accounts are invisible; (3) **ESG/climate disclosure mandates** (California SB 253, CSRD, CDP) demanding audit-grade Scope 2 data; (4) **new sustainability or CEM hire** (software typically purchased within first 90 days); (5) **discovered billing error or fraud incident**; (6) **BPS compliance mandates** (LL97, BERDO, DC BEPS); (7) **CFO cost-pressure initiative** when utilities become the 2nd–4th largest controllable expense.

FTE benchmarks: IFMA North America cites **1 maintenance FTE per ~47,000 RSF** (general O&M, not energy-specific). Dedicated energy FTEs at 100–500 location operators are typically **0–1 internal**; effective locations-per-energy-FTE ratio is **200–500:1** or fully outsourced. GWT2Energy benchmarks suggest an in-house energy team is justified only at **1,000+ corporate-owned locations** (3–4 person team).

Sources: AEE 2015 Jobs Report; Hilton CDP 2023; Wawa/CSTORE Decisions; UtiliSave; ENGIE Impact; Facilio; EnergyCAP pricing/marketing pages; IFMA O&M benchmarks; GWT2Energy; Tallyfy/Procurify/Moxo approval-matrix data.

Source quality flags: Wawa data is 2008-era. Approval thresholds synthesized across SaaS procurement vendor blogs (directional, not survey-validated). "7.5% savings" and "1M errors flagged" are EnergyCAP marketing claims, not independently verified. No authoritative survey publishes "energy FTE per 100 locations" for QSR/c-store.

So what for Clear Current

The ICP is a **VP of Operations or Director of Facilities at a 150–400 location operator** (QSR, casual dining, c-store, fitness, small hotel groups) currently stitching together spreadsheets, AP clerks, and a legacy bill-audit consultant — below the ~1,000-unit threshold where a dedicated Energy Manager is justified. The three sharpest triggers — **double-digit rate shock, M&A-driven portfolio chaos, and new ESG/finance mandates** — push decisions into the **$15–75K ARR VP/CFO band**, below board scrutiny. Position against the *status quo* (spreadsheets + Cass/ENGIE/SpyGlass outsourcing) with a land wedge of "bill-shock prevention + GL-coded ERP-ready data" — not head-on against EnergyCAP in its entrenched government/education/healthcare base.

Thread 2 — Vertical SaaS investor benchmarks (seed → Series A)

Key findings

NRR compressed meaningfully from 2021 peaks. Median NRR across private B2B SaaS is now **101–104%** (Benchmarkit 2025, n≈900–1,600; Pavilion 2025), down from ~110% in 2021. By ACV tier: SMB (<$12K) ~100% median; mid-market ($25K–100K) **103–108%**; enterprise (>$100K) **110–115%**. Energize Capital benchmarks climate/energy software separately at **~130%+ NRR** at Series B (strong land-and-expand via multi-BU replication). Hybrid subscription+usage pricing delivers the best NRR in the 2025 dataset (110% median).

GRR median is 88% across B2B SaaS; the Series A bar is 90%+, best-in-class 95%+. For ACVs above $100K, median GRR reaches 90%+ with top-quartile 95%+. GRR below 85% is a documented red flag for Series A in vertical/enterprise SaaS. Proptech-specific benchmarks from MetaProp/Fifth Wall cite **92%+ annual logo retention** and **<8% annual churn** as Series A readiness thresholds.

CAC payback has lengthened substantially. All-company median jumped from 12–14 months historically to **18–20 months in 2024** (Benchmarkit, KeyBanc 15th Annual). At enterprise ACV (>$100K), median is **~24 months** and can exceed 30. The NRR-adjusted guidance (Kyle Poyar): <100% NRR → target <12mo; 100–120% NRR → 12–18mo acceptable; 150% NRR → 18–24mo acceptable. Magic Number median fell to 0.90 in 2024; New CAC Ratio median is $2.00 of S&M spent per $1 of new ARR.

Series A ARR bar has risen to $1.5–2M median for B2B SaaS in 2025 (some at $1M; crowded categories demanding $2–3M). Growth expectations: 2–3x YoY minimum; ICONIQ 2025 reports top-quartile $1–10M ARR companies grew 515% YoY (AI-skewed). Series B has reset from $3–5M pre-correction to **$5–8M ARR** today. **Energize Capital explicitly rejects T2D3/T3D3 for climate SaaS** — median climate Series B YoY growth is **~70%**, gross margins **60–80%** (not 80%+), but growth decay is <5% and capital efficiency is 8× better than public climate peers ($100M gross profit on $88M capital vs. $704M).

LTV:CAC: 3:1 minimum healthy; 2024 median 3.6:1; best-in-class 5:1+. Less diligenced at seed (too noisy); CAC payback + NRR + GRR combo preferred.

Gross margins: Target 75–80%+ subscription for traditional SaaS; **climate/energy SaaS accepted at 60–80%** with explicit justification (data integration, implementation services, hosting). Below 60% subscription GM triggers "is this really software?" concern. Services revenue median is 15% of total.

Red flags (cited repeatedly across 2025 diligence sources): customer concentration >10–15% from any single logo; services revenue bundled into ARR; CAC payback >18mo mid-market / >24mo enterprise; burn multiple >2.5x at Series A (median now 1.6x); GM below 70% total / 60% subscription; GRR <85%; NRR <100%; lengthening sales cycles without cohort improvement. **Green flags**: net-negative churn, multi-year auto-renewal contracts, 3–5 referenceable enterprise logos pre-A, burn multiple <1.5x, hybrid pricing, clean ARR bridge.

Sources: Bessemer Cloud 100 Benchmarks 2024 (no "State of the Cloud 2025/2026" published); ICONIQ State of Software 2025; SaaS Capital 2023/2025; Benchmarkit 2025; KeyBanc 15th Annual; High Alpha 2025 (OpenView-successor dataset, n=800+); Energize Capital "10 Ways to Win in Climate Software"; MetaProp/Fifth Wall; CFO Advisors 2025.

Flags: Energize's climate benchmarks are ~3 years old but still authoritative. ICONIQ's 515% top-quartile growth is AI-skewed — traditional top-quartile is closer to 150–200%. Proptech numbers from Qubit/Ellty are database-derived, not primary publications.

So what for Clear Current

Clear Current should privately target **$500K–$1M ARR, 3–5 referenceable enterprise logos, 90%+ GRR, 110%+ trending NRR, and burn multiple <1.5x** to credibly reach Series A in 12–18 months at the $1.5–2M median bar — but pitch against the **Energize climate-SaaS benchmark set** (70% YoY growth, 60–80% GM, <5% growth decay), not T2D3. The highest-leverage metrics to overperform on are **NRR (target 115%+), CAC Payback (<18mo despite enterprise ACVs), and GRR (95%+)** — these three drive the most variance in 2025 Series A valuations. Preempt the three deal-killers: **no customer >15% of ARR, zero services revenue bundled into ARR, subscription GM above 60%**.

Thread 3 — Churn, switching costs, and competitive displacement

Key findings

EnergyCAP ownership context (critical for trigger mapping): Founded 1980 (FASER); acquired by Enron 1996; repurchased post-bankruptcy; renamed EnergyCAP in 2010. **Resurgens Technology Partners took a strategic equity investment in March 2021** (not "Resolute" — this was a common misattribution; Resolute Capital Partners is an unrelated Nashville PE firm). Resurgens' explicit thesis includes **"execute acquisitions to expand product breadth"** — meaning EnergyCAP is an active roll-up platform, creating integration disruption opportunities for challengers. EnergyCAP reports 2,100+ organizations and 10,000+ energy managers, claims "top-decile customer satisfaction and retention."

Switching friction is high but manageable. Primary costs: (1) **historical utility bill data migration** (often 3–5 years of meter-level bills); (2) meter inventory re-mapping; (3) retraining of Facilities + AP users; (4) re-integration with ERP (NetSuite, SAP, Oracle, Sage Intacct) and ESG/carbon platforms (Persefoni, Watershed, Workiva Envizi); (5) multi-year auto-renewal contract terms.

Published churn/NRR data is sparse — none of EnergyCAP, Gridium, WatchWire, Tango, or Brightly publish specific retention metrics. Triangulation: Capterra gives EnergyCAP 4.7/132 reviews, WatchWire 4.8/16 reviews — high satisfaction consistent with EnergyCAP's "top-decile retention" claim. Implied annual logo retention across category likely sits in the **90–95% band** for enterprise/mid-market energy SaaS, consistent with MetaProp/Fifth Wall proptech norms.

Contract terms: Multi-year is standard. EnergyCAP pricing model is **per meter-year** (~$40–52/meter); WatchWire uses per-account/per-meter with enterprise custom; Gridium roughly **$5,000–$6,000 per facility/year**; EnergyCAP base deployments start ~$5K/yr and scale to $50K+ for large portfolios. Implementation is a separate $5K–$100K+ one-time fee.

Switch triggers:

1. **PE-driven price escalation post-acquisition.** Resurgens' roll-up thesis at EnergyCAP creates predictable pricing pressure at renewal — a standard PE SaaS playbook.
2. **Version EOL / forced cloud migration.** EnergyCAP's transition from legacy on-prem (EnergyCAP 7) to cloud UtilityManagement created re-implementation pain for a portion of the base.
3. **Feature gaps in AI/billing error detection.** Parsepoint explicitly notes EnergyCAP lacks built-in OCR — "teams either enter data manually or purchase a separate OCR tool and build an integration."
4. **Personnel change** (new energy manager or CFO with a preferred tool).
5. **Mid-market mismatch.** Parsepoint: "For teams managing 10 to 50 sites, the cost-per-location math often doesn't work. EnergyCap's value proposition is built around large portfolios where the per-site cost is absorbed across hundreds or thousands of locations."

Winning displacement pitch vs. EnergyCAP specifically: (a) AI-native billing-error detection built in, not bolted on; (b) cloud-native architecture without legacy migration baggage; (c) shorter implementation than "months before teams see value"; (d) pricing that works for 100–400 sites rather than 1,000+. Competitors on G2/Capterra most frequently positioned as alternatives: WatchWire, Measurabl, Spacewell Energy, Energy Elephant, Brightly Energy Manager, Facilio.

Sources: EnergyCAP Wikipedia; Resurgens Tech Partners portfolio page; G2; Capterra; SourceForge; Parsepoint comparison guide; Software Equity Group transaction advisor announcement.

Flags: No public NRR/GRR for energy SaaS vendors — triangulated from review counts, vendor claims, and category norms. "Resolute" reference in the task brief was incorrect — **Resurgens** is the correct investor.

So what for Clear Current

The displacement motion against EnergyCAP is strongest at **100–400 site mid-market operators** where EnergyCAP's per-meter pricing math breaks down, its lack of native OCR forces bolt-on spend, and Resurgens' roll-up/re-pricing thesis creates renewal-window openings. Position on three wedges: **(1) AI-native billing-error**

detection replaces the EnergyCAP + separate OCR + audit-consultant stack; (2) cloud-native shortens implementation from "months" to weeks; (3) transparent all-in pricing under $75K ARR versus opaque per-meter escalators. The highest-probability switch window is **12–18 months after a PE re-pricing event or forced cloud migration** — build a renewal-cohort monitoring capability into demand generation.

Thread 4 — NG Insight / ENGIE Impact and the bill-pay ecosystem

Key findings

"NG Insight" is almost certainly a misreference to ENGIE Insight (now ENGIE Impact, formerly Ecova). Two unrelated entities use similar naming: (1) **Northrop Grumman "NG InSight"** — defense/C5ISR product line, irrelevant; (2) **ENGIE Insight / ENGIE Impact** — the utility-bill-management business. The latter is the only plausible target for Clear Current's market mapping.

ENGIE Impact profile: Acquired by GDF SUEZ (now ENGIE) from Avista in 2014 for **$335M**. At acquisition: 700,000+ client sites, 1,450 employees, $20B in utility expenses managed, ~$180M 2013 revenue. By 2018 rebrand: 900+ multi-site clients, $3.2B claimed savings delivered over 5 years, **25%+ of Fortune 500 as customers**. 1,000+ employees at Spokane HQ. Divested utility solutions (retrofits) to CLEAResult Q1 2018; retained software + managed services (Data Management/20/20 Platform, now ENGIE Ellipse). **Model:** full-service outsourced utility expense management — bill ingestion, audit, rate optimization, payment initiation, sustainability reporting.

Major players in the bill-pay / utility expense management ecosystem:

Vendor	Position	Scale
Cass Information Systems (CASS)	Publicly traded; St. Louis; largest U.S. transport + facility/ utility bill processor	**~$90B annual payments volume; ~50M invoices/yr; ~$130M TTM fee revenue; 1,027 FTE**; acquired AcuAudit Q4 2024; facility dollar volume ~$5–6B/qtr with **+23–26% YoY growth** in 2024
ENGIE Impact	Enterprise outsourced (software + managed service)	900+ multi-site clients; 25%+ F500
Schneider Electric EcoStruxure Resource Advisor/RA+	Bundled with energy procurement; launched agentic AI "Sera" in 2025	**Processes 39M+ utility bills/year**; ~400 AI professionals globally
Arcadia (Arc platform, acquired Urjanet May 2022)	Largest utility data aggregator — not direct bill-pay but the dominant data-API layer	**>95% of US residential + commercial utility accounts**; 9,500+ utilities in 52 countries; Arcadia valued ~$1.45B
EnergyCAP	Utility accounting SaaS + Bill CAPture / Bill Pay / Vendor Mgmt Services	2,100+ customers
Vervantis	Mid-market bill processing for C&I	1M+ invoices/yr; $3B+ managed
RealPage Utility Management	Multifamily-focused	2.2M+ submetered devices across 4,000+ properties
Yardi Energy Suite	Bundled with Voyager property mgmt	Multifamily/CRE
Paymentus (PAY)	B2C utility payment platform	20+ years in utility; 450+ integrations

Additional niche players: Cost Control Associates, UMC Solutions (EnergyCAP dealer), Commercial Water & Energy, Spring Global, SpyGlass Group.

Market size (wide variance; source methodologies differ):

- Market Research Future: **$2.89B (2024) → $6.86B (2035), 8.18% CAGR**
- Zion Market Research: $5.83B (2023) → $12.41B (2032), 7.8% CAGR
- Verified Market Reports: $1.2B (2024) → $2.5B (2033), 8.9% CAGR
- Research and Markets: $4.49B (2025) → $6.43B (2031), 6.17% CAGR

North America holds ~38% share. **Treat these as directional ranges, not authoritative.**

Key 2024–26 trends: AI-integrated bill audit (Schneider RA+, EnergyCAP Watts AI, Vervantis, Oracle Opower); carbon/ESG reporting as primary demand driver; consolidation (Arcadia/Urjanet 2022, Cass/AcuAudit 2024); post-American Water Works ransomware (Oct 2024, 14M customers, 6-week disruption) raising cybersecurity concerns; smart-meter rollouts driving data volumes.

Integration path for Clear Current: Three options:

1. **Channel partner layer** on top of Cass/ENGIE/Schneider (they provide bill-pay transaction processing; Clear Current provides AI error detection)
2. **Direct competitor to audit function** within bill-pay bundles (riskier, longer sales cycle)
3. **API/data partner** to bill-pay vendors who lack AI

Sources: Cass 10-K filings (SEC); ENGIE press releases; Arcadia/Axios; Verdantix; market research firms.

So what for Clear Current

The likely "NG Insight" reference is **ENGIE Impact** — a **potential channel partner, not a pure competitor**, since their model is full-service outsourcing where AI billing-error detection would strengthen their audit economics. The cleanest GTM pattern is a **hybrid: compete directly at mid-market 100–400 site operators currently using spreadsheets or EnergyCAP, while pursuing API/data-partnership with Cass (50M invoices/yr) and Schneider (39M bills/yr)** — both are bill-pay aggregators whose audit layers are still largely rules-based and would value AI anomaly detection. Avoid ENGIE Impact and Cass as direct replacements; their enterprise customer relationships are too sticky — instead, integrate under them for the top 25% of F500 logos while winning the mid-market directly.

Thread 5 — Healthcare energy management

Key findings

Spend & intensity benchmarks:

- **U.S. hospital aggregate energy spend: ~$8.3B/year** (DOE); broader healthcare sector 4.1B sq ft spends **>$5B/year** on energy (DOE Better Buildings).
- **Energy = 1–3% of hospital operating budget**; an estimated **15%+ of profit margin** (DOE Better Buildings). The widely-misquoted "50% of budget" figure is actually **~50% of the facilities-department budget**, not organization-wide (ASHE/HFM Magazine).
- **Per-bed**: ~$10,900–$13,611/year electricity+gas (E SOURCE/DOE/EIA).
- **Per square foot**: **~$2.84/sf electricity + $0.94/sf gas ≈ $3.78/sf** (DOE/E SOURCE, 2003 CBECS basis — likely conservative today).

- **Order-of-magnitude 100-facility health system spend**: $3.78/sf × 100 facilities × ~300K sf avg ≈ **$110M/year** — fits HCA, CommonSpirit, Ascension, Providence profiles.

Energy intensity (ENERGY STAR Portfolio Manager, Aug 2024 national median table): General Medical/Surgical Hospital **site EUI 234.3 kBtu/sf, source EUI 426.9**; Medical Office 97.7 site / 232.8 source; Office 52.9 site / 116.4 source. **Hospitals use ~2.5–3× the energy intensity of commercial offices.** Older DataTrends data (n=4,939) showed hospital median source EUI 467, range <100 to >1,400.

Ownership: Default is **Facilities/Engineering (Director → VP Facilities → COO or Chief Facilities Officer)**. Exemplars: HCA runs facility mgmt via HealthTrust Supply Chain; Ascension via wholly-owned Medxcel Facilities Management; UPMC has a centralized Energy Management & Engineering Department. **Kaiser Permanente has a Chief Energy Officer + Executive Director of Environmental Stewardship — best-in-class, not typical.** Parkland (Dallas) hired a dedicated in-house energy manager in 2021 for ENERGY STAR certification. Practice Greenhealth 2023 benchmark: **~66% of respondents have appointed someone to lead sustainability at the facility level**; ~90% of top-performing institutions have an executive-level sustainability champion.

Regulatory/accreditation drivers:

- **Joint Commission Environment of Care standards**: EC.02.05.01 (utility systems) accounts for ~28% of all EC findings per OXMaint analysis; >65% of 2025-surveyed hospitals received EC citations (flag: vendor-sourced stat).
- **CMS Conditions of Participation §482.41**: Physical environment compliance tied to Medicare reimbursement — the biggest regulatory stick.
- **HHS Health Sector Climate Pledge** (launched Earth Day 2022): voluntary but reputationally significant; **143 organizations/960 hospitals signed by Nov 2024**; with federal (VHA, IHS, MHS) >1,200 hospitals (~19% of U.S. hospitals). Targets: 50% Scope 1+2 reduction by 2030, net zero 2050, Scope 3 inventory by end 2024. After 2025 federal admin change, Health Care Without Harm relaunched the **Health Sector CARES Pledge** to preserve the commitment pathway.
- **NYC LL97**: Hospitals NOT exempt. Nonprofits get an **Article 321 adjustment** (limit = 85% of 2018 emissions for 2024–2029; 70% for 2030–2034); adjustment applications closed **January 1, 2025**. Envigilance estimates a 200,000 sf hospital exceeding 2030 limit by ~1,100 tons could face **~$295K/year** in penalties.
- **California SB 253/261** (effective 2026 for large entities): pulls in large health systems with CA operations (Kaiser, Sutter, Providence, CommonSpirit/Dignity) for Scope 1/2/3 reporting.

Tools: Brightly (Siemens, formerly Dude Solutions) dominates healthcare CMMS — built-in ICRA/PCRA/ISLM workflows, Joint Commission compliance binder, 12,000+ clients, $321B in assets managed. **EnergyCAP** explicitly markets a healthcare vertical ($5K entry → $50K+ portfolios); emphasizes the "finance-facilities divide." Nuvolo (ServiceNow-based) for medical equipment/EAM. Accruent for space/lease. FM:Systems,

Archibus as IWMS generalists. **Spreadsheets + contingency-fee bill-audit firms (UtiliSave, ENGIE Impact, OEO, Honeydew, Emergent) remain dominant outside the top-50 systems.** Energy analytics / AI-driven optimization is a white space.

Billing error/overpay benchmarks (all vendor-marketed; flag heavily):

- UtiliSave (16,000+ properties audited): **overcharges for large institutions average 2.5–10%** of spend; **93% hit rate** on finding mistakes — most credible published figure.
- EnergyPrint: "~17% of utility invoices contain an error."
- OEO/National Utilities Refund: "80% of commercial invoices contain errors" — self-serving contingency marketing.
- Emergent Energy: "5–15% overpayment."
- **Implied healthcare overpay pool: $8.3B × 3% = ~$250M/yr recoverable; at 5% = ~$415M/yr.**

Pain points specific to hospital billing: multi-commodity complexity (electric/gas/fuel oil/steam/chilled water/water-sewer/medical gases); 24/7 demand charges ratcheted by ICU/OR/ER load (demand charges = 30–70% of commercial electric bills per Inertia Resources); tenant chargeback to physician MOBs; rate-schedule mismatches; frequent sales-tax/franchise-fee exemption errors on nonprofit accounts.

Market size: Verdantix: global smart-buildings software market **$6.9B (2022)**. Space/workplace software $1.1B (2022) → $2.1B (2027), 15% CAGR. Healthcare is a top-3 buyer vertical. **Narrow U.S. hospital energy/utility SaaS TAM: ~$60–300M/yr** (~6,000 hospitals × $10–50K ACV), expanding to **~$500M+** bundled with ambulatory/MOB/compliance modules.

Sources: ENERGY STAR Portfolio Manager PDF; DOE Better Buildings; ASHE; Practice Greenhealth; HHS OCCHE; Joint Commission; Brightly; EnergyCAP healthcare page; Verdantix; UtiliSave.

So what for Clear Current

Healthcare is **accessible but contested**: the pain is large and quantifiable (**~$8.3B U.S. hospital energy spend; 1–3% of operating budget; 2.5–10% overpay on large-institution bills; 30–40% EUI gap between median and top quartile hospitals**), and regulatory tailwinds are strong (960+ HHS Pledge hospitals, LL97 at $268/ton, Joint Commission EC.02.05 creating a compliance spine) — but Brightly/Siemens, EnergyCAP, and Nuvolo are entrenched. **ICP**: mid-to-large systems (20–200 facilities, $30–200M energy spend) that (a) signed HHS/CARES Pledge or operate in a BPS jurisdiction, (b) have a sustainability exec + facilities director, and (c) have fragmented utility data across legacy systems — **CommonSpirit, Providence, Advocate Health, NYU Langone, Mass General Brigham** rather than Kaiser (too mature) or small community hospitals (budget-constrained). The dollarized pitch is straightforward: sq ft × $3.78/sf × (3% billing-error recovery + EUI-gap consumption savings) → $2–10M/year identifiable per mid-size system, plus avoided LL97-style penalties — but expect **12–24 month sales cycles** and incumbent displacement as the real challenge.

Thread 6 — NYC LL97 and CRE energy compliance calendar 2025–2026

Key findings

NYC LL97 exact deadlines and penalties:

Date	Obligation
May 1, 2025	First LL97 compliance report due (for CY2024 emissions) — registered design professional-certified
June 30, 2025	60-day grace period end — unpenalized filing window
Aug 29, 2025	Extension application deadline ($60 fee)
Dec 31, 2025	Extended filing deadline for those granted extension
May 1, 2026	CY2025 LL97 report due
June 30, 2026	60-day grace period end for CY2025 report
Aug 29, 2026	Extension deadline for CY2025 report
May 1, 2027	First LL97 report due for buildings with 1 to <35% rent-regulated units
May 1, 2036	First LL97 report due for specific rent-regulated housing pathways
2030–2034	Stricter emissions caps (70% of 2018 baseline for nonprofit hospitals)
2050	Net-zero requirement

Fine escalation:

- **Emissions exceedance: $268/metric ton CO_2e over limit per year** (Class 2 violation)
- **Late filing: $0.50/sf/month** (applies even during grace period if filing later)
- **Failure to file: $10,000 plus additional $10,000 for non-compliance**
- **False filing: up to $500,000 and/or imprisonment**
- **Extension fee: $60** via BEAM Portal

Coverage: Approximately **50,000 buildings** covered (buildings >25,000 sf, or multiple buildings on same tax lot totaling >50,000 sf). Buildings account for **~70% of NYC GHG emissions**. In the first 2024 compliance period, roughly **70% of covered buildings are projected to comply without action**; ~25–30% require intervention. Article 321 pathway covers affordable housing and houses of worship with prescriptive ECM options.

LL84 (benchmarking, annual) and LL88 (lighting/submetering) run alongside LL97. LL84 annual due May 1; missing = $500/quarter up to $2,000/yr. LL88: $1,500/yr for missing reports, $500 per non-compliant tenant space. LL87 (energy audits every 10 years for >50K sf): $3,000 starting penalty.

Utility billing data feeds LL97 via ENERGY STAR Portfolio Manager (via LL84 benchmarking) → certified RDP report → DOB BEAM portal. Data accuracy failures expose owners to the $500K false-filing penalty. **Tenant cost pass-throughs** are contractually negotiated; tenants are responsible for 50–70% of building energy use but landlords own the obligation.

Other major BPS and CRE reporting deadlines 2025–2026:

Jurisdiction	Key 2025–26 Deadline	Penalty
GRESB 2026	Submission **July 1, 2026 (23:59 PDT)**; prelim results Sept 1; final Oct 1	Score impact only (competitive)
Boston BERDO 2.0	2025 emissions compliance began; **Aug 15, 2026 extended reporting deadline**; five-year verification cycles 2026/2031/2036/2041	$234/MT over limit; $1,000/day for >35K sf non-compliant; $300/day reporting violation (large) / $150/day (small); $1,000–$5,000 for unresolved discrepancies
Washington DC BEPS	First compliance cycle ends 2026	Up to **$10/sf** gross floor area — **$1M exposure for 100,000 sf building**
Denver Energize Denver	Annual benchmarking June 1; 30% reduction target by 2030	**$0.30/kBtu** over threshold
Washington State Clean Buildings	**June 1, 2026: Tier 1 compliance deadline** (>50,000 sf commercial)	Up to $5,000 + $1/sf/yr
Oregon statewide BPS	Benchmarking began Jan 2025; compliance 2028–2030	Up to $0.85/sf early-adopter incentive
Maryland BEPS (Climate Solutions Now Act)	2025 benchmarking began; **June 1, 2026 third-party verification of 2025 data**; 2030 compliance	**$230/MT CO_2e**
Chicago Energy Benchmarking	Annual reporting	Fines per ordinance
Seattle BEPS	Active	GHG-based, tiered
Philadelphia Building Tune-Ups	Annual benchmarking + periodic audits	**$300/day late**
Montgomery County MD BEPS	Active	Tiered
St. Louis BEPS	Active	Tiered
California AB 802	Annual benchmarking June 1 (>50K sf)	Enforcement via Energy Commission
California SB 253/261	**Effective 2026** for large entities — Scope 1/2/3 reporting	SEC-like attestation required
SEC Climate Disclosure Rule	**Stayed / effectively dormant as of April 2026**	TBD
CDP	July–August annual submission	Scoring only

40+ U.S. cities will have active BPS by end of 2026 (JLL research), up from 13 in early 2024. JLL reports **penalties increase an average of 82% between first and second compliance periods**.

How CRE managers are preparing: Most use ENERGY STAR Portfolio Manager for LL84/benchmarking baseline. Tool gaps: jurisdiction-specific metric conversions (GHG-based for NYC/Boston/Seattle/Chicago vs. EUI-based for Denver/Philadelphia/DC/Washington State) require separate tracking unless using a multi-BPS platform. Retro-commissioning agents and RDPs (Bright Power, ERS, Steven Winter Associates, Cozen O'Connor) are dominant compliance consultants. **Data quality is the binding constraint** — Boston BERDO program manager Aidan Callan: "Building owners have to be absolutely confident in the data that they're looking at if they're going to make well-informed decisions."

Sources: NYC Mayor's Office / NYC Accelerator; NYC DOB LL97 BEAM portal; Urban Green Council; Cozen O'Connor; Harris Beach Murtha; Phillips Lytle; Greenberg Traurig E2 Law Blog; NYC HPD LL97 FAQs (July 2025); RAND Engineering; CooperatorNews; Boston.gov BERDO; GRESB.com; GBCI.org; Envigilance BPS tracker; Environ Energy; Facilities Dive; JLL ESG & Risk.

Source quality: NYC primary sources (DOB, HPD, Mayor's Office, Urban Green Council) are authoritative. Envigilance/Environ Energy/Deepki/JLL citations are consistent but secondary. Boston BERDO details align across FirstService Residential, Facilities Dive, and Boston.gov.

So what for Clear Current

The GTM calendar has three clear annual pressure windows: **Q1 (Jan–Mar) for LL97 RDP engagement and data gathering; Q2 (Apr–Jun) for filing season (LL97 May 1, LL84/LL88 May 1, California AB 802 June 1, WA State Clean Buildings June 1, MD verification June 1, GRESB submission late June)**; and **Q3 (Jul–Aug) for GRESB July 1 and extension deadlines**. Clear Current should time demand generation around **late-February through April** (the panic window when owners realize their utility data is unreliable) and again **June–July** for GRESB participants. Product-wise, **jurisdiction-specific conversion logic (GHG-based vs. EUI-based) and an audit-trail for LL97 RDP-certification defensibility are table-stakes module features** — the $500K false-filing penalty makes data-accuracy attestation a CFO-level value prop, not a facilities-level one.

Cross-thread synthesis for the five deliverables

Engagement Trigger Map (Thread 1 + 3 + 6): The top-5 triggers ranked by recency and buying-intent signal strength are: (1) utility rate-shock event (double-digit YoY), (2) M&A/portfolio acquisition creating utility account chaos, (3) BPS filing panic window (Q1–Q2 annually), (4) EnergyCAP/incumbent post-acquisition renewal re-pricing, (5) new sustainability/CEM hire.

Seasonal Energy Management Calendar (Thread 6): Q1 planning/data-gathering; **Q2 filing crunch (May 1 LL97 + LL84, June 1 CA/WA/MD, July 1 GRESB)**; Q3 grace-period/extension handling + GRESB review; Q4 budget planning + ENERGY STAR recert.

User Journey Maps (Threads 1 + 5): Build 4–6 around (a) QSR/c-store VP Ops 200–400 units, (b) Director of Facilities casual dining 100–300 units, (c) Hotel Chief Engineer + corporate VP ESG split, (d) Health system Director of Facilities + Chief Sustainability Officer, (e) Optional: CRE asset manager in NYC/Boston/DC BPS jurisdiction.

Opportunity Heatmap (Thread 2 + 5): Healthcare TAM $60–500M narrow; Multi-site QSR/c-store biggest white space (spreadsheet-dominant); CRE in BPS jurisdictions has regulatory urgency but incumbent saturation (WatchWire, Measurabl).

Product Module Recommendations (Threads 3 + 4 + 6): (1) AI-native billing-error detection (wedge vs. EnergyCAP's missing OCR), (2) BPS-jurisdiction compliance module with GHG/EUI conversions, (3) ESG-ready Scope 2 data export (Persefoni, Watershed, Envizi connectors), (4) Cass/ENGIE Impact API integration to bolt on as an audit layer, (5) CFO-grade audit trail for LL97 $500K false-filing defense.